

UNIVERSITY DIGITAL SYSTEM



University of engineering and technology

Submitted by

Dua Ahmed

Roll No : 2024-SE-49

Arooj Rafique

Roll No : 2024-SE-05

Fizza Naeem

Roll No : 2024-SE-23

Submitted to

Sir Zeeshan



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1. Vision

To provide a single digital platform that enhances the efficiency, accessibility and management of university operations by countering the shortcomings of current systems.

2. Scope

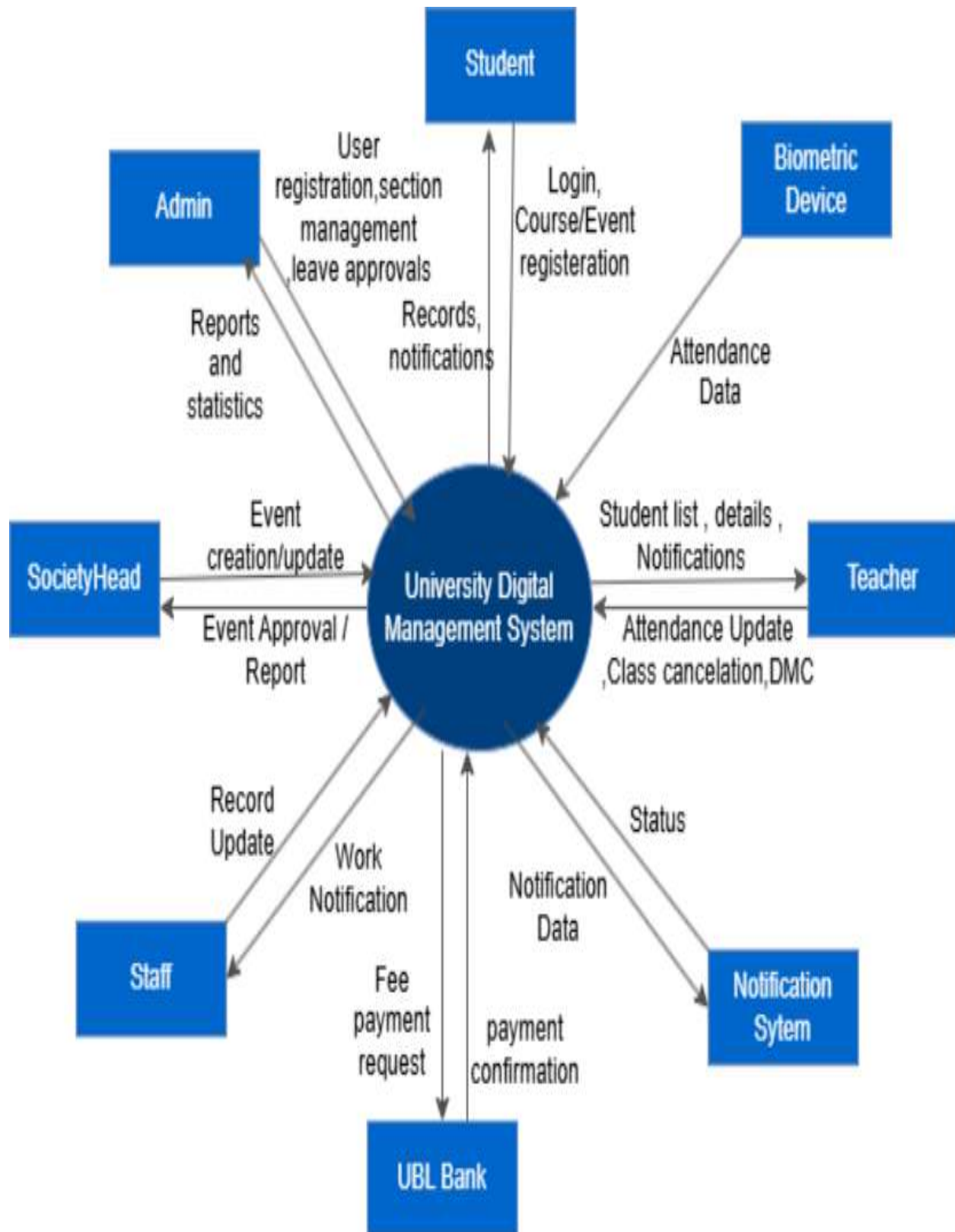
The UET Digital Management system encompasses the area of automation and integration of a number of academic, administrative and general university processes under a single digital application. The system has been arranged into two big parts, a general system and a role-based special system. The general system offers standard services, which do not require any logging in, and are intended to assist in how the campuses run and make the services convenient to users. It has features dealing with biometric attendance, university societies, teacher availability, emergency alerts, campus navigation, classroom availability as well as schedule viewing. These characteristics improve accessibility and experience in the campuses. The specialized system allows role-based access to the users such as students, teachers, administrators, society heads, and staff members, with each role being assigned particular functionalities. The student portal enables the analysis of attendance, course registration, leave management, viewing of results, fee management, society participation, submission of feedback and notification. The teacher portal also allows marking of attendance, managing results, scheduling meetings, managing leaves and communication. The administration portal is concerned with system management, such as user management, timetable preparation, attendance management, fee management, leave management, and society management. The society head portal will handle members, events, task allocation, and communication and the staff portal will assist with the operational tasks and designated duties. The system has an overall benefit of increased efficiency, communication, and data management in the university. Its coverage is, however, restricted in its internal digital operations but not external systems and physical infrastructure control.

3. Stakeholders and Requirement Gathering Techniques

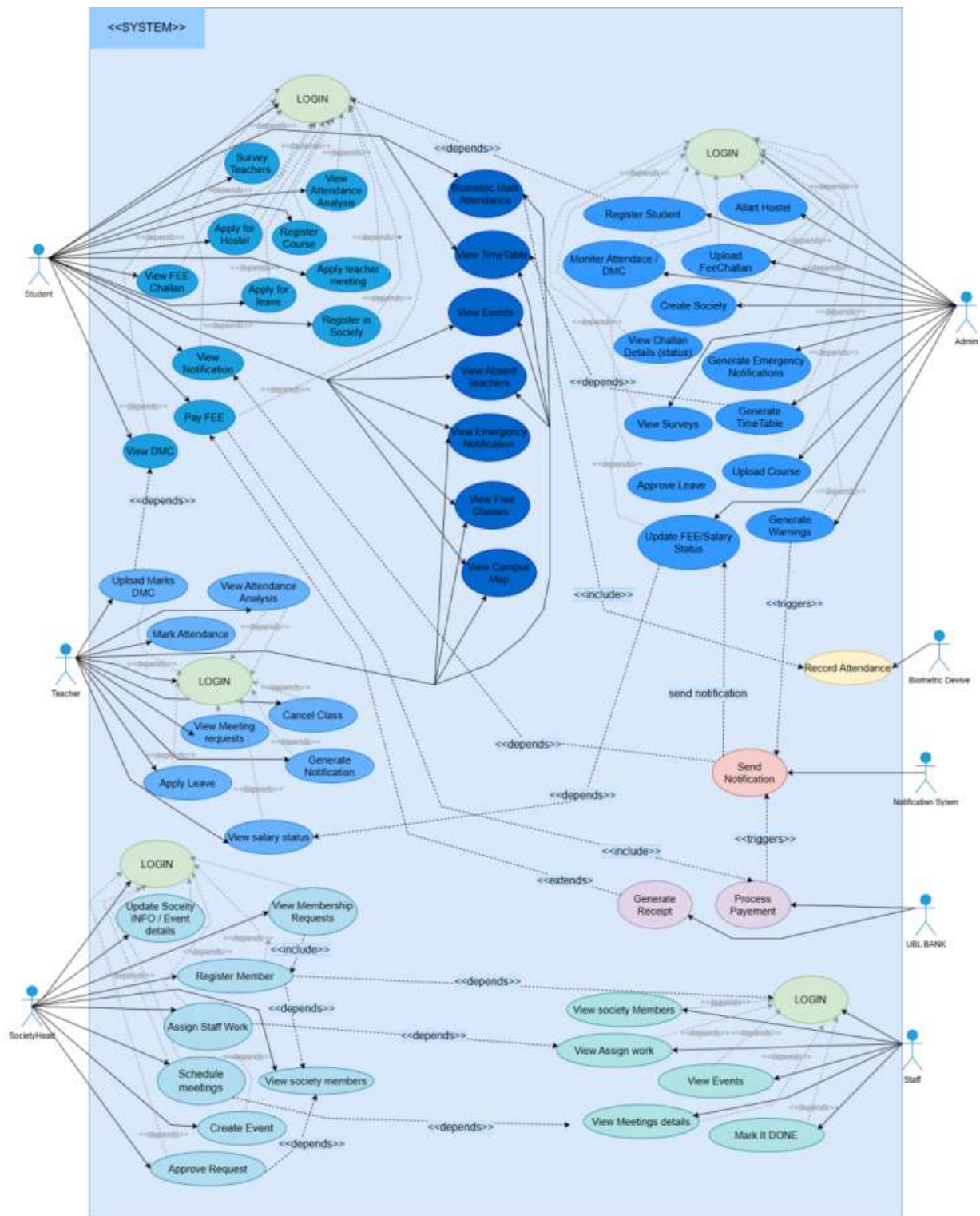
Stake holder	Classification	Potential Gathering methods	Best choice	Justification
Student	Primary	Questionnaires, Focus groups	Questionnaires	We shall use the Questionnaire/Survey method because it allows me to collect data from a massive number of students at once.
Teacher	Primary	Interviews, Prototyping, workshops	Interviews	We shall use the Structured Interview method because it allows for a deep, one-on-one conversation.
Admin	Primary	User observation, Interviews, Questionnaires	User Observations	We shall use the User Observation method because by watching them perform their daily tasks (like generating fee challans)
Staff	Secondary	User observation, Interviews, Questionnaires	User Observation	We shall use the User Observation method because it is the most accurate way to see their daily responsibilities.
Society Head	Primary	Workshops, Brain storming, Role play	Workshops	We shall use the Requirements Workshop method because it brings all the heads into one room.
UBL bank	Tertiary	Interface analysis, Document analysis	Interface analysis	We shall use the Interface Analysis method because We need to understand the technical "handshake" between our software and the bank's server.

Biometric device	Tertiary	Interface analysis, Document analysis	Document analysis	We shall use the Document Analysis method because the device comes with a technical "rulebook" (data sheet).
Notifications system	Tertiary	Interface analysis, Prototyping	Interface analysis	We shall use the Interface Analysis method because it allows us to define the technical requirements for sending bulk messages.

4. Context Diagram



5. Use Case Diagram



6. Detailed Requirements Section

6.1 Functional Requirements

The following requirements define the specific actions and services the Digital University Management System must provide. They are categorized by Actor ID (starting with 01 for Students and ending with 06 for System/All Users) and ranked by priority within each group.

01: Student Requirements

- **FR-01-001:** The system shall allow the student to register and select courses for the semester.
- **FR-01-002:** The system shall allow the **student** to process fee payments through bank integration and update payment status accordingly.
- **FR-01-003:** The system shall allow the **student** to view their results and DMC records.
- **FR-01-004:** The system shall allow the **student** to view their respective timetables and also allow students to view teacher schedules.
- **FR-01-005:** The system shall allow the **student** to view attendance records, percentage, and graphical analysis on a weekly and monthly basis.
- **FR-01-006:** The system shall allow the **student** to apply for full or half leave by entering required details such as date.
- **FR-01-007:** The system shall allow the **student** to request meetings with teachers based on available timetable slots using student ID.
- **FR-01-008:** The system shall allow the **student** to apply for society membership and notify them upon approval or rejection.
- **FR-01-009:** The system shall allow the **student** to receive notifications regarding results, leave status, fee updates, and society membership.
- **FR-01-010:** The system shall allow the **student** to submit feedback and surveys for teachers.

02: Teacher Requirements

- **FR-02-001:** The system shall allow the **teacher** to upload student marks and DMC records.
- **FR-02-002:** The system shall allow the **teacher** to mark class attendance and identify students who were present in university but absent in class.
- **FR-02-003:** The system shall allow the **teacher** to approve or reject meeting requests from students.
- **FR-02-004:** The system shall allow the **teacher** to view their attendance records and analysis.

- **FR-02-005:** The system shall allow the **teacher** to view their assigned academic timetable.
- **FR-02-006:** The system shall allow the **teacher** to apply for full or half leave.
- **FR-02-007:** The system shall allow the **teacher** to send notifications specifically to students.

03: Admin Requirements

- **FR-03-001:** The system shall allow the **admin** to register students, teachers, and societies.
- **FR-03-002:** The system shall allow the **admin** to create and manage timetables for sections and teachers.
- **FR-03-003:** The system shall allow the **admin** to generate fee challans for students.
- **FR-03-004:** The system shall allow the **admin** to view and monitor attendance records of students and teachers and issue warnings if attendance is below threshold.
- **FR-03-005:** The system shall allow the **admin** to view an automatically generated list of teachers who have not marked their attendance for the day.
- **FR-03-006:** The system shall allow the **admin** to approve or reject leave requests.
- **FR-03-007:** The system shall allow the **admin** to create and manage university societies.
- **FR-03-008:** The system shall allow the **admin** to generate and manage system notifications.
- **FR-03-009:** The system shall allow the **admin** to calculate fees and apply scholarships to student accounts.
- **FR-03-010:** The system shall allow the **admin** to upload and distribute digital fee challans to users.

04: Society Head Requirements

- **FR-04-001:** The system shall allow the **society head** to create, update, and manage events.
- **FR-04-002:** The system shall allow the **society head** to approve or reject student membership requests.
- **FR-04-003:** The system shall allow the **society head** to assign tasks to staff members.
- **FR-04-004:** The system shall allow the **society head** to send notifications to members and general users.

05: Staff Requirements

- **FR-05-001:** The system shall allow the **staff** to view assigned tasks and update their completion status.
- **FR-05-002:** The system shall allow the **staff** to assist in society operations and event management.

06: All Users / General System Functions

- **FR-06-001:** The system shall allow **all users** to log in using valid credentials and captcha verification.
- **FR-06-002:** The system shall allow **all users** to record attendance using biometric fingerprint verification along with user ID.
- **FR-06-003:** The system shall allow **all users** to view information about university societies, including members, activities, and event details.
- **FR-06-004:** The system shall allow **all users** to search for classrooms and access location details, visual maps, and shortest path navigation.
- **FR-06-005:** The system shall allow **all users** to view classrooms that are currently free for scheduling makeup lectures.
- **FR-06-006:** The system shall allow **all users** to view emergency notifications generated by authorized actors (admin, teachers, and society heads).

6.2 Business Requirements

The University Digital Management System **wants** to reduce manual academic and administrative processes by 60% and decrease the time required for tasks like attendance and registration by 50% to achieve high operational efficiency. Furthermore, the administration **wants** to improve communication efficiency by 70% through a centralized system and increase attendance accuracy by 90% via biometric verification to eliminate proxy issues. To promote a digital environment, the client **wants** to reduce paper-based workflows by 80% and decrease student dependency on physical office visits by 70% to enhance student convenience. Additionally, the university **wants** to boost administrative productivity by 60% and improve data accessibility by 75% to ensure quick information retrieval for all stakeholders. Finally, the university **wants** to enhance the overall campus experience by 50% by providing smart digital services across the university.

6.3 Business Rules

BRL-01: User Authentication Rule

All users must provide valid credentials to access role-based system functionalities.

BRL-02: Biometric Attendance Rule

Attendance shall only be marked through biometric verification and cannot be manually altered without admin authorization.

BRL-03: Attendance Threshold Rule

Students must maintain a minimum of **70% attendance**; otherwise, they may receive warnings or restrictions.

BRL-04: Leave Approval Rule

Leave requests (full or half) must be approved by the authorized authority before being considered valid.

BRL-05: Fee Payment Rule

Students must clear all dues before the specified deadline to remain eligible for academic activities.

BRL-06: Course Registration Rule

Students are only allowed to register for courses within the defined registration period.

BRL-07: Society Membership Rule

Students can only become members of a society after approval from the society head.

BRL-08: Event Management Rule

All society events must be officially registered and approved within the system before execution.

BRL-09: Timetable Constraint Rule

Class schedules must not overlap for students or teachers.

BRL-10: Notification Authorization Rule

Only authorized roles (admin, teacher, society head) can generate notifications.

BRL-11: Data Modification Rule

Only authorized users are allowed to modify critical data such as attendance, results, and fee records.

BRL-12: Payment Verification Rule

Fee status shall only be updated after successful confirmation from the bank system.

BRL-13: GPA Requirement Rule

Students must maintain a minimum **GPA of 2.0 (as per defined by university policy)** to continue their academic program without restrictions.

6.4 Non- Functional Requirements

NFR-001: Response Time

The system must respond to user requests within **2–3 seconds** under normal operating conditions.

NFR-002: Security

1. The system must ensure secure authentication using **user credentials and captcha verification**, and protect user data from unauthorized access.
2. The system security must be ensured by providing access to only registered users.

NFR-003: Throughput

The system must handle 10,000 user requests per second simultaneously without significant performance degradation.

NFR-004: Resource Utilization

The system must efficiently utilize system resources such as memory, CPU, and network bandwidth, not more than 1GB of resources

NFR-005: Scalability

The system must be scalable so that it can handle 10,000 of users and data without affecting performance, in the next 1 year.

NFR-006: Capacity

The system must support a large number of users (students, teachers, and staff) and maintain extensive academic and administrative records by using 1TB of data.

NFR-007: Availability

The system must be available **24/7** using UPS or generator , except during scheduled maintenance periods.

NFR-008: Reliability

1. The system must operate consistently without failure and ensure accurate processing of data and transactions.
2. The system's uptime must be more than 1000 hours.

NFR-009: Recoverability

The system must store **restore point** every 2 weeks so it can be recovered from errors.

NFR-010: Serviceability

1. The system must provide functionality for remote diagnostics.
2. The system must be able to clean after one year

NFR-011: Manageability

The system must provide a dashboard to the admin for manageability using proper **GRAPHS**.

NFR-012: Regulatory Compliance (Not applicable)

The system must comply with university policies provided by UET and applicable rules related to academic and administrative data handling.

NFR-013: Maintainability

The system must be designed in a modular manner to allow easy updates within 2 hours of down time, enhancements, and bug fixes.

NFR-014: Environmental Constraints

The system must operate effectively on standard computing environments such as university computer labs and personal devices, by maintaining temperature of 50°F.

NFR-015: Data Integrity (Not applicable for our system)

The system must ensure that all stored data remains accurate, consistent, and free from unauthorized modification using checksum.

NFR-016: Usability

1. The system must provide a **user-friendly interface** that is easy to learn and use for all types of users, by providing easy to understand text, images, headings.
2. The system must provide shortcuts to make user work easy.

NFR-017: Interoperability

The system must be capable of interacting with external systems such as banking and notification services.

6.5 External Interfaces

1. User Interface

The system shall provide a **simple, user-friendly, and role-based graphical user interface** for all users including students, teachers, admin, society heads, and staff.

- ❖ The interface shall support **login authentication with captcha**.
- ❖ The system shall display **dashboards according to user roles**.
- ❖ The interface shall include modules for **attendance, courses, results, notifications, and societies**.
- ❖ The system shall provide **clear navigation menus** and responsive design for ease of use.

2. Hardware Interface

The system shall interact with hardware components required for its operation.

- ❖ The system shall integrate with a **biometric fingerprint device** for attendance marking.
- ❖ The system shall be accessible through **computers, laptops, and mobile devices**.

- ❖ The system shall support standard input/output devices such as **keyboard, mouse, and display screens**.

3. Software Interface

The system shall interact with external software systems to perform specific operations.

- ❖ The system shall integrate with a banking system such as **UBL Bank** for fee payment processing.
- ❖ The system shall interact with a **notification system** for sending alerts and messages.
- ❖ The system shall use a **database management system (DBMS)** to store and manage data.
- ❖ The system shall be compatible with modern **web browsers**.

4. Communication Interface

The system shall support communication between different components and users.

- ❖ The system shall use **internet connectivity** for data exchange.
- ❖ The system shall support **secure data transmission protocols**.
- ❖ The system shall allow communication through **notifications and alerts**.
- ❖ The system shall ensure reliable communication between users and external systems.

6.6 Physical Product Requirements

1. End User (Terminal & Client Access)

- **Biometric Fingerprint Device:** Integrated hardware for authenticating attendance for students, teachers, and staff.
- **Mobile Devices & Tablets:** Used by students and teachers for portable access to portals, navigation, and notifications.
- **Keyboards & Mice:** Standard input peripherals for terminal interactions.

2. Admin End (Central Management)

- **Laptops/Workstations:** High-performance machines for administrators to manage user registration, timetable generation, and fee processing.
- **Central Processing Unit (CPU):** Robust processing units capable of handling administrative dashboard multitasking and system control.
- **Display Screens:** High-resolution monitors for monitoring attendance analysis and system-wide statistics.

3. Developer & Infrastructure End (Backend & Support)

- **Network Server:** High-capacity servers to host the web application and manage real-time data exchange.
- **Database Management System (DBMS):** Dedicated storage environment for user records, biometric data, and academic results.
- **UPS (Uninterruptible Power Supply):** Essential for ensuring the server and biometric terminals remain operational during power fluctuations.
- **Backup Generator:** Secondary power source to maintain 24/7 availability of critical digital university services.

- **RAM (Memory):** Optimized server-side memory to support concurrent role-based access for thousands of students and faculty members

6.7 Development Constraints

1. Frontend & Backend (Language & Frameworks)

- **Frontend:** Built with modern web technologies (HTML, CSS, JavaScript) to ensure compatibility with all major web browsers.
- **Backend:** A server-side language (e.g., C#) to handle the logic for role-based specialized systems and database interactions.
- **Integration:** Must support secure data transmission protocols for external interfaces like the **UBL Bank** system for fee payments.

2. Architecture View & Data Relation

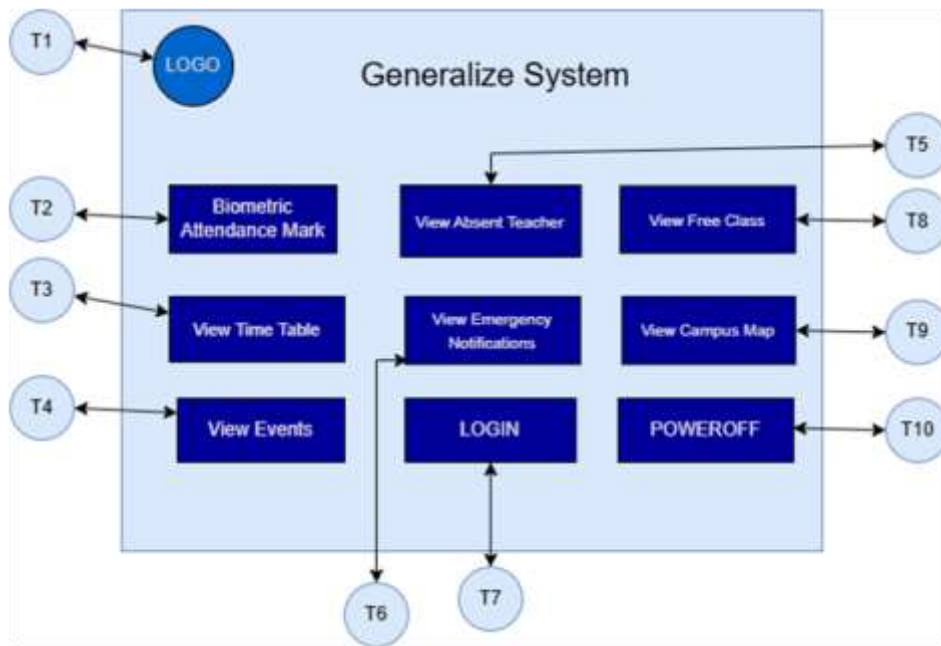
- **Unified Digital Platform:** A centralized architecture that integrates academic, administrative, and general campus services.
- **Role-Based Access Control (RBAC):** Data relations are structured so that users (Students, Teachers, Admins, etc.) only access functionalities mapped to their specific roles.
- **External Data Hooks:** The system maintains a data relationship with a separate **Notification System** and **Banking API** for real-time alerts and payment confirmations.

3. OS & Software Environment

- **Operating Systems:**
 - **Windows/Linux:** Servers can be hosted on either OS, provided they support the chosen DBMS and web server environment.

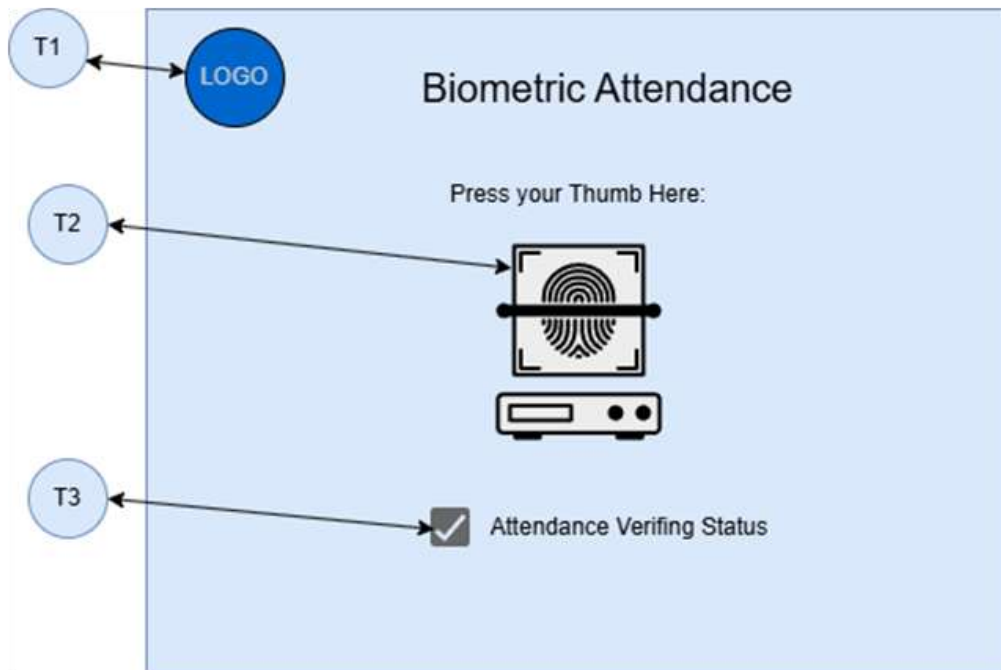
- o **Cross-Platform Clients:** The user interface is designed to be responsive, working across various operating systems on laptops and mobile devices.
- **Security Software:** Includes integrated **Captcha** verification for login authentication to prevent automated attacks.

6.8 Wireframes

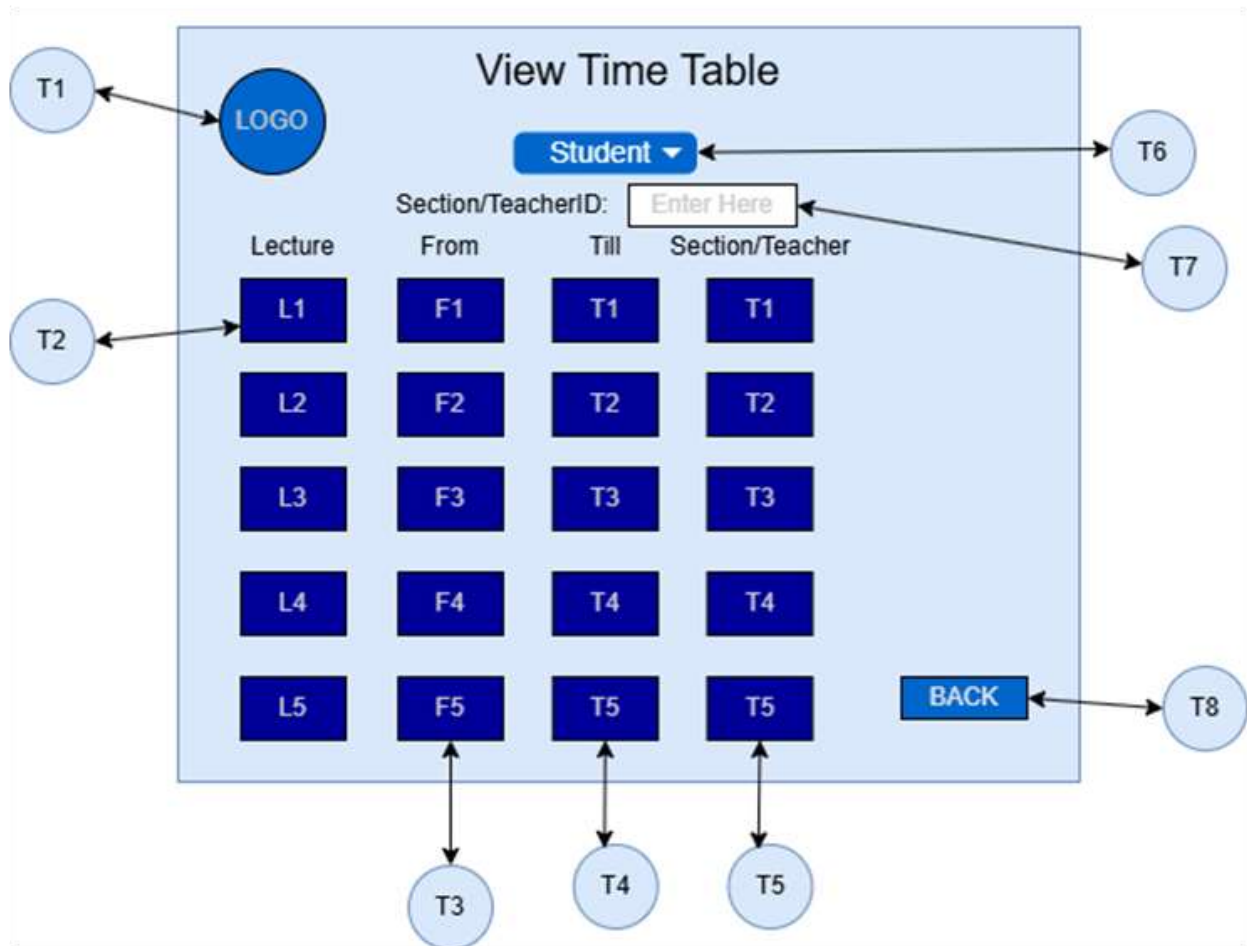


ID	Requirements(User)	Requirements(Admin)
T1	As a user, I shall jump to the homepage when I click the logo image.	As an admin, I shall update or change the institution's logo.
T2	As a user, I shall scan my finger to mark my attendance via the biometric system.	As an admin, I shall manage biometric records and generate attendance reports.
T3	As a user, I shall view the daily or weekly timetable for classes.	As an admin, I shall upload, edit, or modify the class schedules.
T4	As a user, I shall view upcoming campus events and details.	As an admin, I shall add new events or remove expired ones from the list.
T5	As a user, I shall view the list of teachers who are absent today.	As an admin, I shall update the teacher attendance status in real-time.
T6	As a user, I shall receive and view urgent emergency notifications/alerts.	As an admin, I shall broadcast emergency alerts to all system terminals.

T7	As a user, I shall enter my credentials to access my private dashboard.	As an admin, I shall manage user accounts, reset passwords, and control access.
T8	As a user, I shall search for and view currently vacant classrooms.	As an admin, I shall update classroom availability and room assignments.
T9	As a user, I shall view the interactive campus map for navigation.	As an admin, I shall update the map locations or labels for new buildings.
T10	As a user, I shall safely shut down the terminal session.	As an admin, I shall remotely restart or power off system terminals for maintenance.

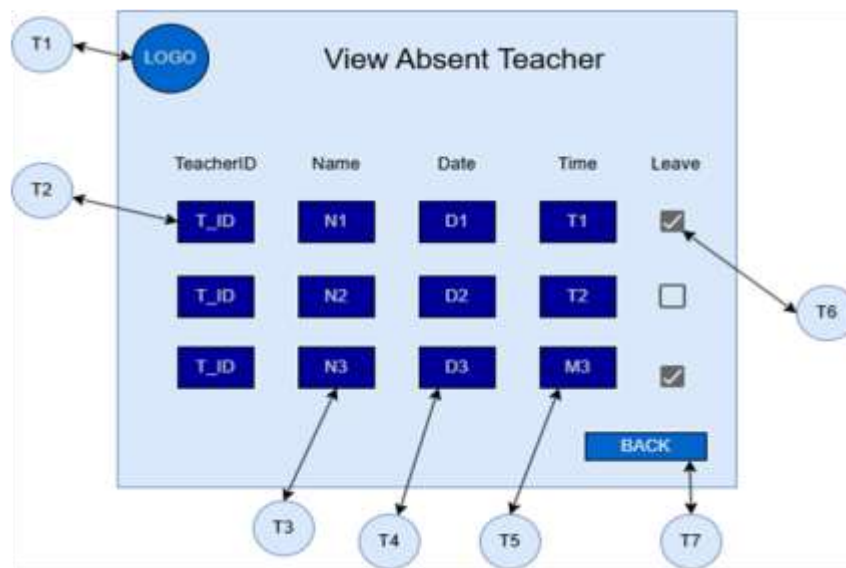


ID	Requirements(User)	Requirements(Admin)
T1	As a user, I shall click the logo to return to the main "Generalize System" dashboard.	As an admin, I shall configure the logo link to ensure it redirects to the correct homepage.
T2	As a user, I shall place my thumb on the scanner to initiate the biometric verification process.	As an admin, I shall calibrate the scanner sensitivity and manage the biometric database.
T3	As a user, I shall view the status (Success/Fail) to confirm if my attendance was recorded.	As an admin, I shall view logs of verification attempts to troubleshoot any system errors.



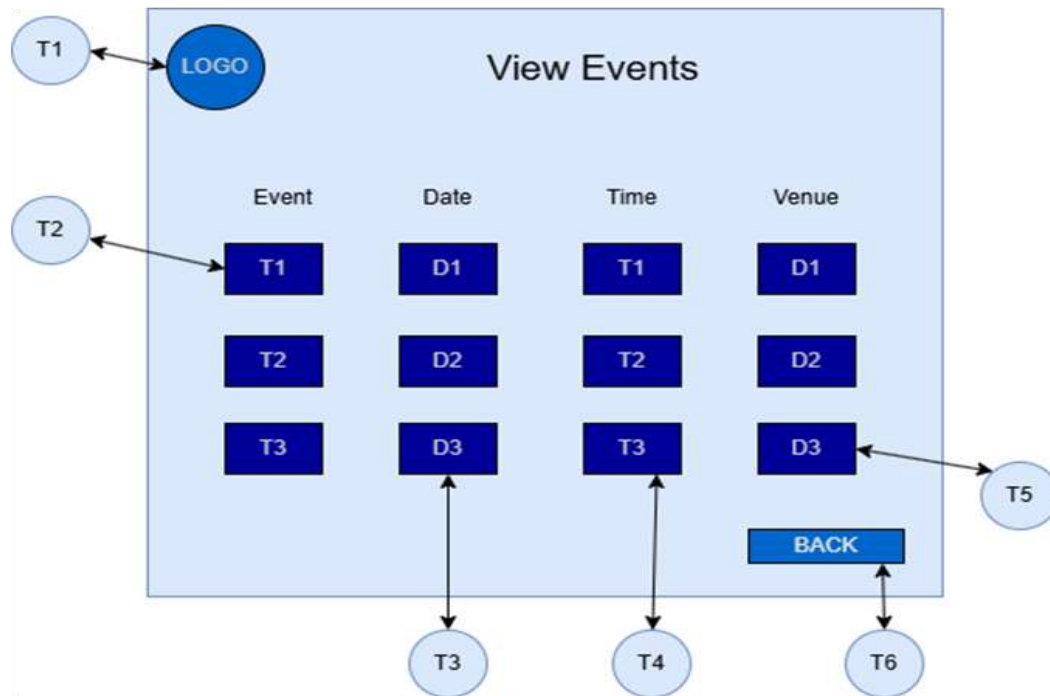
ID	Requirements(User)	Requirements(Admin)
T1	As a user, I shall click the logo to return to the main "Generalize System" dashboard.	As an admin, I shall configure the logo link to the homepage.
T2	As a user, I shall view the lecture name or number (L1–L5) in the schedule.	As an admin, I shall update lecture names and course codes.
T3	As a user, I shall view the starting time (From) of each lecture.	As an admin, I shall set and modify the lecture start times.
T4	As a user, I shall view the ending time (Till) of each lecture.	As an admin, I shall set and modify the lecture end times.
T5	As a user, I shall view the assigned	As an admin, I shall assign specific

	section or teacher for each slot.	teachers or sections to time slots.
T6	As a user, I shall select my role (Student or Teacher) from the dropdown menu.	As an admin, I shall manage the roles available in the dropdown.
T7	As a user, I shall enter my Section or Teacher ID to filter the timetable.	As an admin, I shall ensure the search database is synced with valid IDs.
T8	As a user, I shall click the "BACK" button to return to the previous screen.	As an admin, I shall ensure the navigation path is correctly mapped.

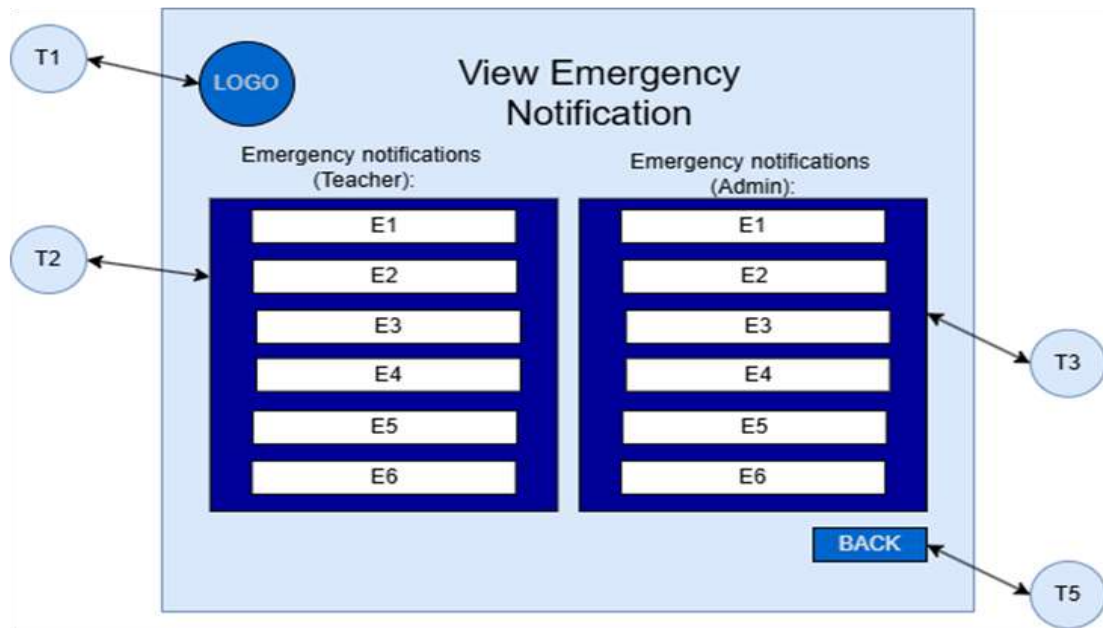


ID	Requirements(User)	Requirements(Admin)
T1	As a user, I shall click the logo to return to the main "Generalize System" dashboard.	As an admin, I shall configure the logo link to ensure proper home navigation.
T2	As a user, I shall view the unique Teacher ID of the absent staff member.	As an admin, I shall link teacher IDs to the central employee database.
T3	As a user, I shall view the full name (N1–N3) of the teacher who is absent.	As an admin, I shall update teacher profiles and names in the system.

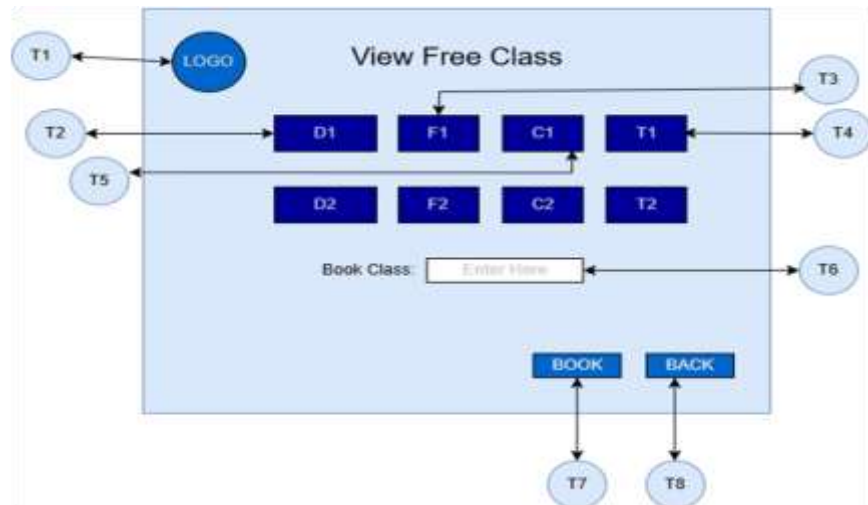
T4	As a user, I shall view the specific date (D1–D3) of the teacher's absence.	As an admin, I shall select or schedule the absence date in the record.
T5	As a user, I shall view the time or shift (T1–M3) affected by the absence.	As an admin, I shall specify which lecture slots or hours the teacher is away.
T6	As a user, I shall see a checkmark status indicating if the leave was formally approved.	As an admin, I shall toggle the leave status (Checked/Unchecked) based on approval.
T7	As a user, I shall click the "BACK" button to return to the previous screen.	As an admin, I shall ensure the back button redirects to the correct system menu.



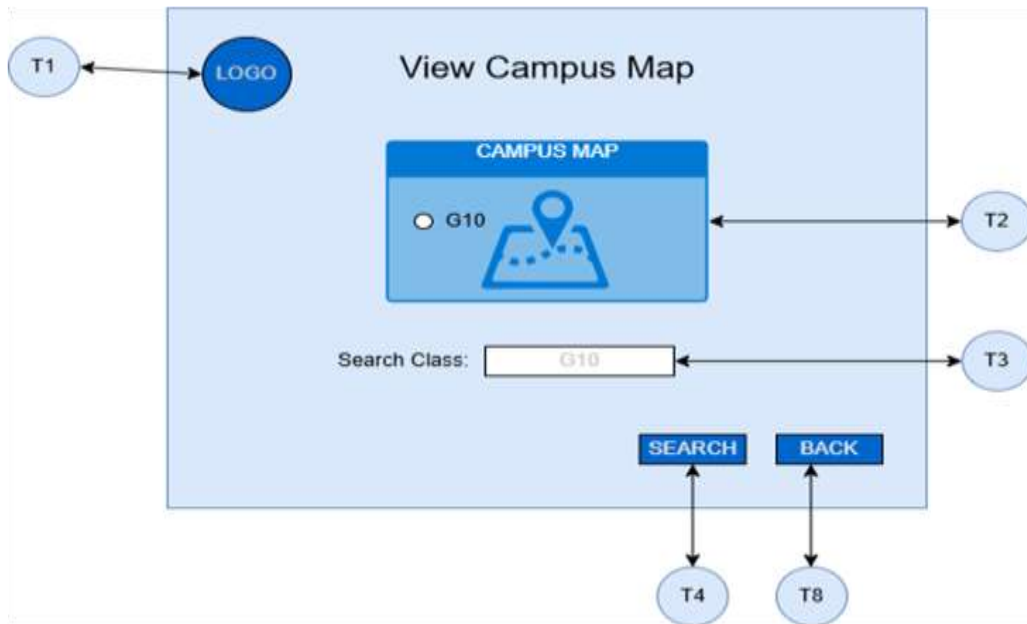
ID	Requirements(User)	Requirements(Admin)
T1	As a user, I shall click the logo to return to the main dashboard.	As an admin, I shall maintain the homepage link for the logo.
T2	As a user, I shall view the titles (T1–T3) of upcoming campus events.	As a Society Head, I shall enter the names of new events into the system.
T3	As a user, I shall view the scheduled date (D1–D3) for each event.	As a Society Head, I shall set and update event dates in the database.
T4	As a user, I shall view the specific start and end times for the events.	As a Society Head, I shall manage the timing for event sessions.
T5	As a user, I shall view the venue or location where the event is being held.	As a Society Head, I shall assign and book rooms or halls for specific events.
T6	As a user, I shall click the "BACK" button to go back to the main menu.	As an admin and society head, I shall ensure the "BACK" button leads to the correct previous page.



ID	Requirements(User)	Requirements(Admin)
T1	As a user, I shall click the logo to return to the main "Generalize System" dashboard.	As an admin, I shall maintain the homepage link for the logo.
T2	As a student, I shall view emergency notifications (E1–E6) specifically meant for teachers (e.g., room changes).	As a admin, I shall view notification that are generated by teacher.
T3	As a student, I shall view general emergency notifications (E1–E6) issued by the Admin.	As an admin, I shall create, broadcast, and delete critical emergency alerts for the entire campus.
T5	As a user, I shall click the "BACK" button to return to the previous screen.	As an admin, I shall ensure the "BACK" button navigation is functioning correctly.



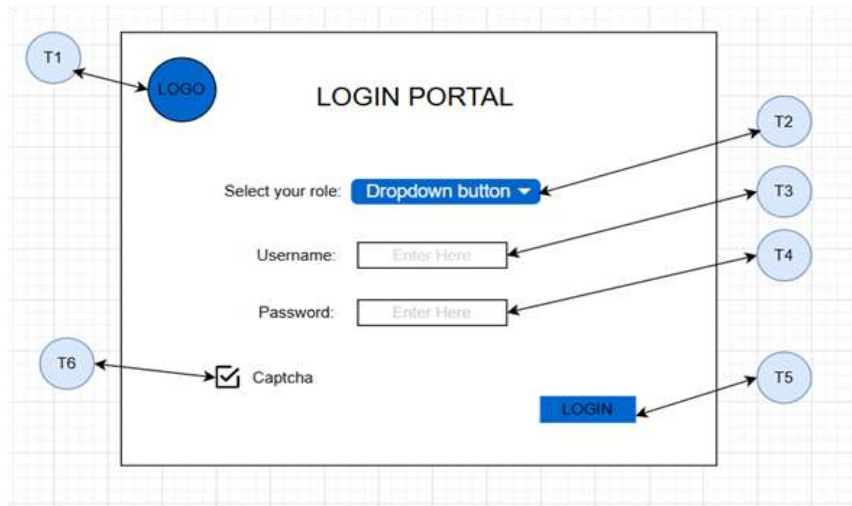
ID	Requirements(User)	Requirements(Admin)
T1	As a user, I shall click the logo to return to the main dashboard.	As an admin, I shall manage the home navigation link for the system logo.
T2	As a user, I shall view the Department or Block (D1–D2) where the room is located.	As an admin, I shall update and manage the list of campus departments.
T3	As a user, I shall view the Floor number (F1–F2) of the available classroom.	As an admin, I shall categorize rooms based on their floor level.
T4	As a user, I shall view the specific Time slot (T1–T2) the class is vacant.	As an admin, I shall synchronize room availability with the master timetable.
T5	As a user, I shall view the Classroom number (C1–C2) that is currently free.	As an admin, I shall manage the inventory of all physical classrooms.
T6	As a user, I shall enter the specific Class ID or Name I wish to book in the input field.	As an admin, I shall validate the entered room ID against the vacant list.
T7	As a user, I shall click the "BOOK" button to reserve the selected free classroom.	As an admin, I shall receive and approve or log the room booking request.
T8	As a user, I shall click the "BACK" button to return to the previous screen.	As an admin, I shall ensure the back button redirects to the correct menu.



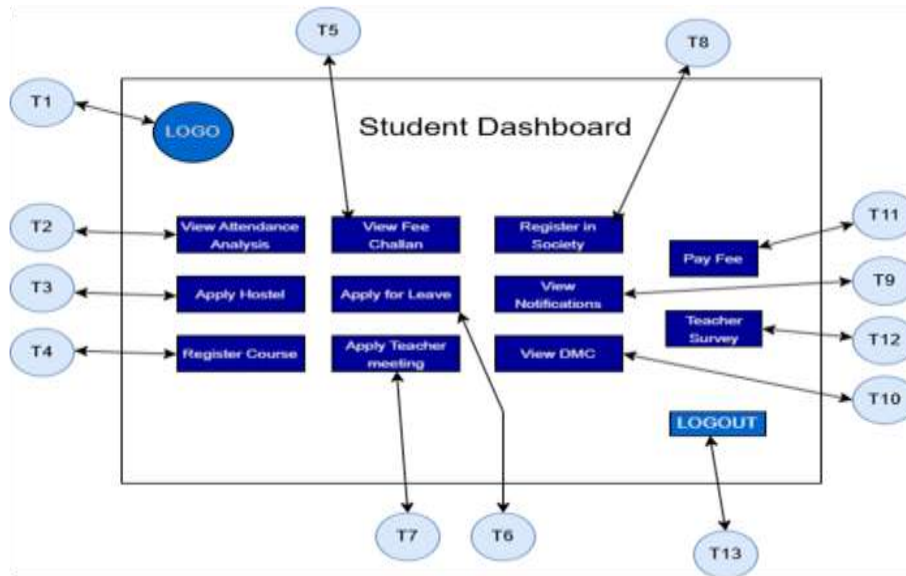
ID	Requirements(User)	Requirements(Admin)
T1	As a user, I shall click the logo to return to the main dashboard.	As an admin, I shall ensure the logo link redirects correctly to the homepage.
T2	As a user, I shall view the interactive campus map showing the location of classrooms (e.g., G10).	As an admin, I shall upload and update the floor plans and building layouts.
T3	As a user, I shall enter the Classroom ID or Name in the search box to find a location.	As an admin, I shall maintain a database of all campus locations and coordinates.
T4	As a user, I shall click the "SEARCH" button to highlight the searched room on the map.	As an admin, I shall configure the search logic to ensure accurate room mapping.
T8	As a user, I shall click the "BACK" button to return to the previous screen.	As an admin, I shall ensure the back navigation leads to the correct menu.

Specialize

Student:

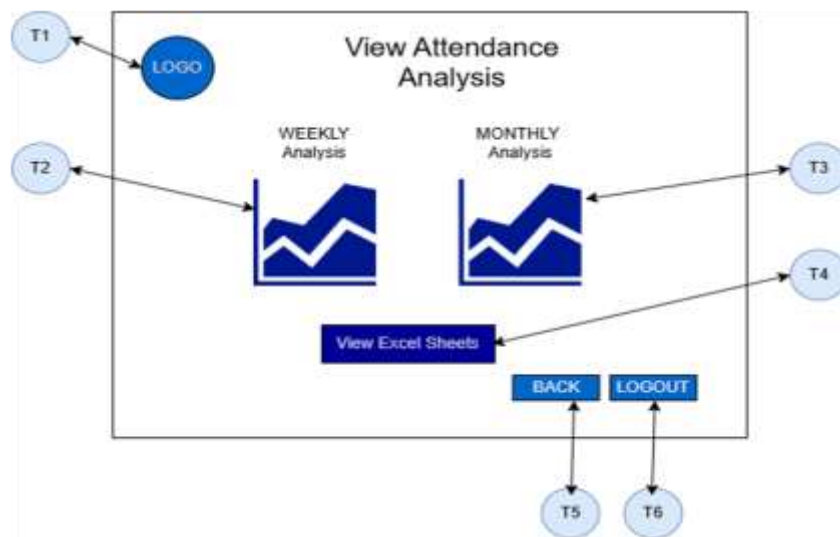


ID	Requirements(User)
T1	As a user, I shall click the logo to return to the landing page.
T2	As a user, I shall select my specific role (Student, Teacher, Staff, or Society Head) from the dropdown.
T3	As a user, I shall enter my assigned username.
T4	As a user, I shall enter my secure password to authenticate my identity.
T5	As a user, I shall click the "LOGIN" button to access the dashboard corresponding to my role.
T6	As a user, I shall complete the Captcha verification to prove I am not a bot.



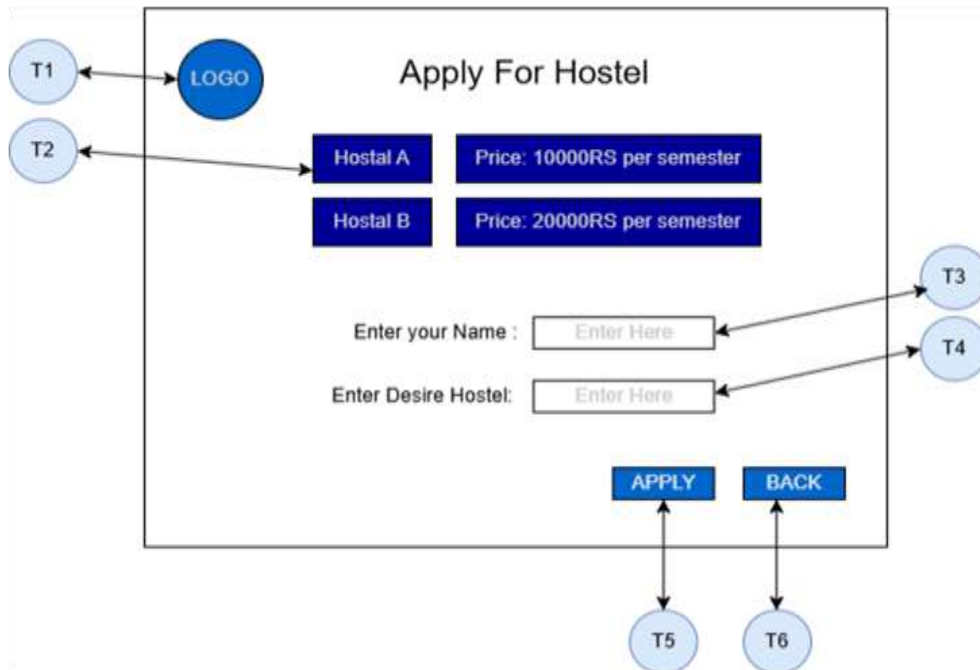
ID	Requirements(Student)	Requirements(Admin)
T1	As a student, I shall click the logo to return to the homepage.	As an admin, I shall update or manage the institution logo.
T2	As a student, I shall view my attendance analysis and percentages.	As an admin, I shall upload and verify attendance records
T3	As a student, I shall apply for hostel accommodation.	As an admin, I shall process hostel applications and assign rooms.
T4	As a student, I shall register for my upcoming semester courses.	As an admin, I shall manage course offerings and enrollment limits.
T5	As a student, I shall view and download my fee challan.	As an admin, I shall generate fee challans and update payment statuses.
T6	As a student, I shall submit an application for leave.	As an admin, I shall review, approve, or reject leave requests.
T7	As a student, I shall request a meeting with a specific teacher.	As an admin, I shall coordinate schedules between students and faculty.
T8	As a student, I shall register as a member of a university society.	As an admin, I shall manage society memberships and approvals.
T9	As a student, I shall view my personal and academic notifications.	As an admin, I shall broadcast specific alerts to the student dashboard.

T10	As a student, I shall view my Detailed Marks Certificate (DMC).	As an admin, I shall upload exam results and verify DMC accuracy.
T11	As a student, I shall initiate the online fee payment process.	As an admin, I shall manage the integration with the bank/payment gateway.
T12	As a student, I shall complete the teacher evaluation survey.	As an admin, I shall design survey forms and analyze feedback results.
T13	As a student, I shall securely log out of the dashboard.	As an admin, I shall monitor active sessions and manage user security.



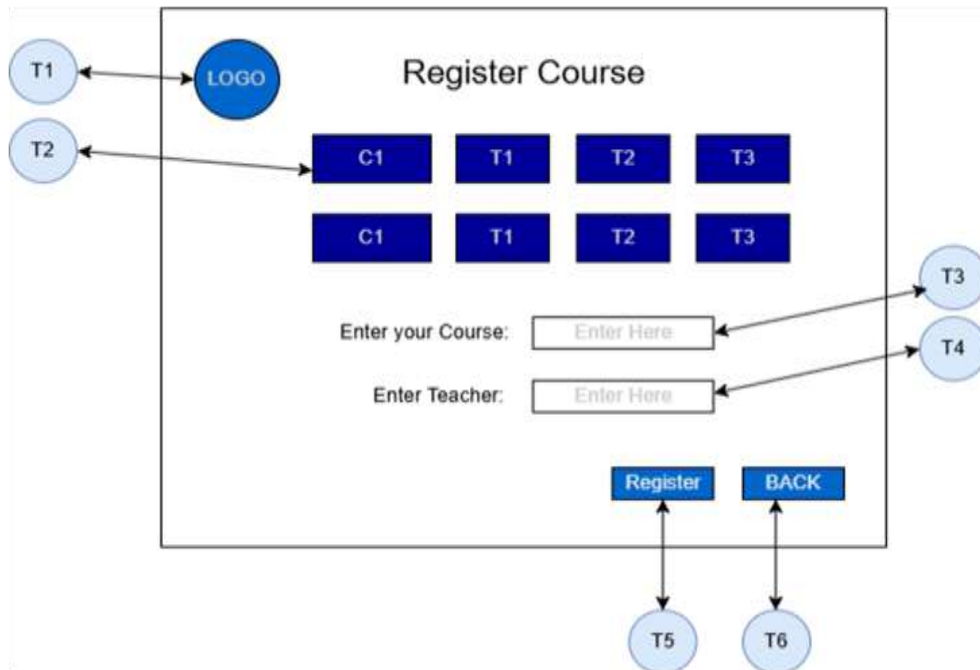
ID	Requirements(Student)	Requirements(Admin)
T1	As a student, I shall click the logo to return to the dashboard.	As an admin, I shall manage the institutional logo and its link.
T2	As a student, I shall view my Weekly Analysis graph to track recent attendance trends.	As an admin, I shall monitor weekly attendance patterns across the department.

T3	As a student, I shall view my Monthly Analysis graph for a long-term attendance summary.	As an admin, I shall generate monthly attendance summaries for academic records.
T4	As a student, I shall click "View Excel Sheets" to see a detailed tabular breakdown of my presence/absence.	As an admin, I shall export and manage the raw Excel data for the entire class.
T5	As a student, I shall click the "BACK" button to return to the previous screen.	As an admin, I shall ensure the back navigation maintains the correct system flow.
T6	As a student, I shall click "LOGOUT" to end my secure session.	As an admin, I shall click "LOGOUT" to exit the administrative portal.



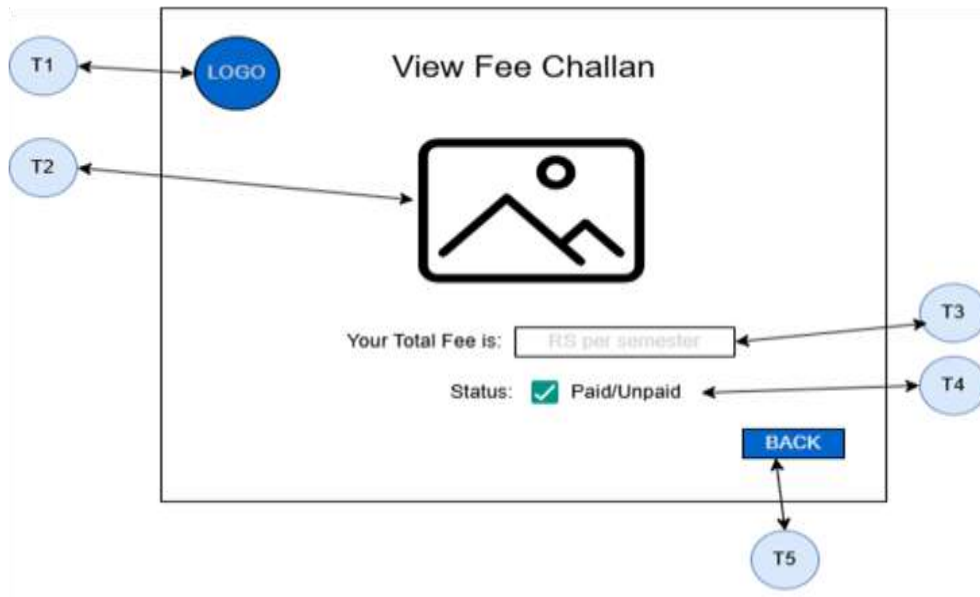
ID	Requirements(Student)	Requirements(Admin)
T1	As a student, I shall click the logo to return to the student dashboard.	As an admin, I shall manage the homepage redirection link.
T2	As a student, I shall view the available hostel options (Hostel A, Hostel B) and their respective prices per semester.	As an admin, I shall update hostel names, availability, and pricing details.
T3	As a student, I shall enter my full name in the designated input field.	As an admin, I shall verify student names against the enrollment database.

T4	As a student, I shall enter my desired hostel choice (e.g., Hostel A) to indicate my preference.	As an admin, I shall collect and organize student preferences for room allocation.
T5	As a student, I shall click the "APPLY" button to submit my hostel application.	As an admin, I shall receive, review, and approve or reject the submitted applications.
T6	As a student, I shall click the "BACK" button to return to the previous screen.	As an admin, I shall ensure the back button correctly redirects to the main dashboard.



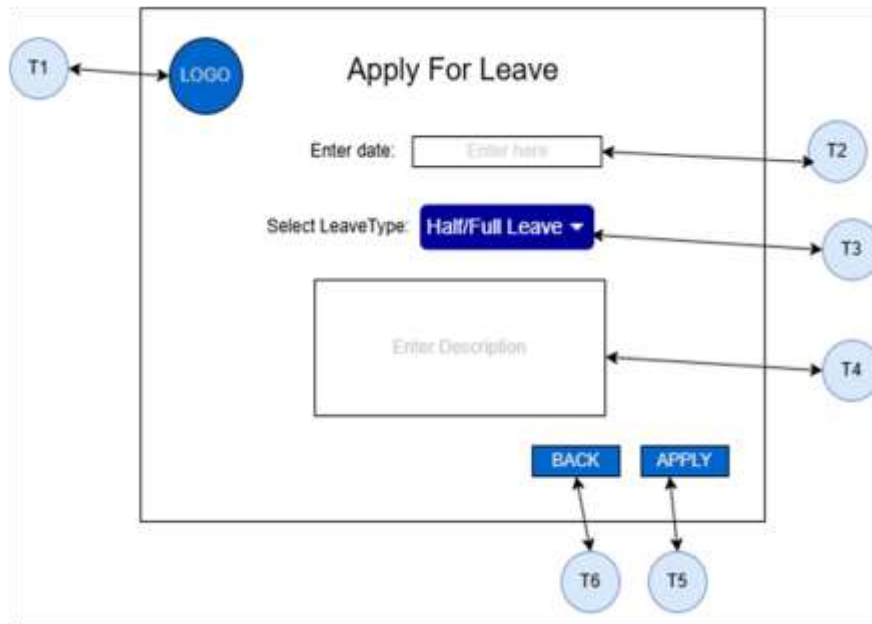
D	Requirements(Student)	Requirements(Admin)
T1	As a student, I shall click the logo to return to the main student dashboard.	As an admin, I shall configure the logo redirection to the correct portal landing page.
T2	As a student, I shall view the list of available courses (C1) and their associated timings (T1–T3).	As an admin, I shall update course catalogs, lecture timings, and room availability.
T3	As a student, I shall enter the	As an admin, I shall validate that

	specific Course Name or ID I wish to enroll in.	the entered course ID exists and is open for registration.
T4	As a student, I shall enter the name of the preferred Teacher for the selected course.	As an admin, I shall manage faculty assignments and ensure teacher sections are correctly mapped.
T5	As a student, I shall click the "Register" button to finalize my course enrollment.	As an admin, I shall process the registration, check for credit hour limits, and update the student's schedule.
T6	As a student, I shall click the "BACK" button to return to the previous screen without saving changes.	As an admin, I shall ensure the back button navigation maintains the correct user flow.



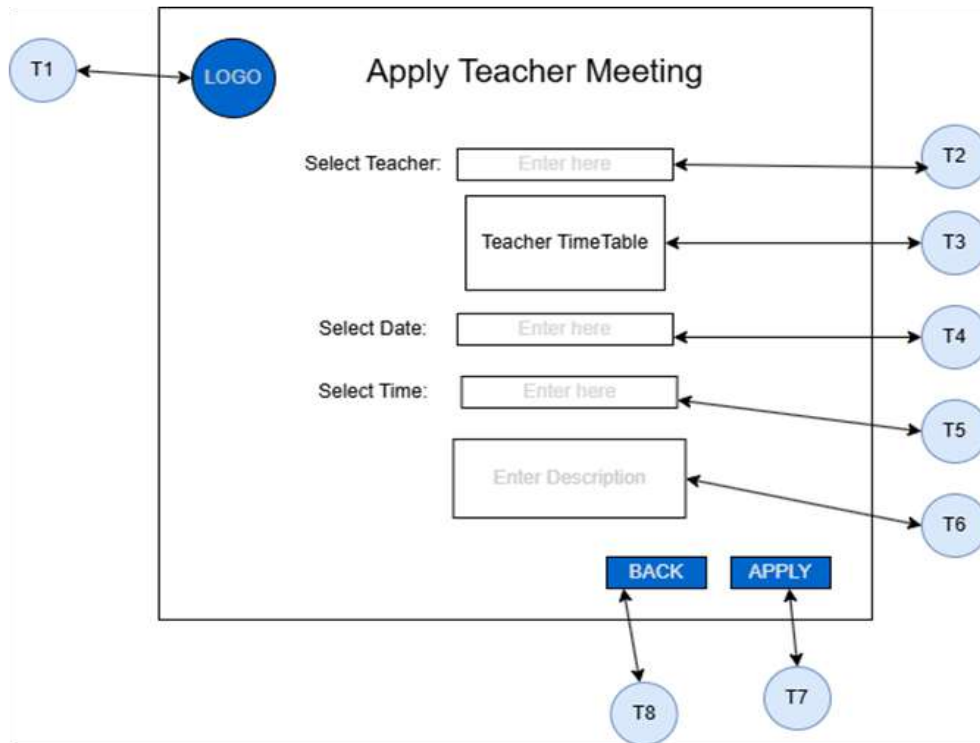
ID	Requirements(Student)	Requirements(Admin)
T1	As a student, I shall click the LOGO to return to the main student dashboard.	As an admin, I shall configure the logo redirection to the correct portal landing page.
T2	As a student, I shall view the "View Fee Challan" option to initiate the fee payment process.	As an admin, I shall ensure that the fee challan is generated correctly and is accessible to the student.
T3	As a student, I shall view my Total Fee amount displayed clearly for the current semester.	As an admin, I shall update and manage the fee structures and ensure the correct amount is displayed per

		student and semester.
T4	As a student, I shall view the Status of my fee (Paid/Unpaid) and have the ability to check/uncheck or mark it.	As an admin, I shall update and maintain the accuracy of the student's fee payment status in the system.
T5	As a student, I shall click the "BACK" button to return to the previous screen.	As an admin, I shall ensure the back button navigation maintains the correct user flow without causing data errors.



ID	Requirements(Student)	Requirements(Admin)
T1	As a student, I shall click the LOGO to return to the main student dashboard.	As an admin, I shall configure the logo redirection to ensure it leads to the correct institutional landing page.
T2	As a student, I shall Enter date to specify when the leave is required.	As an admin, I shall define the "Date" constraints
T3	As a student, I shall select the Leave Type (Half/Full) from	As an admin, I shall manage the leave categories and ensure the

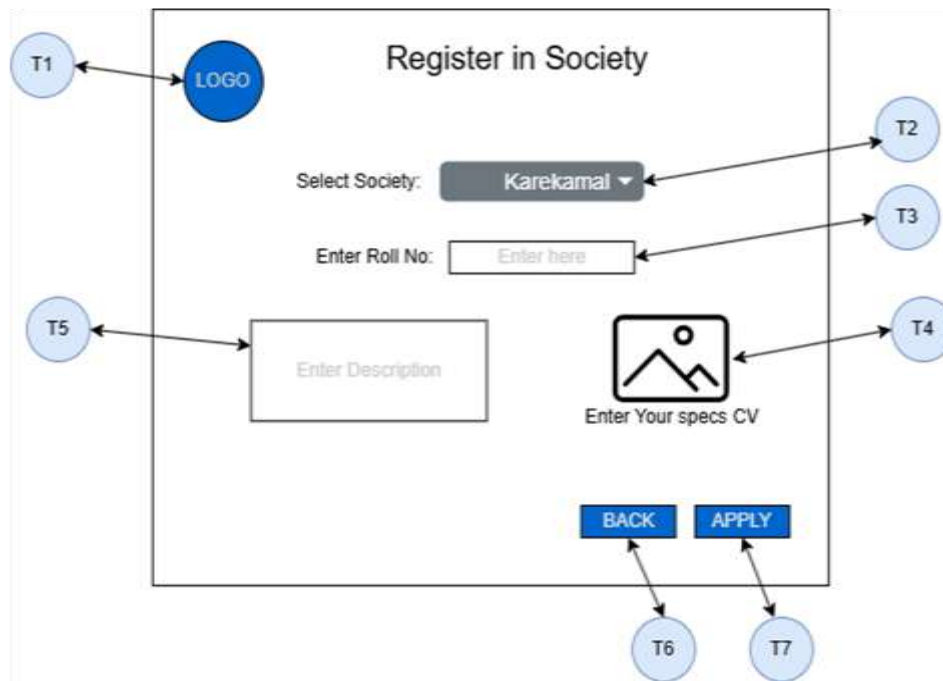
	the dropdown menu.	dropdown options align with university policy.
T4	As a student, I shall Enter Description to provide a valid reason or medical evidence for the leave request.	As an admin, I shall review the submitted description to approve or reject the application based on the provided details.
T5	As a student, I shall click the APPLY button to submit my request to the department for formal approval.	As an admin, I shall receive a notification of the submission and update the student's attendance record once approved.
T6	As a student, I shall click the BACK button to discard the current form and return to the previous module.	As an admin, I shall ensure that clicking 'Back' does not trigger a partial save or create duplicate "Pending" entries in the database.



ID	Requirements(Student)	Requirements(Admin)
T1	As a student, I shall click the LOGO to return to the student portal home screen.	As an admin, I shall ensure the logo link is correctly mapped to the student dashboard.
T2	As a student, I shall Select Teacher from a searchable list to initiate a meeting request.	As an admin, I shall maintain an updated database of all faculty members available for meetings.
T3	As a student, I shall view the Teacher TimeTable to identify	As a teacher, I shall update my weekly schedule and "free hours" to

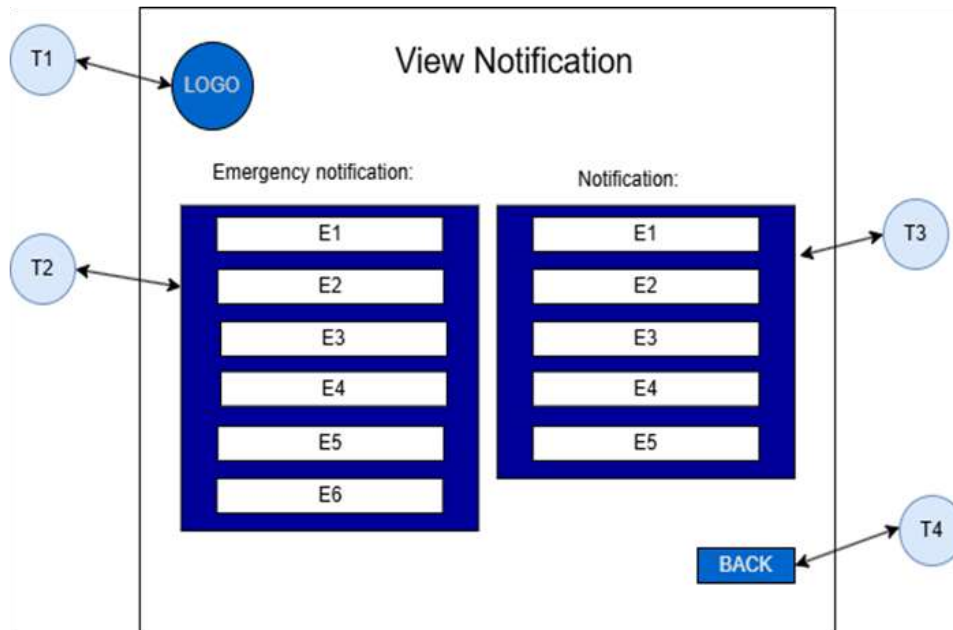
	available slots before choosing a time.	be visible to students.
T4	As a student, I shall Select Date for the meeting based on the teacher's availability.	As an admin, I shall ensure the calendar restricts selections to working days only.
T5	As a student, I shall Select Time for the appointment that does not conflict with my classes.	As a teacher, I shall receive a notification once a specific time slot is requested.
T6	As a student, I shall Enter Description to state the purpose of the meeting (e.g., project query, grade review).	As a teacher, I shall review the description to prepare for the meeting or request more info.
T7	As a student, I shall click APPLY to send the meeting request for the teacher's approval.	As a teacher, I shall have the authority to "Accept" or "Reschedule" the incoming request.
T8	As a student, I shall click BACK to cancel the request and return to	As an admin, I shall ensure that clicking 'Back' clears any temporary

	the previous navigation menu.	input to protect data privacy.
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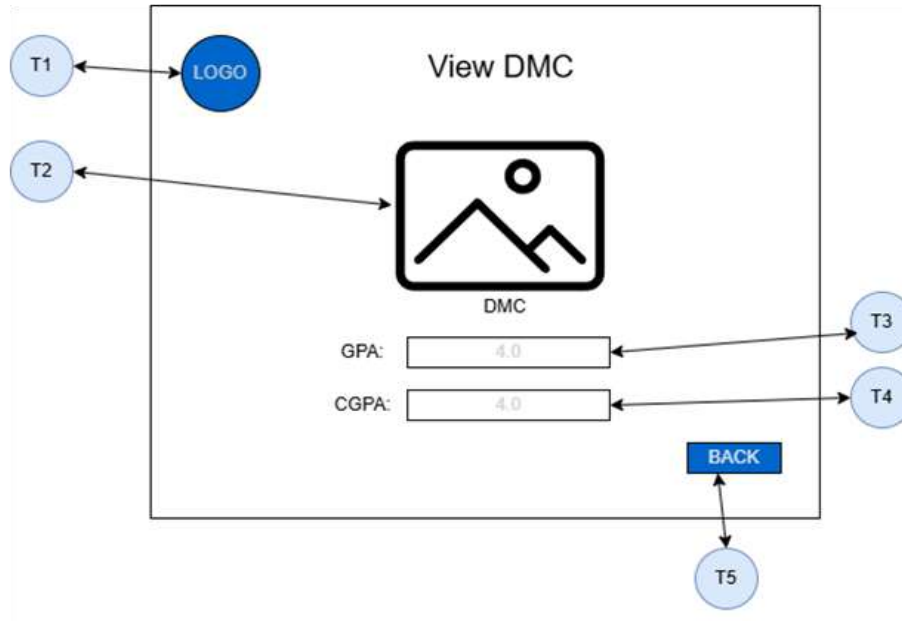


ID	Requirements(Student)	Requirements(Admin)
T1	As a student, I shall click the LOGO to go back to the home dashboard.	As an admin, I shall ensure the logo redirects to the student portal home.
T2	As a student, I shall Select Society (e.g., Karekamal) from the list.	As an admin, I shall manage the list of available societies in the system.
T3	As a student, I shall Enter Roll No to identify myself for registration.	As an admin, I shall verify that the Roll No exists in the student database.
T4	As a student, I shall upload my Specs	As an admin, I shall set the file size and

	CV (image/file) as part of the application.	type limits for CV uploads.
T5	As a student, I shall Enter Description to explain my skills or interests.	As an admin, I shall review the student's description during the selection process.
T6	As a student, I shall click BACK to cancel and exit the registration form.	As an admin, I shall ensure no data is saved if the student clicks 'Back'.
T7	As a student, I shall click APPLY to submit my registration to the society.	As an admin, I shall receive the application and store it in the registration log.

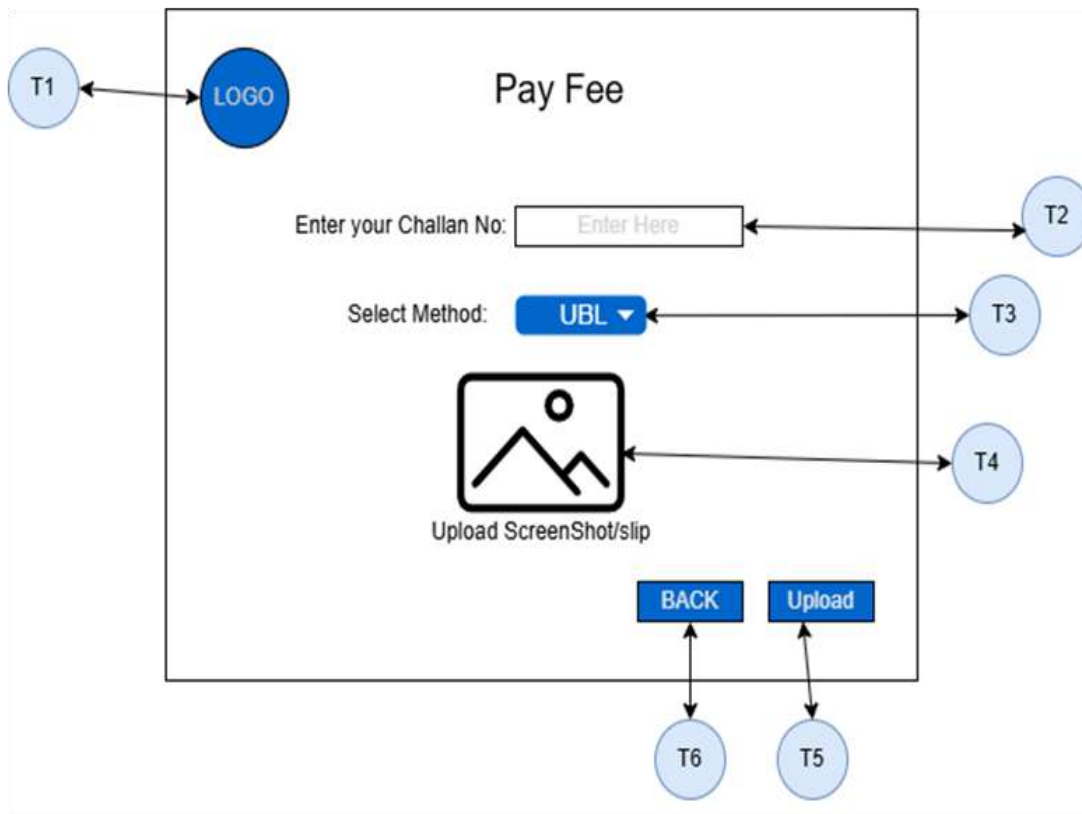


ID	Requirements(Student)	Requirements(Admin)
T1	As a student, I shall click the LOGO to return to the student portal home.	As an admin, I shall ensure the logo redirection link is functional and accurate.
T2	As a student, I shall view Emergency notifications (E1-E6) for urgent campus alerts.	As an admin, I shall post and prioritize high-priority alerts in the emergency section.
T3	As a student, I shall view general Notifications (E1-E5) regarding routine campus updates.	As an admin, I shall update and manage general announcements and academic news.
T4	As a student, I shall click the BACK button to return to the previous navigation screen.	As an admin, I shall ensure the back button preserves the correct user session flow.



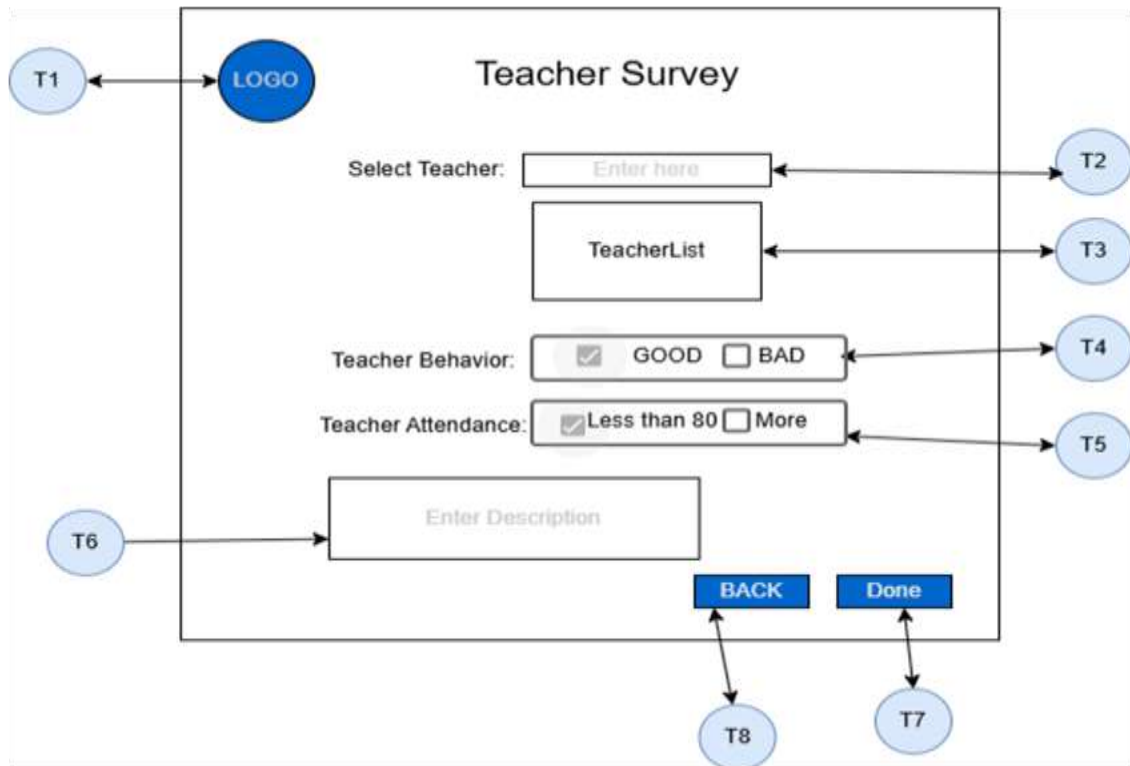
ID	Requirements(Student)	Requirements(Admin)
T1	As a student, I shall click the LOGO to return to the student portal home screen.	As an admin, I shall ensure the logo link remains active and points to the correct dashboard.
T2	As a student, I shall view my official DMC image or scanned document for the semester.	As an admin, I shall upload the correct scanned DMC file for each student in the database.
T3	As a student, I shall view my current semester GPA displayed in a read-only field.	As an admin, I shall ensure the GPA is calculated and updated based on the student's latest marks.

T4	As a student, I shall view my overall CGPA to track my cumulative academic performance.	As an admin, I shall maintain the accuracy of the CGPA by aggregating all previous semester results.
T5	As a student, I shall click the BACK button to return to the academic records menu.	As an admin, I shall ensure the back button navigation is smooth and does not log the user out.



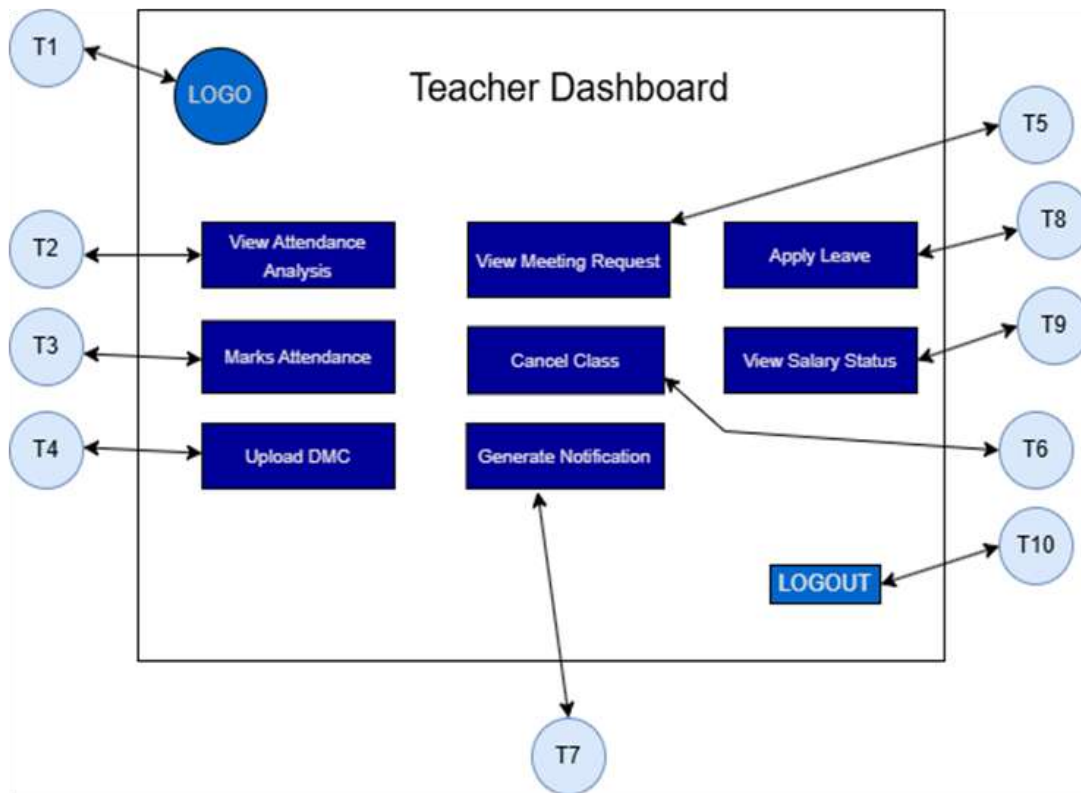
ID	Requirements(Student)	Requirements(Admin)
T1	As a student, I shall click the LOGO to return to the student portal home screen.	As an admin, I shall ensure the logo link points to the correct main dashboard.
T2	As a student, I shall Enter your Challan No to identify the specific fee record for payment.	As an admin, I shall verify that the entered Challan No exists and matches the student's records.
T3	As a student, I shall Select Method (e.g., UBL) from the	As an admin, I shall manage the list of supported banks and payment

	dropdown to specify the bank used.	methods.
T4	As a student, I shall Upload ScreenShot/slip as digital proof of my bank transaction.	As an admin, I shall view and download the uploaded slip to cross-verify with the bank statement.
T5	As a student, I shall click Upload to submit my payment proof for verification.	As an admin, I shall receive a notification of the submission to start the approval process.
T6	As a student, I shall click BACK to exit the payment screen without submitting.	As an admin, I shall ensure the system does not record incomplete attempts if the student clicks 'Back'.



ID	Requirements(Student)	Requirements(Admin)
T1	As a student, I shall click the LOGO to return to the student portal home.	As an admin, I shall ensure the logo redirection is correctly mapped to the home dashboard.
T2	As a student, I shall Select Teacher by typing the name to find the specific instructor for evaluation.	As an admin, I shall provide a searchable list of all active faculty members in the system.
T3	As a student, I shall view the TeacherList to confirm and select	As an admin, I shall update the faculty list at the start of each

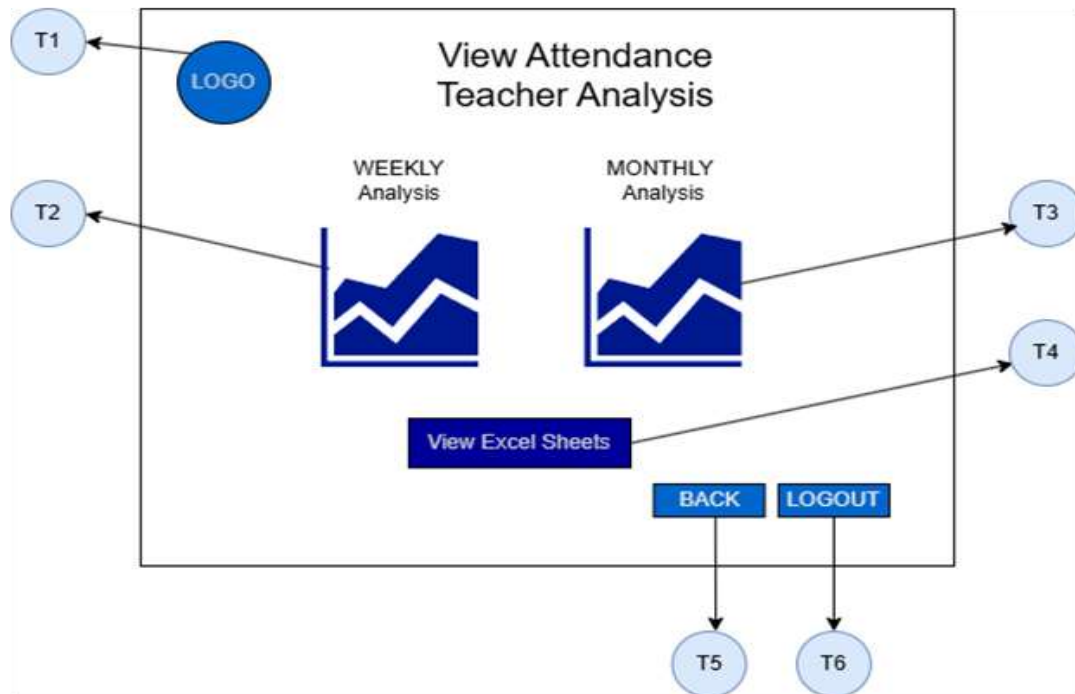
	the correct faculty member.	semester for accurate surveys.
T4	As a student, I shall select Teacher Behavior (Good/Bad) to provide feedback on professionalism.	As an admin, I shall aggregate behavior ratings to generate faculty performance reports.
T5	As a student, I shall select Teacher Attendance (Less than 80 / More) to report on punctuality.	As an admin, I shall monitor attendance feedback to ensure compliance with university standards.
T6	As a student, I shall Enter Description to provide detailed comments or specific suggestions.	As an admin, I shall review student comments for qualitative assessment during appraisals.
T7	As a student, I shall click Done to submit my survey response to the department.	As an admin, I shall ensure that survey submissions are saved securely and remain anonymous.
T8	As a student, I shall click BACK to exit the survey without saving any data.	As an admin, I shall ensure the back button resets all fields to prevent accidental submissions.



ID	Requirements(Teacher)	Requirements(Admin)
T1	As a teacher, I shall click the LOGO to refresh or return to the main dashboard view.	As an admin, I shall ensure the logo link points correctly to the Teacher Portal home.
T2	As a teacher, I shall click View Attendance Analysis to see student presence trends.	As an admin, I shall ensure the system pulls real-time attendance data from the database.

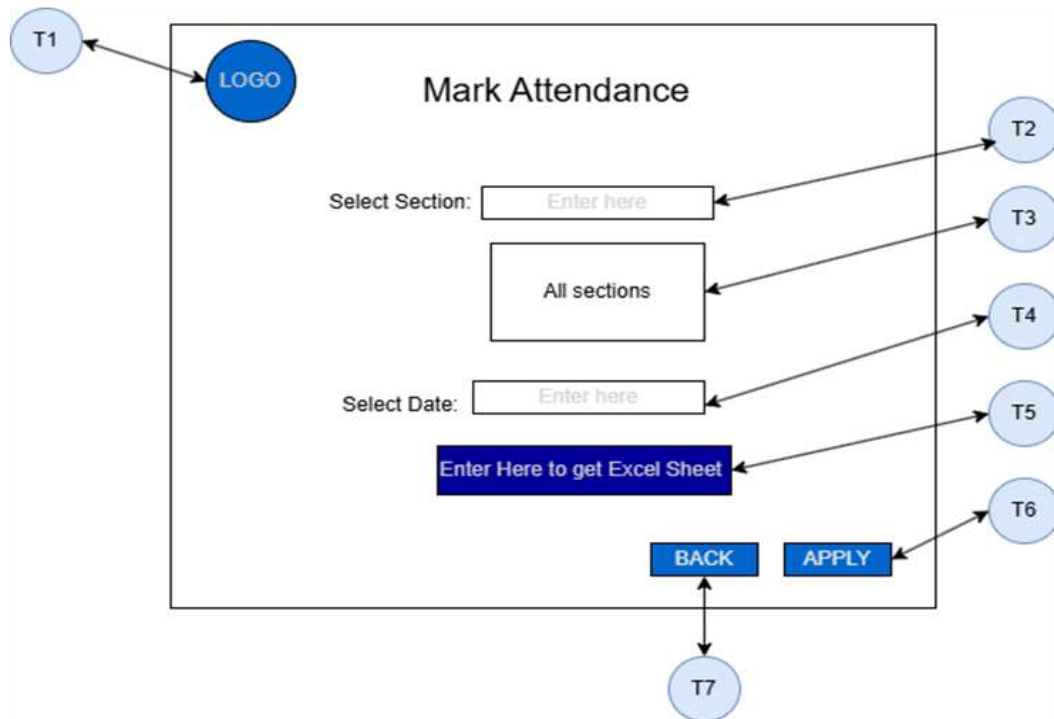
T3	As a teacher, I shall click Marks Attendance to record student presence for my classes.	As an admin, I shall verify that the teacher can only mark attendance for their assigned subjects.
T4	As a teacher, I shall click Upload DMC to submit student semester result sheets.	As an admin, I shall set file size and format restrictions for the uploaded result documents.
T5	As a teacher, I shall click View Meeting Request to respond to student appointment invites.	As an admin, I shall provide a notification badge when a new meeting request is received.
T6	As a teacher, I shall click Cancel Class to notify students and the office of a cancellation.	As an admin, I shall ensure an automated email/SMS is sent to students when a class is canceled.
T7	As a teacher, I shall click Generate Notification to post announcements for my students.	As an admin, I shall ensure notifications are correctly tagged with the teacher's ID and timestamp.

T8	As a teacher, I shall click Apply Leave to submit my own leave request to the HOD.	As an admin, I shall route the leave application to the relevant department head for approval.
T9	As a teacher, I shall click View Salary Status to check my monthly payment details.	As an admin, I shall integrate the dashboard with the Finance module to fetch secure payroll data.
T10	As a teacher, I shall click LOGOUT to securely end my current session.	As an admin, I shall ensure all temporary session data and cache are cleared upon logout.



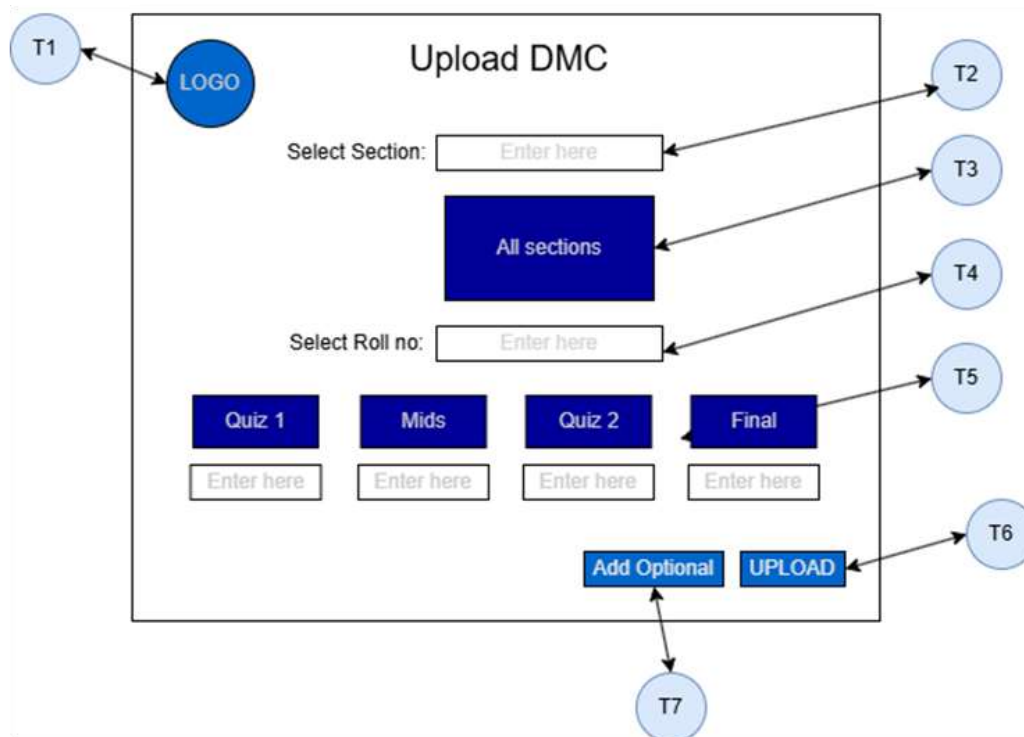
ID	Requirements(Teacher)	Requirements(Admin)
T1	As a teacher, I shall click the LOGO to return to the main teacher dashboard.	As an admin, I shall maintain the redirection link to the teacher portal home.
T2	As a teacher, I shall view the WEEKLY Analysis graph to monitor short-term attendance trends.	As an admin, I shall ensure the system pulls and calculates weekly attendance percentages accurately.
T3	As a student, I shall view the MONTHLY Analysis graph to track long-term attendance	As an admin, I shall update the monthly data aggregates at the end of every calendar month.

	patterns.	
T4	As a teacher, I shall click View Excel Sheets to access raw data for detailed record-keeping.	As an admin, I shall ensure the data is exportable in .xlsx or .csv format without errors.
T5	As a teacher, I shall click BACK to return to the previous navigation screen.	As an admin, I shall ensure the back button preserves the current state of the dashboard session.
T6	As a teacher, I shall click LOGOUT to safely exit the system and protect sensitive student data.	As an admin, I shall terminate the session and clear the browser cache upon logout.



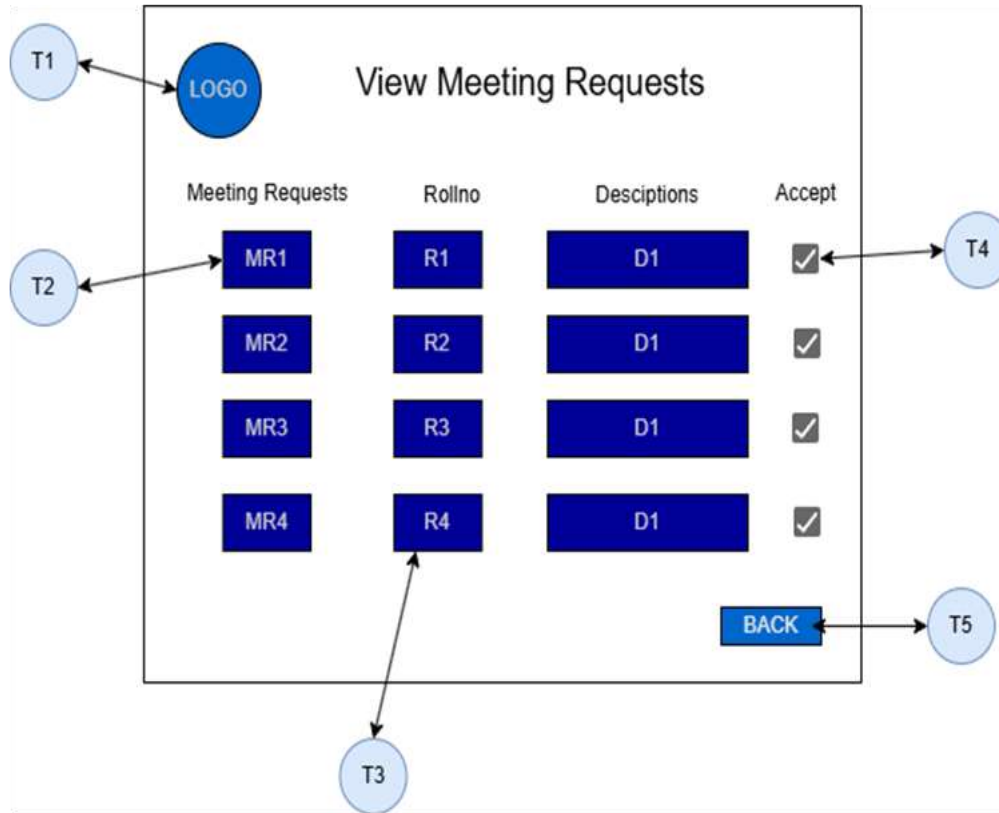
ID	Requirements(Teacher)	Requirements(Admin)
T1	As a teacher, I shall click the LOGO to return to the teacher portal home.	As an admin, I shall ensure the logo redirection link is functional and accurate.
T2	As a teacher, I shall Select Section by typing to find the specific class for marking attendance.	As an admin, I shall provide a list of all active class sections in the system.
T3	As a teacher, I shall view All sections to verify and select the	As an admin, I shall update the section list at the start of each

	correct student group.	semester for accurate record-keeping.
T4	As a teacher, I shall Select Date to specify the day for the attendance record.	As an admin, I shall ensure the calendar restricts selections to valid working days.
T5	As a teacher, I shall click Enter Here to get Excel Sheet to download a template or report of the class.	As an admin, I shall ensure the system generates and provides the file in .xlsx or .csv format.
T6	As a teacher, I shall click APPLY to save and submit the attendance record to the database.	As an admin, I shall receive the submission and update the students' attendance logs accordingly.
T7	As a teacher, I shall click BACK to cancel and return to the main dashboard.	As an admin, I shall ensure that clicking 'Back' clears any temporary input to protect data integrity.



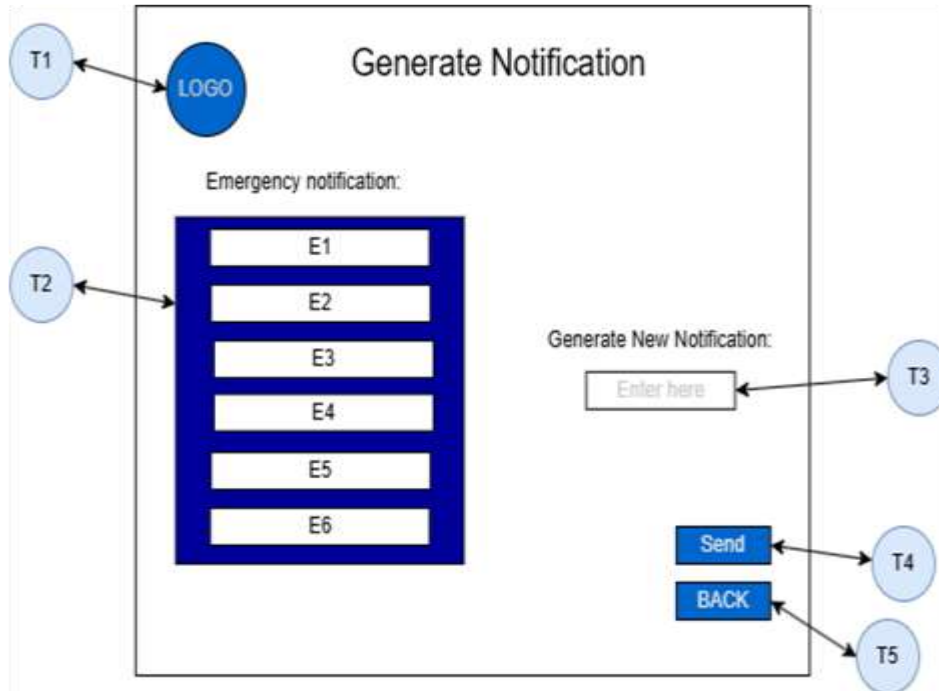
ID	Requirements(Teacher)	Requirements(Admin)
T1	As a teacher, I shall click the LOGO to return to the teacher portal home screen.	As an admin, I shall ensure the logo link points to the correct main dashboard.
T2	As a teacher, I shall Select Section to filter students for marks entry.	As an admin, I shall provide a list of class sections assigned to the specific teacher.
T3	As a teacher, I shall view All sections to verify the chosen class group.	As an admin, I shall ensure the section list is updated based on the current semester.

T4	As a teacher, I shall Select Roll no to specify the individual student for grade entry.	As an admin, I shall verify that the Roll No exists within the selected class section.
T5	As a teacher, I shall Enter marks for Quiz 1, Mids, Quiz 2, and Final exams.	As an admin, I shall set the minimum and maximum marks allowed for each assessment field.
T6	As a teacher, I shall click UPLOAD to save the student's marks certificate to the system.	As an admin, I shall generate a digital DMC for the student once the teacher submits the data.
T7	As a teacher, I shall click Add Optional to include extra fields like assignments or lab marks.	As an admin, I shall allow dynamic field creation to accommodate different course structures.



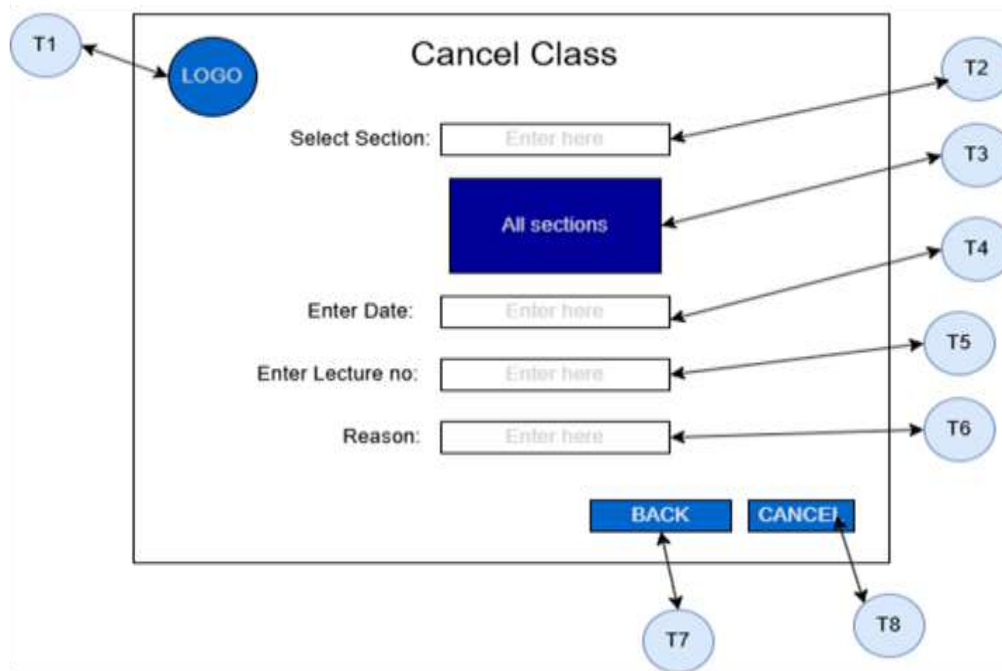
ID	Requirements(Teacher)	Requirements(Admin)
T1	As a teacher, I shall click the LOGO to navigate back to the main portal.	As an admin, I shall ensure the logo link is consistent across all faculty pages.
T2	As a teacher, I shall view the Meeting Requests (MR1-MR4) sent by students.	As a system, I shall pull all pending meeting requests from the database for the teacher.

T3	As a teacher, I shall view the Roll no of the student who requested the meeting.	As a system, I shall link the Roll No to the student's profile for the teacher's reference.
T4	As a teacher, I shall click the Accept checkbox to confirm my availability for the meeting.	As a system, I shall update the status to "Accepted" and notify the student immediately.
T5	As a teacher, I shall click BACK to return to the previous dashboard view.	As an admin, I shall ensure the navigation doesn't lose the teacher's place in the list.



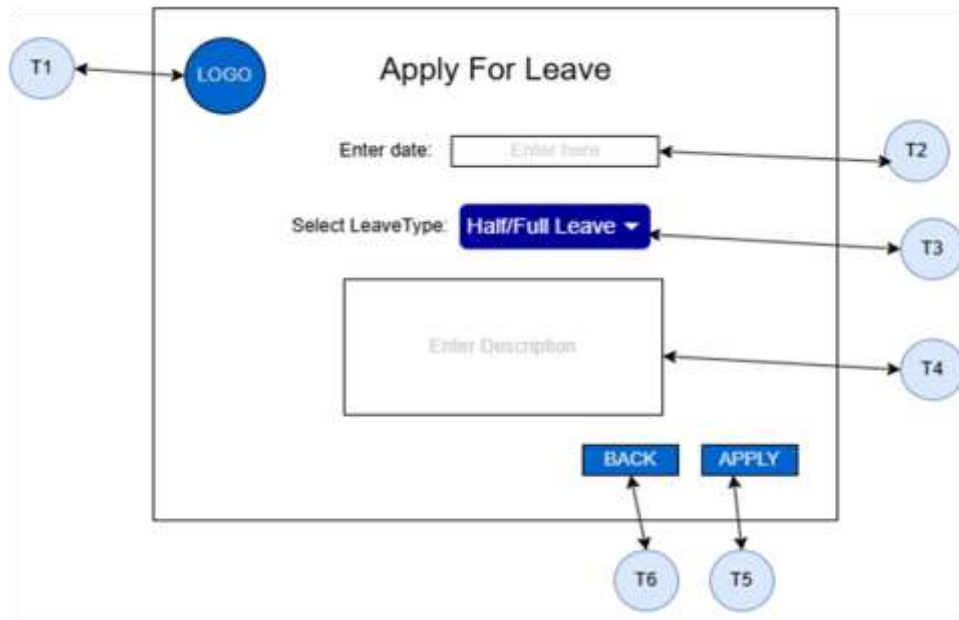
ID	Requirements(Teacher)	Requirements(Admin)
T1	As a teacher/admin, I shall click the LOGO to return to my main dashboard.	As a teacher/admin, I shall click the LOGO to return to my main dashboard.
T2	As a teacher/admin, I shall view existing Emergency notifications (E1-E6) to avoid duplicate alerts.	As a teacher/admin, I shall view existing Emergency notifications (E1-E6) to avoid duplicate alerts.
T3	As a teacher/admin, I shall Generate New Notification by typing the announcement details.	As a teacher/admin, I shall Generate New Notification by typing the announcement details.
T4	As a teacher/admin, I shall click Send to broadcast the notification to the selected student groups.	As a teacher/admin, I shall click Send to broadcast the notification to the selected student groups.

T5	As a teacher/admin, I shall click BACK to discard the draft and return to the dashboard.	As a teacher/admin, I shall click BACK to discard the draft and return to the dashboard.
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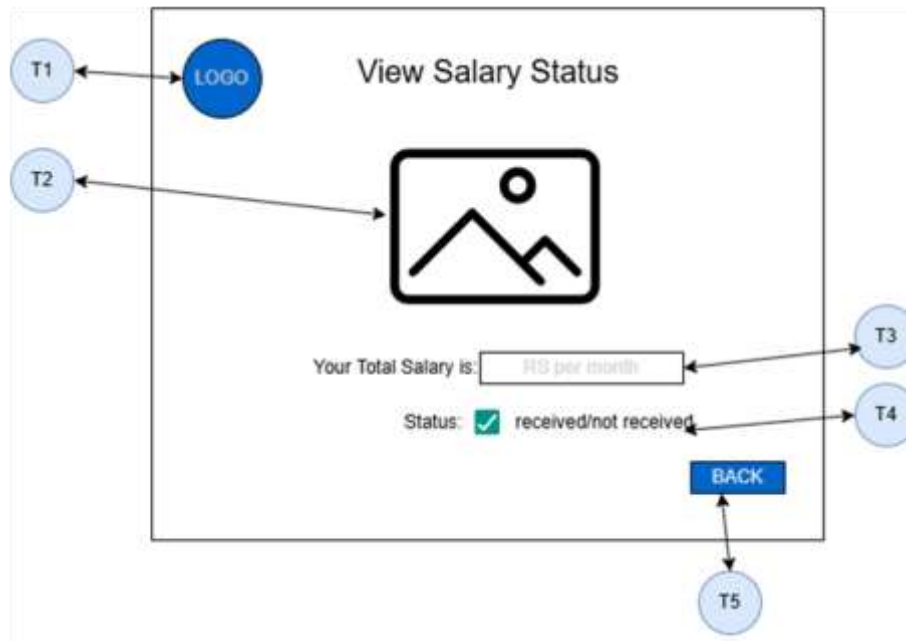
ID	Requirements(Teacher)	Requirements(Admin)
T1	As a teacher, I shall click the LOGO to return to the teacher portal home.	As an admin, I shall ensure the logo redirection link is functional and accurate.
T2	As a teacher, I shall Select Section to specify which class is being canceled.	As an admin, I shall provide a list of all class sections assigned to that specific teacher.
T3	As a teacher, I shall view All sections to verify and select the correct student group.	As an admin, I shall update the section list at the start of each semester.

T4	As a teacher, I shall Enter Date to specify when the lecture was scheduled.	As a system, I shall restrict date selection to current or future dates only.
T5	As a teacher, I shall Enter Lecture no to identify which specific session is canceled.	As a system, I shall verify the lecture number against the existing class schedule.
T6	As a teacher, I shall provide a Reason for the cancellation for student/admin review.	As an admin, I shall review these reasons for departmental record-keeping.
T7	As a teacher, I shall click BACK to exit the form without canceling the class.	As an admin, I shall ensure no "Cancellation" alert is triggered if back is clicked.
T8	As a teacher, I shall click CANCEL to finalize the action and notify the class.	As a system, I shall automatically send an alert to all students enrolled in that section.



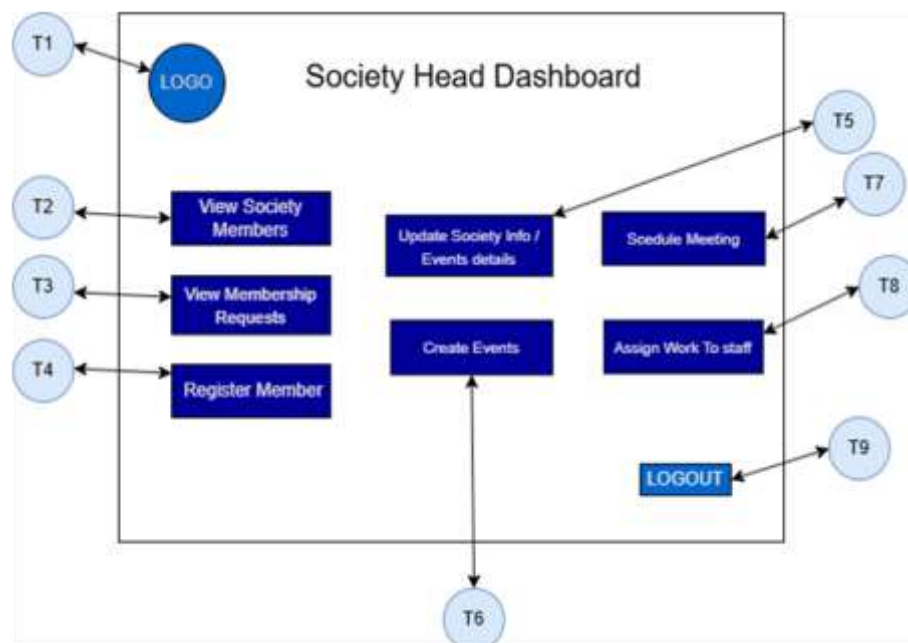
ID	Requirements(Teacher)	Requirements(Admin)
T1	As a teacher, I shall click the LOGO to return to my main dashboard view.	As an admin, I shall ensure the logo redirection maintains the teacher's session.
T2	As a teacher, I shall Enter date to specify the exact day I will be away from duties.	As an admin, I shall restrict date selection to prevent backdating leave requests.
T3	As a teacher, I shall select the Leave Type (Half/Full) from the dropdown menu.	As an admin, I shall manage leave categories and their impact on the payroll system.
T4	As a teacher, I shall Enter	As an admin, I shall review the

	Description to provide the official reason for my leave.	reason before approving or rejecting the request.
T5	As a teacher, I shall click APPLY to submit my leave request for approval.	As an admin, I shall receive a system notification to take action on the new application.
T6	As a teacher, I shall click BACK to exit the form without saving any data.	As an admin, I shall ensure no "pending" entry is created if the teacher clicks back.



ID	Requirements(Teacher)	Requirements(Admin)
T1	As a teacher, I shall click the LOGO to navigate back to the portal home.	As an admin, I shall ensure the logo link points to the secure home view.
T2	As a teacher, I shall view the Salary Slip/Image for a visual confirmation of the record.	As a finance admin, I shall upload the digital salary slips at the start of every month.
T3	As a teacher, I shall view Your Total Salary in a read-only field (RS per month).	As a finance admin, I shall ensure the salary amount is pulled from the secure payroll database.

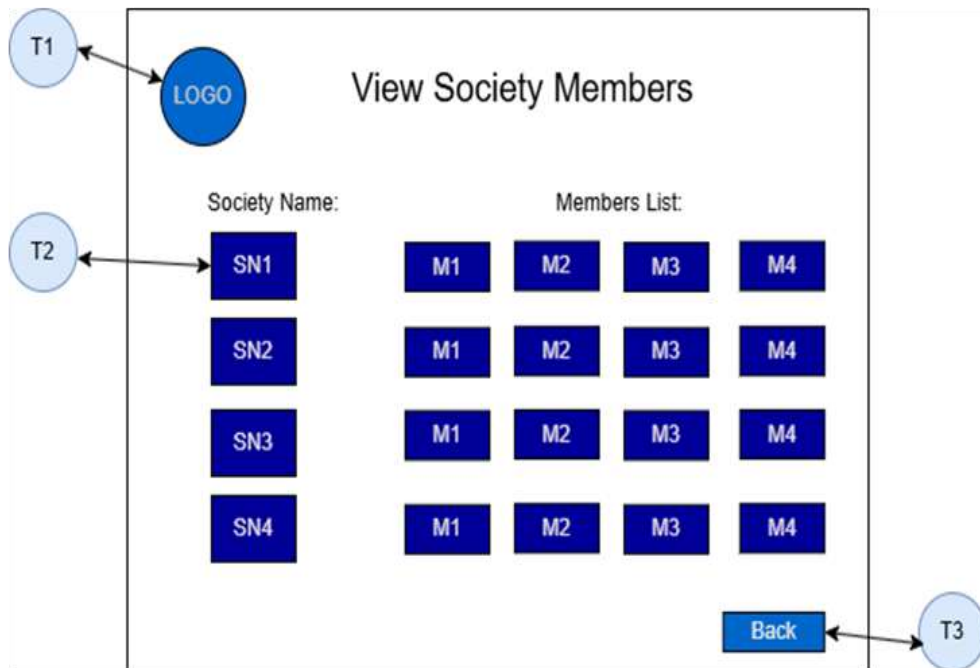
T4	As a teacher, I shall check the Status (received/not received) for payment confirmation.	As a finance admin, I shall update the status checkbox once the transaction is completed.
T5	As a teacher, I shall click BACK to exit the financial view.	As an admin, I shall ensure that sensitive financial views are cleared from the screen upon exit.



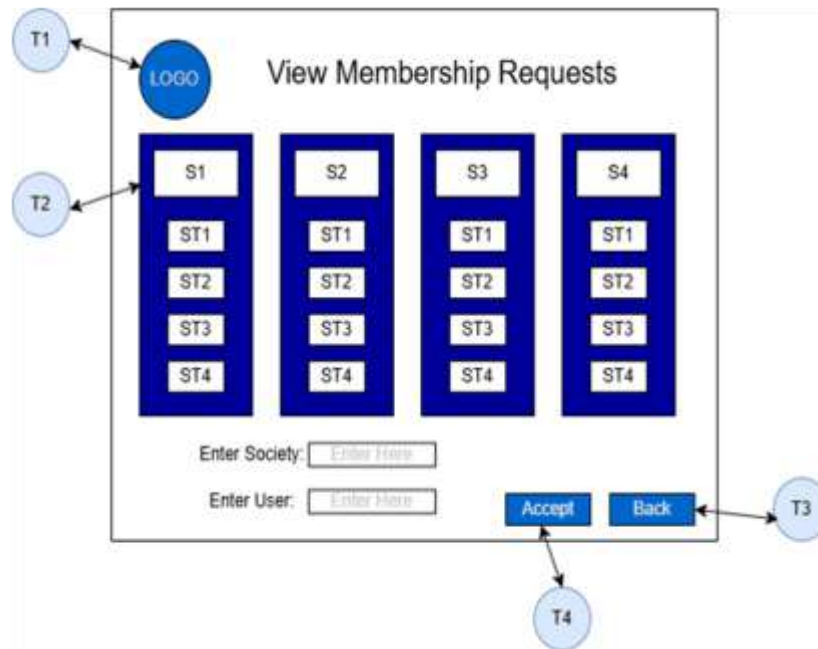
ID	Requirements(SocietyHead)	Requirements(Admin)
T1	As a teacher, I shall click the LOGO to return to the teacher portal home screen.	As an admin, I shall ensure the logo redirection maintains the teacher's session.

T2	As a teacher, I shall Enter date to specify the day I will be absent from duties.	As a system, I shall restrict date selection to prevent backdating leave requests.
T3	As a teacher, I shall select the Leave Type (Half/Full) from the dropdown menu.	As an admin, I shall manage these categories and their impact on faculty payroll.
T4	As a teacher, I shall Enter Description to provide the official reason for my leave.	As an admin/HOD, I shall review this reason before approving or rejecting the request.
T5	As a teacher, I shall click APPLY to formally submit the request for approval.	As a system, I shall push a notification to the HOD/Admin to take action on the new application.
T6	As a teacher, I shall click BACK to discard the form and return to the dashboard.	As a system, I shall ensure no "pending" record is created if the teacher clicks back.
T7	As a teacher, I shall click BACK to discard the form and return to the dashboard.	As an admin/HOD, I shall review this reason before approving or rejecting the request.

T8	As a teacher, I shall click BACK to discard the form and return to the dashboard.	As an admin/HOD, I shall review this reason before approving or rejecting the request.
T9	As a teacher, I shall click BACK to discard the form and return to the dashboard.	As an admin/HOD, I shall review this reason before approving or rejecting the request.

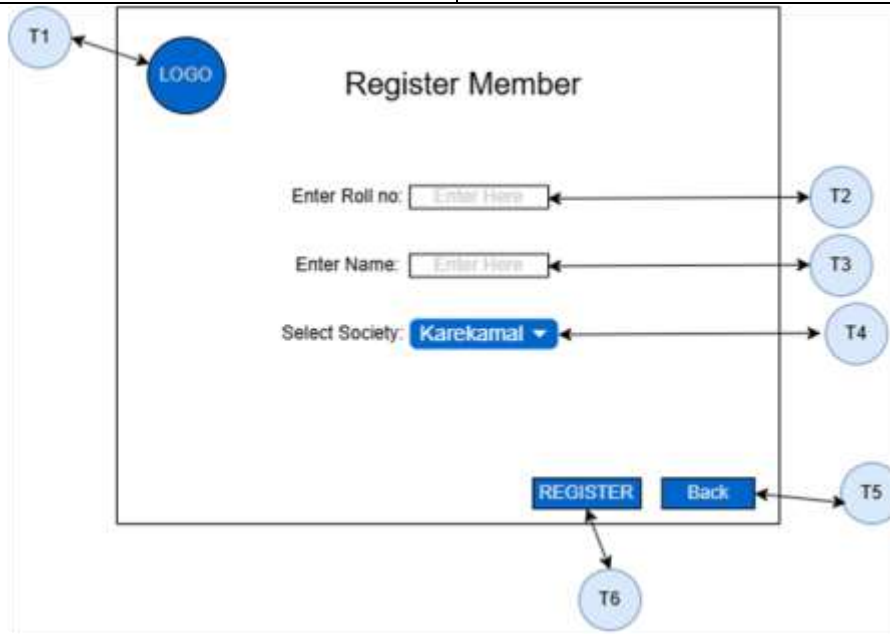


ID	Requirements(Teacher)	Requirements(Admin)
T1	As a teacher, I shall click the LOGO to return to the teacher portal home screen.	As an admin, I shall ensure the logo redirection maintains the teacher's active session and security tokens.
T2	As a teacher, I shall select a Society Name (SN1-SN4) to filter the members displayed on the right.	As a system, I shall dynamically fetch and update the "Members List" grid based on the unique ID of the selected society.
T3	As a teacher, I shall click the BACK button to exit the current view and return to the previous dashboard.	As a system, I shall clear any temporary filter states (selected society) when the user navigates away from this screen.



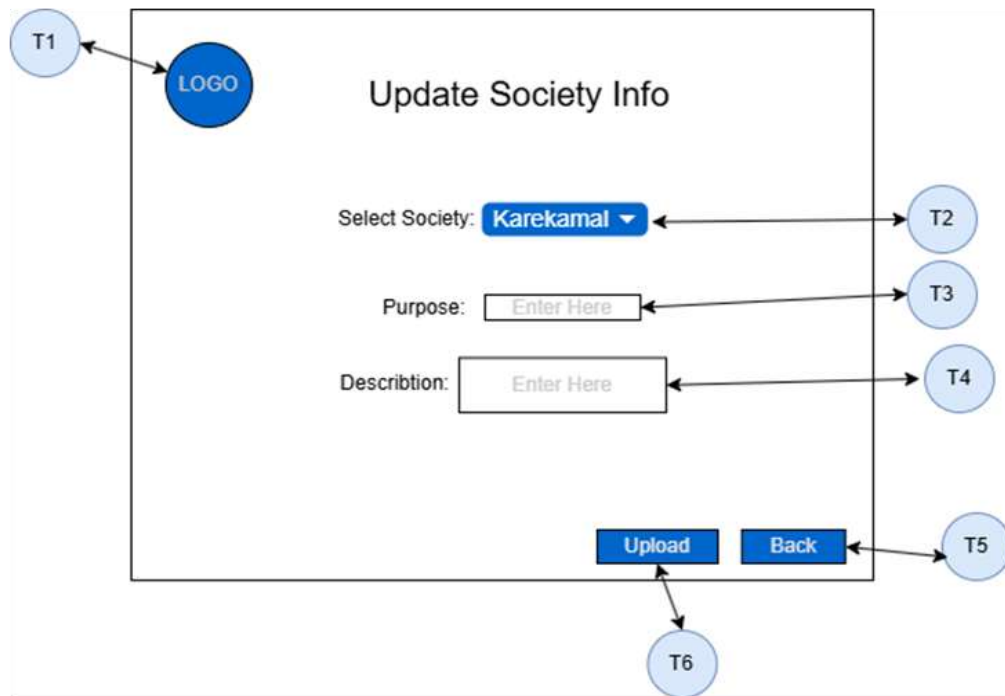
D	Requirements(Teacher)	Requirements(Admin)
T1	As a teacher, I shall click the LOGO to return to the portal home screen.	As a system, I shall ensure that navigating via the logo does not log the user out of their current session.
T2	As a teacher, I shall view the Society columns (S1-S4) and their respective pending student requests (ST1-ST4) to monitor applicant volume.	As a system, I shall dynamically populate these lists from the database, showing only students whose membership status is "Pending."
T3	As a teacher, I shall click the BACK button to return to the main society management dashboard.	As a system, I shall ensure that any unsubmitted text in the "Enter Society" or "Enter User" fields is cleared upon exit.
T4	As a teacher, I shall enter a specific Society and User ID and click ACCEPT to approve the student's membership.	As a system, I shall validate that the entered User ID belongs to the specified Society before updating the status to "Approved."

T5	As a teacher, I shall use the Input Fields to manually type the Society and User names I wish to process.	As a system, I shall provide "placeholder" text (Enter Here) to guide the user on where to input the required data.
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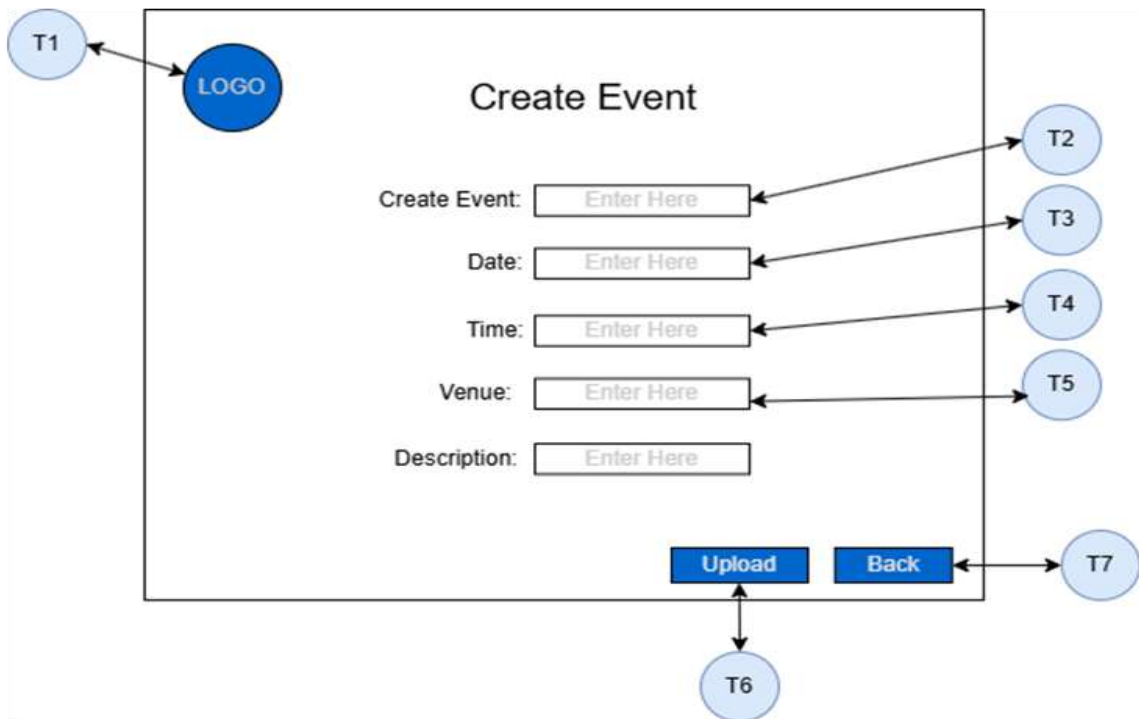
ID	Requirements(SocietyHead)	Requirements(Admin)
T1	As a user, I shall click the LOGO to quickly return to the home screen of the management system.	As a system, I shall ensure that clicking the logo redirects the user to the base URL of their respective role portal.
T2	As a user, I shall enter the Roll no of the student to identify them uniquely in the university database.	As a system, I shall validate that the Roll Number follows the correct format (e.g., 2022-CS-XXX) and exists in the student records.

T3	As a user, I shall enter the Name of the student to be registered for the society.	As a system, I shall ensure the "Name" field accepts only alphabetic characters and is not left blank.
T4	As a user, I shall select the Society (e.g., Karekamal) from the dropdown menu to assign the member correctly.	As a system, I shall fetch the list of active societies from the database to populate the dropdown options.
T5	As a user, I shall click BACK to discard the registration data and return to the previous screen.	As a system, I shall ensure that no data is committed to the database if the back button is pressed before submission.
T6	As a user, I shall click REGISTER to save the student as a formal member of the selected society.	As a system, I shall check for duplicate registrations (if the student is already a member) before creating a new record.



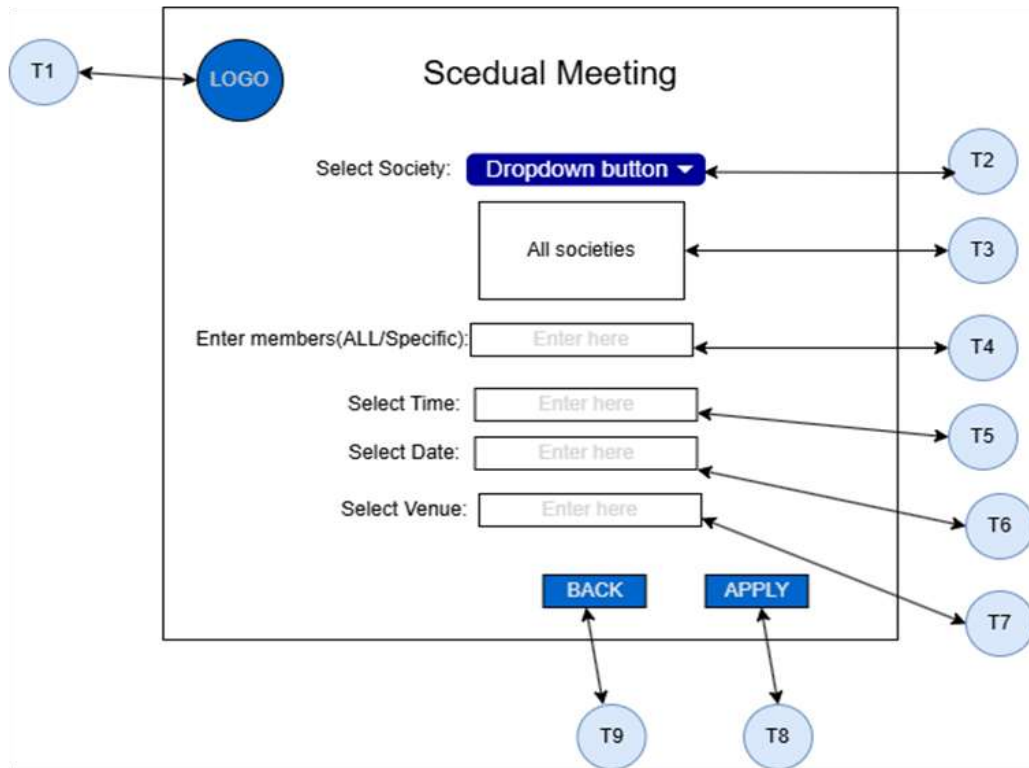
ID	Requirements(SocietyHead)	Requirements(Admin)
T1	As a teacher, I shall click the LOGO to navigate back to the main portal home screen.	As a system, I shall ensure that any unsaved changes in the text fields are discarded when the user clicks the logo.
T2	As a teacher, I shall Select Society (e.g., Karekamal) from the dropdown to choose which society's details to modify.	As a system, I shall load the existing Purpose and Description from the database immediately after a society is selected.

T3	As a teacher, I shall enter the Purpose of the society to define its primary mission or goal.	As a system, I shall restrict the "Purpose" field to a single-line input with a maximum character limit (e.g., 100 characters).
T4	As a teacher, I shall enter a detailed Description to provide more information about society activities.	As a system, I shall provide a multi-line text area for the "Description" field to allow for more comprehensive information.
T5	As a teacher, I shall click BACK to exit the update form and return to the society management dashboard.	As a system, I shall prevent any database "Update" queries from running if the user chooses to go back.
T6	As a teacher, I shall click UPLOAD to save the updated purpose and description to the system.	As a system, I shall perform an UPDATE operation in the database and display a "Successfully Updated" confirmation message.



ID	Requirements(SocietyHead)	Requirements(Admin)
T1	As a user, I shall click the LOGO to navigate back to the portal home screen.	As a system, I shall ensure that no partial event data is saved if the user navigates away using the logo.
T2	As a user, I shall enter the Event Name to specify the title of the upcoming activity.	As a system, I shall validate that the event title is unique and does not exceed a character limit (e.g., 50 characters).
T3	As a user, I shall enter the Date for when the event is scheduled to take place.	As a system, I shall restrict date selection to the present or future to prevent scheduling events in the

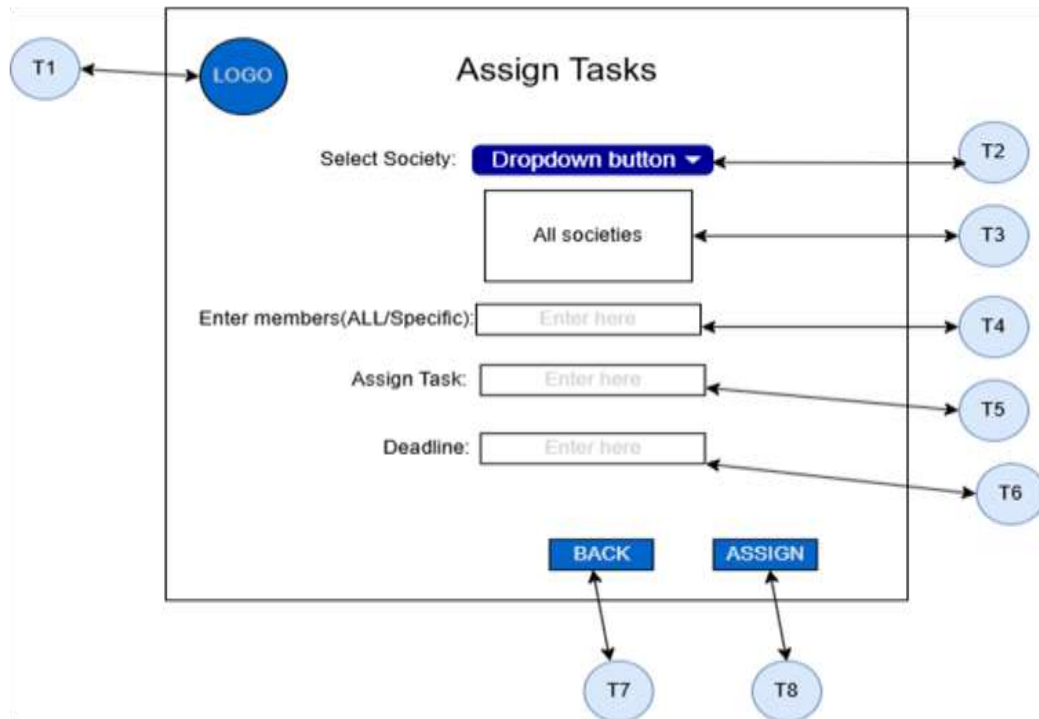
		past.
T4	As a user, I shall enter the Time to inform participants when the event begins.	As a system, I shall ensure the time is entered in a valid format (e.g., HH:MM AM/PM).
T5	As a user, I shall enter the Venue to specify the location (e.g., Seminar Hall, Ground).	As a system, I shall store this location data and link it to the event record for notification purposes.
T6	As a user, I shall click UPLOAD to formally publish the event details to the system.	As a system, I shall save all input fields to the "Events" table and trigger a notification to the relevant society members.
T7	As a user, I shall click BACK to cancel the event creation and return to the previous menu.	As a system, I shall clear all input buffers to ensure the form is empty the next time it is opened.



D	Requirements(SocietyHead)	Requirements(Admin)
T1	As a user, I shall click the LOGO to return to the portal home screen.	As a system, I shall ensure the user's session remains active during the redirection.
T2	As a user, I shall use the Select Society dropdown to choose the specific group for the meeting.	As a system, I shall populate the dropdown with societies the user is authorized to manage.

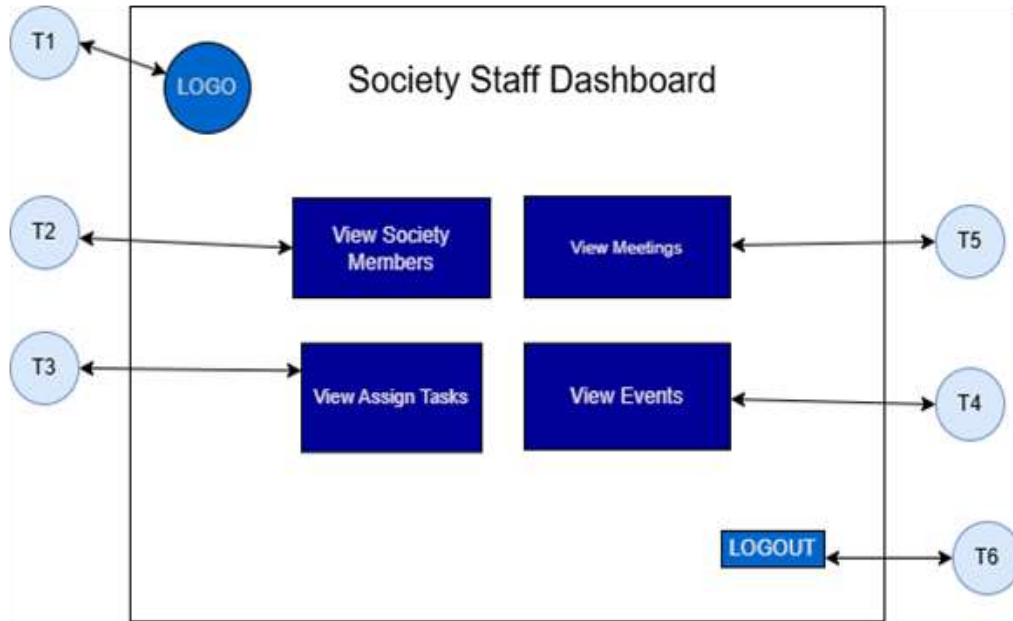
T3	As a user, I shall view the All Societies display area to see a summary of involved groups.	As a system, I shall display selected societies in this area to confirm the meeting scope.
T4	As a user, I shall Enter members (ALL/Specific) to define the attendee list for the meeting.	As a system, I shall provide an autocomplete feature to help the user find specific member IDs quickly.
T5	As a user, I shall Select Time to set the start and end duration of the meeting.	As a system, I shall validate that the time format is correct and does not conflict with existing schedules.
T6	As a user, I shall Select Date to specify when the meeting will occur.	As a system, I shall provide a calendar picker and restrict selection to future dates only.
T7	As a user, I shall Select Venue to designate the physical or virtual location of the meeting.	As a system, I shall check the venue database for availability on the selected date and time.

T8	As a user, I shall click APPLY to finalize the schedule and send invitations.	As a system, I shall create a meeting record and push email/in-app notifications to all listed members.
T9	As a user, I shall click BACK to discard the current inputs and return to the previous screen.	As a system, I shall ensure no "draft" meeting is created in the database upon clicking back.



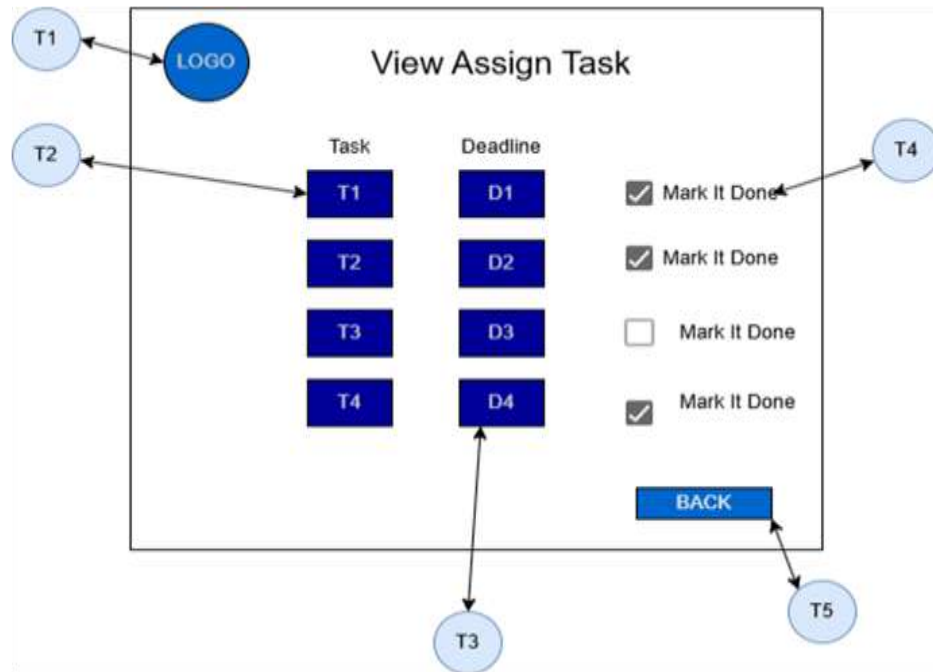
ID	Requirements(SocietyHead)	Requirements(Admin)
T1	As a user, I shall click the LOGO to return to the portal home screen.	As a system, I shall ensure that no partial task data is stored if the user navigates away using the logo.
T2	As a user, I shall select a Society from the dropdown menu to define the scope of the task.	As a system, I shall filter the member list in the background based on the society selected in this dropdown.
T3	As a user, I shall view the All societies box to see which groups are currently being managed.	As a system, I shall display the name of the selected society in this area as a visual confirmation.

T4	As a user, I shall Enter members (ALL/Specific) to designate who is responsible for the task.	As a system, I shall validate the member IDs and ensure they belong to the selected society before proceeding.
T5	As a user, I shall enter the Assign Task details to provide clear instructions on the work required.	As a system, I shall treat this as a mandatory field and prevent submission if the task description is empty.
T6	As a user, I shall enter a Deadline to specify the date and time by which the task must be completed.	As a system, I shall validate that the deadline is set to a future date and time relative to the current system clock.
T7	As a user, I shall click BACK to discard the current task details and return to the previous menu.	As a system, I shall reset all input fields to their default state (empty) upon clicking the back button.
T8	As a user, I shall click ASSIGN to formally issue the task and notify the selected members.	As a system, I shall create a new entry in the "Tasks" table and trigger push notifications for all assigned users.



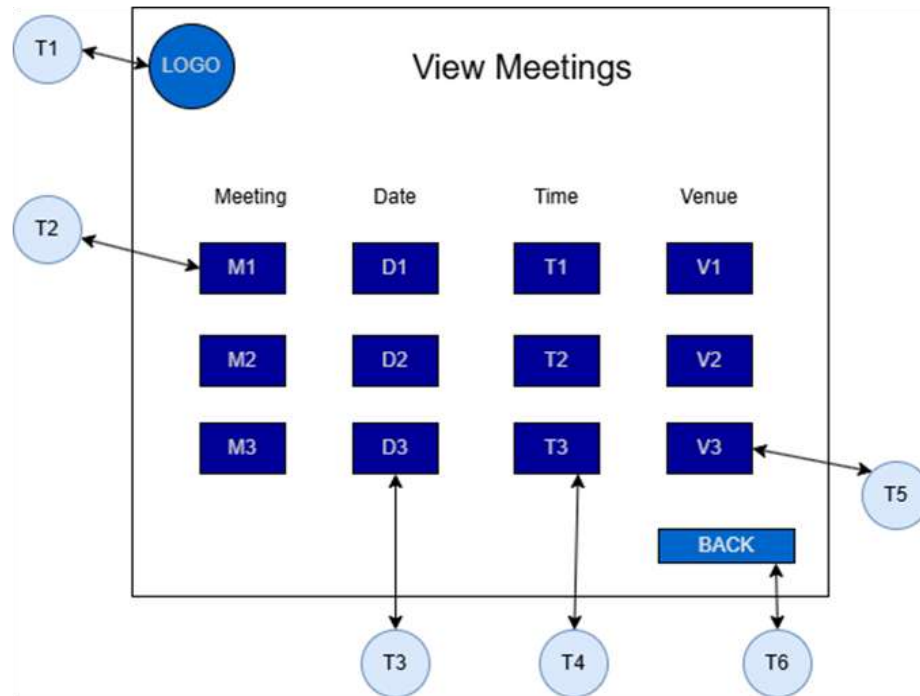
ID	Requirements(SocietyHead)	Requirements(Admin)
T1	As a staff member, I shall click the LOGO to refresh the current dashboard view.	As a system, I shall re-fetch the latest metrics (member counts, pending tasks) from the database upon a logo click.
T2	As a staff member, I shall click View Society Members to see the list of students in my assigned societies.	As a system, I shall redirect the user to the "View Society Members" interface while passing the relevant Staff ID context.
T3	As a staff member, I shall click View Assign Tasks to	As a system, I shall redirect the user to the task management

	check the status of work distributed to students.	module and load the current task-tracking database table.
T4	As a staff member, I shall click View Events to see upcoming scheduled activities and their details.	As a system, I shall ensure that only events occurring after the current date are displayed prominently in the target view.
T5	As a staff member, I shall click View Meetings to access the schedule for upcoming and past meetings.	As a system, I shall verify the user's permissions to ensure they can only view meetings for societies they are authorized to manage.
T6	As a staff member, I shall click LOGOUT to securely terminate my session and exit the portal.	As a system, I shall invalidate the session token, clear cache, and redirect the user back to the login page.



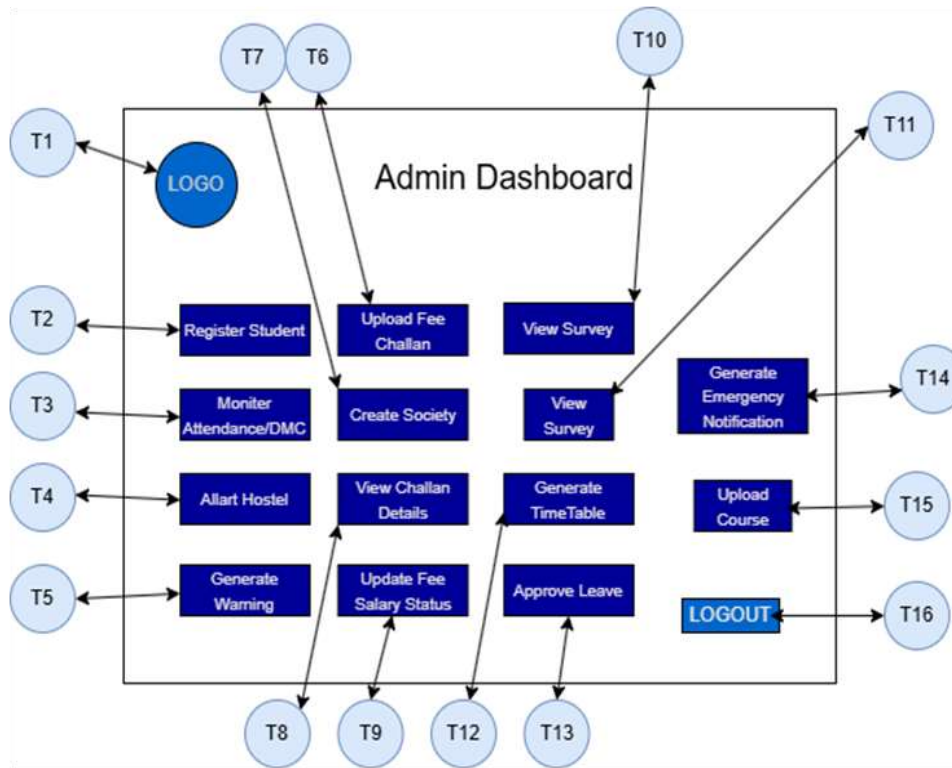
ID	Requirements(SocietyHead)	Requirements(Admin)
T1	As a user, I shall click the LOGO to quickly return to the home screen.	As a system, I shall ensure that clicking the logo redirects the user to their main dashboard view.
T2	As a user, I shall view the Task (T1-T4) column to see the titles of the work I have been assigned.	As a system, I shall fetch only those tasks from the database where the User ID matches the currently logged-in student.

T3	As a user, I shall view the Deadline (D1-D4) for each task to prioritize my work accordingly.	As a system, I shall highlight deadlines in red if the current date has passed the specified deadline.
T4	As a user, I shall click the Mark It Done checkbox to update the status of a completed task.	As a system, I shall update the "Status" field in the database from 'Pending' to 'Completed' upon the checkbox interaction.
T5	As a user, I shall click BACK to exit the task view and return to the previous screen.	As a system, I shall ensure any checked items are saved before the user navigates away from the page.



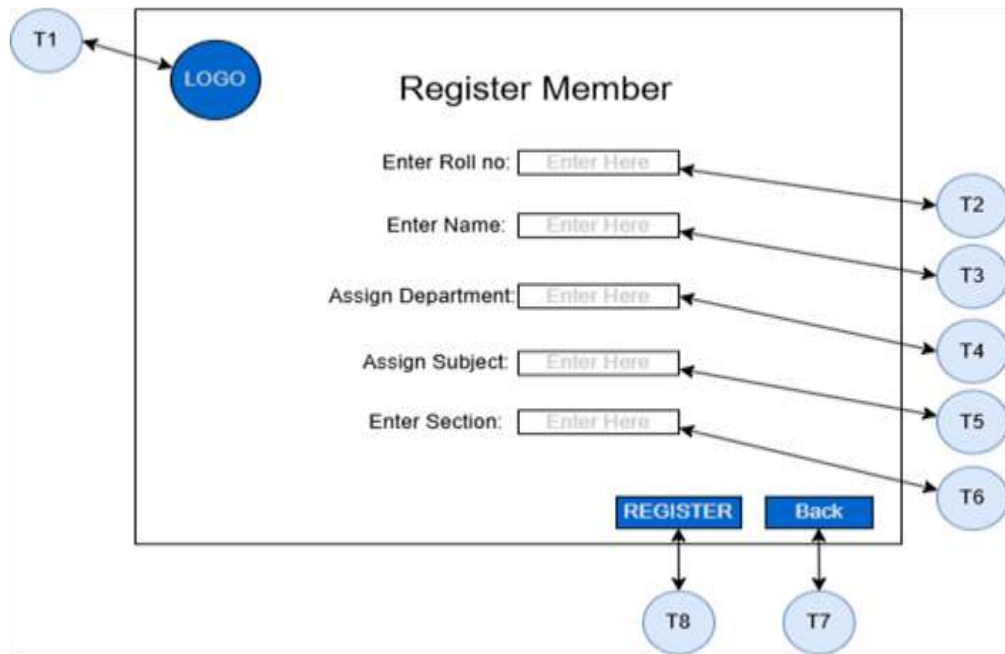
ID	Requirements(SocietyHead)	Requirements(Admin)
T1	As a user, I shall click the LOGO to return to the main dashboard home screen.	As a system, I shall ensure the logo link points to the user's role-based home page (Staff or Student).
T2	As a user, I shall view the Event names (T1-T3) to identify specific upcoming society activities.	As a system, I shall fetch event titles from the database and sort them chronologically by default.
T3	As a user, I shall view the Date (D1-D3) to know when each event is scheduled to take place.	As a system, I shall format dates consistently (e.g., DD-MM-YYYY) to ensure clarity for the user.

T4	As a user, I shall view the Time (T1-T3) to know exactly when the event starts and ends.	As a system, I shall convert stored timestamps into a user-friendly 12-hour or 24-hour format.
T5	As a user, I shall view the Venue (D1-D3) to locate the physical or digital room for the event.	As a system, I shall display the specific room number or hall name as defined in the "Create Event" module.
T6	As a user, I shall click BACK to exit the events list and return to the previous navigation menu.	As a system, I shall redirect the user to the "Society Staff Dashboard" or "Student Dashboard" depending on the session context.



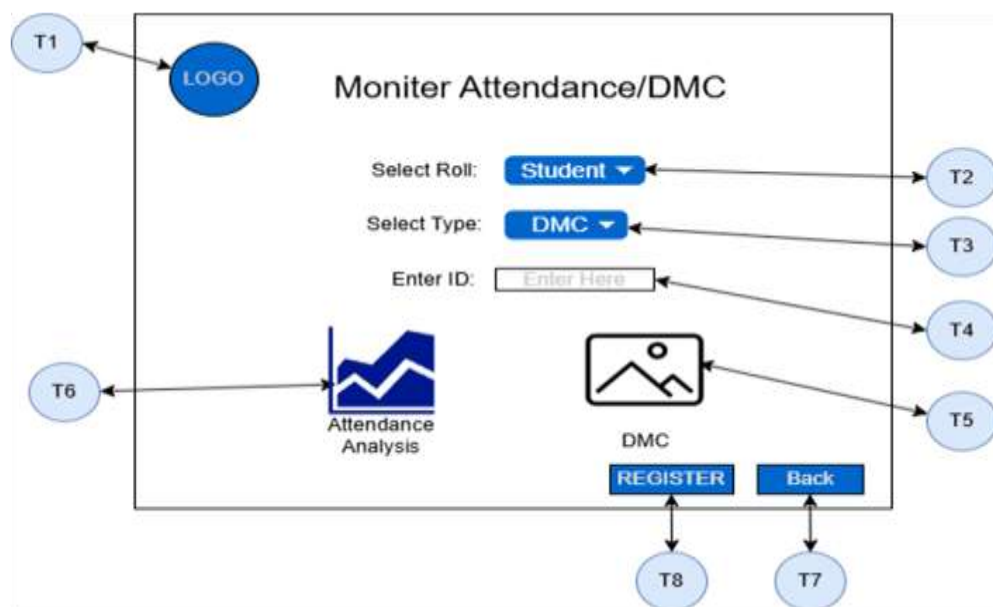
ID	Requirements(Admin)
T1	As an admin, I shall click the LOGO to refresh the dashboard status.
T2	As an admin, I shall click Register Student to add new university enrollments.
T3	As an admin, I shall click Monitor Attendance/DMC to track student performance
T4	As an admin, I shall click Allart Hostel to manage student accommodation and room assignments.
T5	As an admin, I shall click Generate Warning to issue formal notices for low attendance or conduct.
T6	As an admin, I shall click Upload Fee Challan to generate billing for the semester.
T7	As an admin, I shall click Create Society to approve and register new campus organizations
T8	As an admin, I shall click View Challan Details to monitor the status of paid and unpaid fees.
T9	As an admin, I shall click Update Fee Salary Status to manage staff payroll and fee reconciliation.

T10	As an admin, I shall click View Survey to see feedback gathered from students or faculty.
T12	As an admin, I shall click Generate TimeTable to create or modify class schedules.
T13	As an admin, I shall click Approve Leave to review and finalize faculty leave requests.
T14	As an admin, I shall click Generate Emergency Notification to broadcast urgent alerts.
T15	As an admin, I shall click Upload Course to define the curriculum for the semester.
T16	As an admin, I shall click LOGOUT to securely end the administrative session.



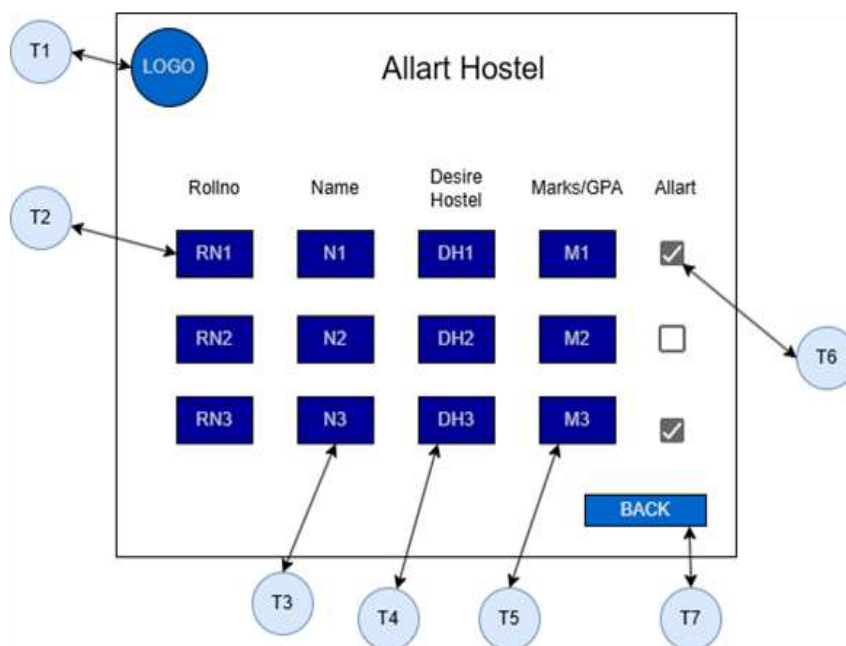
ID	Requirements(Admin)
T1	As a user, I shall click the LOGO to return to the administrative dashboard home screen.
T2	As a user, I shall enter the Roll no to identify the student in the university record system.
T3	As a user, I shall enter the Name of the member for identification and record-keeping.
T4	As a user, I shall Assign Department to categorize the member based on their

	academic faculty.
T5	As a user, I shall Assign Subject to link the member to a specific course or technical society domain.
T6	As a user, I shall Enter Section to specify the member's academic group (e.g., Section A, B, or C).
T7	As a user, I shall click Back to discard the registration and return to the previous menu.
T8	As a user, I shall click REGISTER to formally save the member's details and finalize the enrollment.



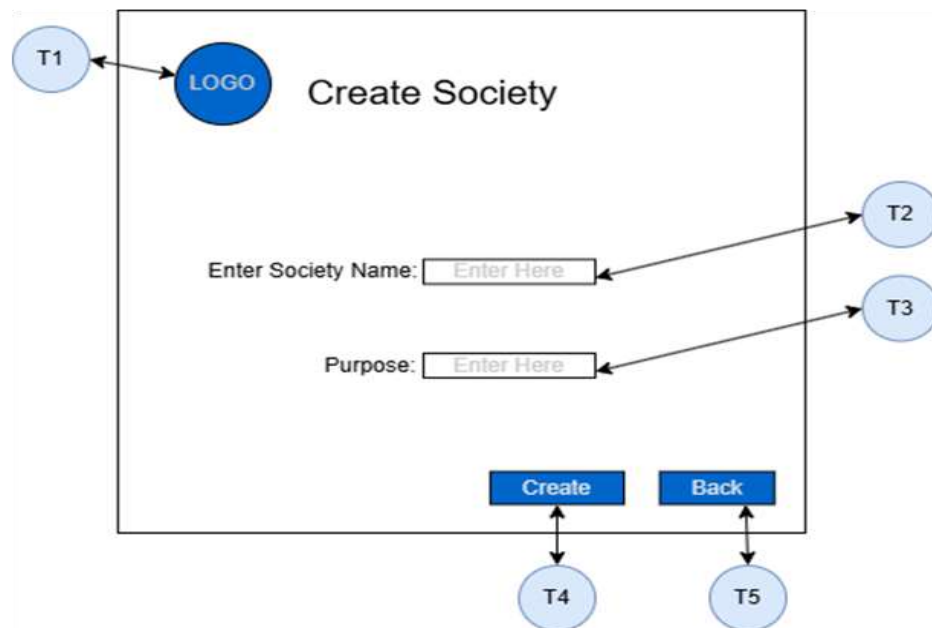
ID	Requirements(Admin)
T1	As a user, I shall click the LOGO to return to the main administrative dashboard home screen.
T2	As a user, I shall select the Role (e.g., Student, Faculty) from the dropdown menu to filter the search criteria.
T3	As a user, I shall select the Type of document (e.g., DMC, Attendance Record) to specify the information to be retrieved.
T4	As a user, I shall input the unique Identification Number (ID) into the text field to fetch the specific record from the database.

T5	As a user, I shall view the DMC placeholder/icon which represents the generated Detail Marks Certificate for the selected ID.
T6	As a user, I shall click the Attendance Analysis icon to view graphical reports and trends regarding student presence.
T7	As a user, I shall click the Back button to discard current selections and return to the previous screen.
T8	As a user, I shall click REGISTER to formally save or submit the attendance/DMC monitoring data to the system.

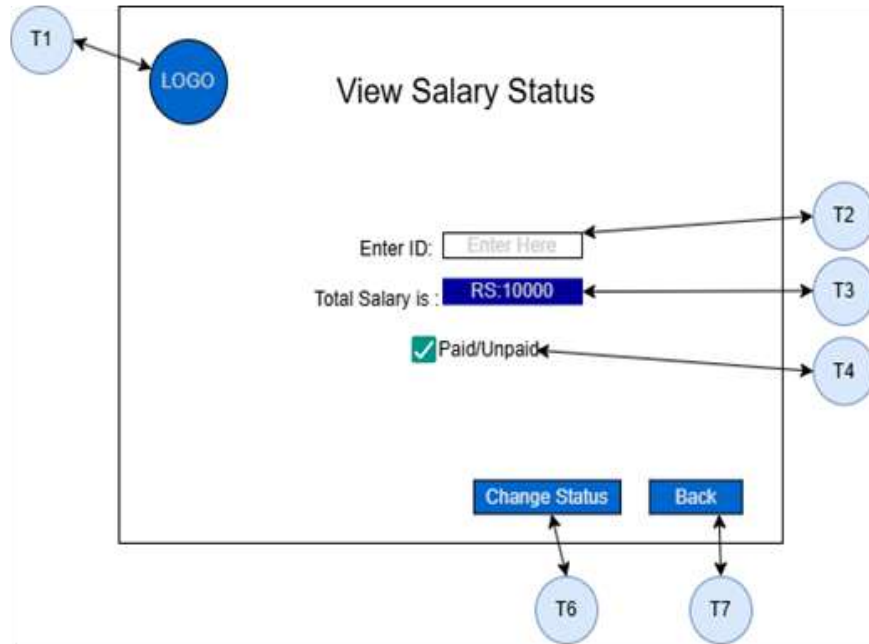


ID	Requirements(Admin)
T1	As a user, I shall click the LOGO to return to the administrative dashboard home screen.
T2	As a user, I shall view the Roll No (RN1, RN2, etc.) to uniquely identify each student candidate in the list.
T3	As a user, I shall view the Name (N1, N2, etc.) to verify the identity of the student applying for a hostel.
T4	As a user, I shall view the Desire Hostel (DH1, DH2, etc.) to see which specific hostel facility the student has requested.

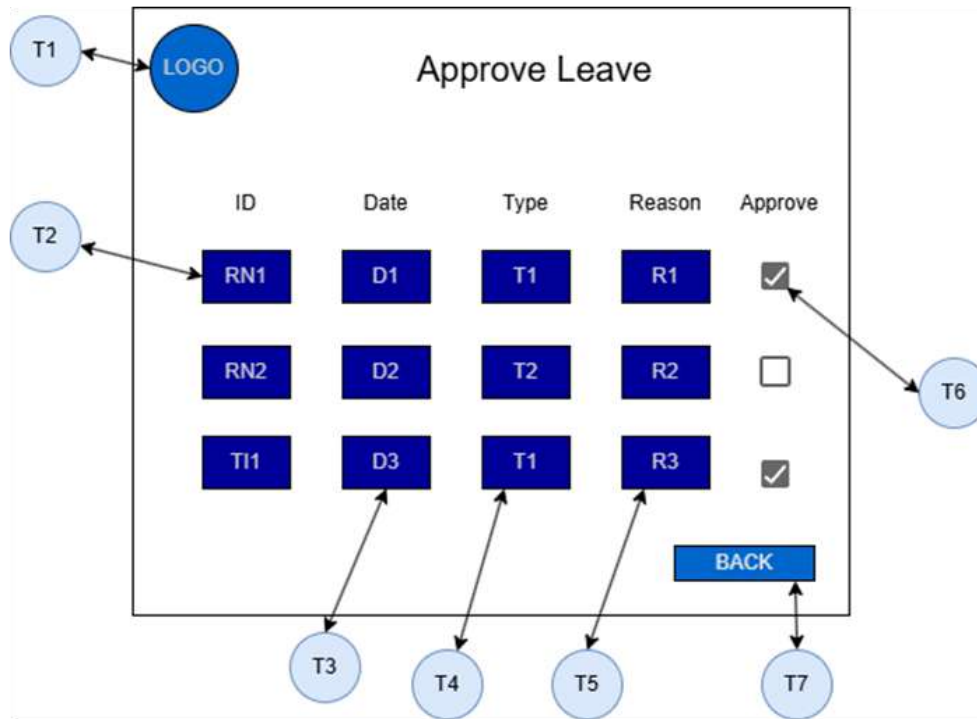
T5	As a user, I shall view the Marks/GPA (M1, M2, etc.) to evaluate the student's eligibility based on academic merit.
T6	As a user, I shall click the Allot Checkbox to formally approve and assign a hostel room to the selected student.
T7	As a user, I shall click the BACK button to return to the previous management menu without saving further changes.



ID	Requirements(Admin)
T1	As a user, I shall click the LOGO to return to the administrative dashboard home screen.
T2	As a user, I shall input the Society Name in the designated text field to define the title of the new organization.
T3	As a user, I shall enter the Purpose or description of the society in the input field to explain its goals or activities.
T4	As a user, I shall click the Create button to register the new society and save the details to the system.
T5	As a user, I shall click the Back button to discard the current entries and return to the previous menu.

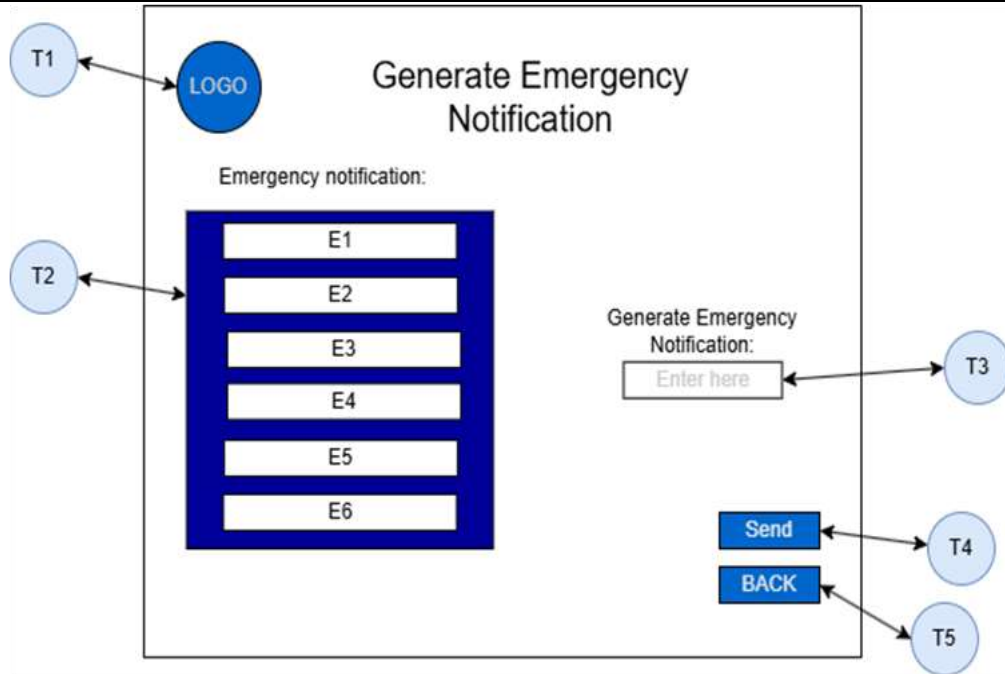


ID	Requirements(Admin)
T1	As a user, I shall click the LOGO to return to the administrative dashboard home screen.
T2	As a user, I shall input the Staff/Employee ID in the text field to retrieve the salary details for that specific person.
T3	As a user, I shall view the Total Salary (e.g., RS:10000) displayed as a read-only value fetched from the payroll records.
T4	As a user, I shall interact with the Paid/Unpaid checkbox to indicate whether the salary for the current period has been disbursed.
T6	As a user, I shall click the Change Status button to update the payment record in the database.
T7	As a user, I shall click the Back button to exit the salary view and return to the previous management menu.



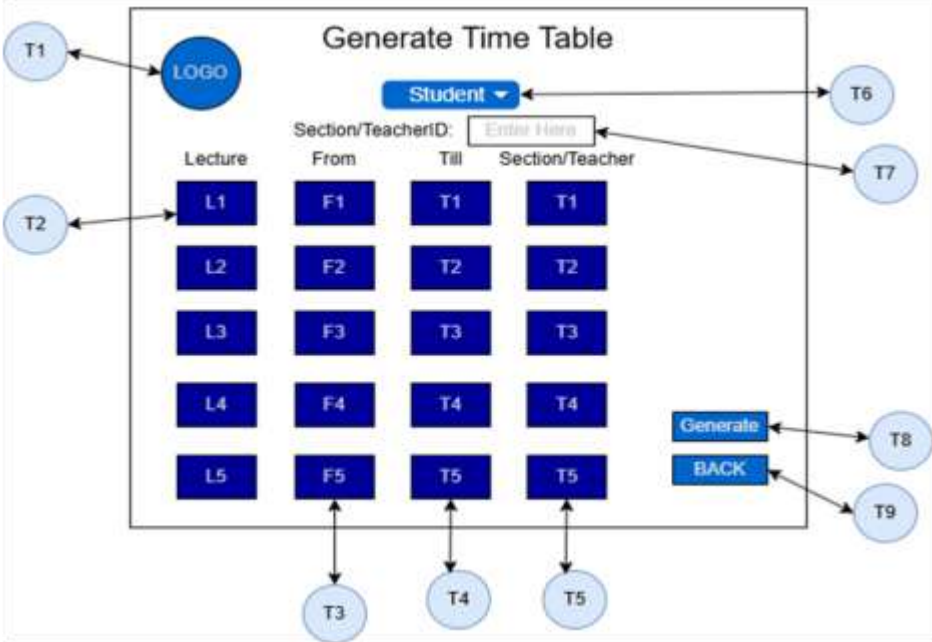
ID	Requirements(Admin)
T1	As a user, I shall click the LOGO to return to the administrative dashboard home screen.
T2	As a user, I shall view the ID (Roll No or Teacher ID) of the individual requesting leave to identify the applicant.
T3	As a user, I shall view the Date (D1–D3) for which the leave has been requested.
T4	As a user, I shall view the Type of leave (e.g., Medical, Casual, Sick) to understand the category of the request.
T5	As a user, I shall view the Reason (R1–R3) provided by the applicant to evaluate the necessity of the leave.

T6	As a user, I shall interact with the Approve Checkbox to grant or deny the specific leave request.
T7	As a user, I shall click the BACK button to return to the previous management menu.



ID	Requirements(Admin)
T1	As a user, I shall click the LOGO to return to the administrative dashboard home screen.
T2	As a user, I shall view and select from a list of Pre-defined Emergency Templates (E1–E6) to quickly populate notification details.
T3	As a user, I shall input a Custom Emergency Message in the text field for specific situations not covered by templates.

T4	As a user, I shall click the Send button to broadcast the notification to all relevant campus users (students and staff).
T5	As a user, I shall click the BACK button to cancel the notification and return to the previous menu.



ID	Requirements(Admin)
T1	As a user, I shall click the LOGO to return to the administrative dashboard home screen.
T2	As a user, I shall view the Lecture list (L1–L5) which identifies the sequence of classes for the day.
T3	As a user, I shall input or select the Start Time (From F1–F5) for each specific lecture slot.

T4	As a user, I shall input or select the End Time (Till T1–T5) for each specific lecture slot.
T5	As a user, I shall assign the Section/Teacher details for each lecture to designate who is teaching or which group is attending.
T6	As a user, I shall use the Dropdown Menu to toggle between "Student" and "Teacher" view to generate the appropriate schedule.
T7	As a user, I shall input a specific Section or Teacher ID in the search field to filter the timetable for a particular entity.
T8	As a user, I shall click the Generate button to process the inputs and create the final timetable record.
T9	As a user, I shall click the BACK button to discard changes and return to the previous management screen.

6.9 Use Cases

Use Case Summary:

Use cases	Actor	Summary	Link
UC-06-001	All Users	User Sign-in / Authentication	UC-06-001: LOGIN
UC-06-002	All Users	Mark Biometric Attendance	UC-06-002: Mark Biometric Attendance
UC-06-003	All Users	View Emergency Notifications	UC-06-003: View Emergency Notifications
UC-01-001	Student	View Timetable	UC-01-001: View Timetable
UC-01-002	Student	View Free Classrooms	UC-01-002: View Free Classrooms
UC-01-003	Student	View Attendance Analysis	UC-01-003: View Attendance Analysis
UC-01-004	Student	Register Courses	UC-01-004: Register Courses
UC-01-005	Student	Apply for Leave	UC-01-005: Apply for Leave
UC-01-006	Student	View DMC	UC-01-006: View DMC
UC-01-007	Student	Manage Fee Challan	UC-01-007: Manage Fee Challan
UC-01-008	Student	Register in Society	UC-01-008: Register in Society
UC-02-001	Teacher	Mark Class Attendance	UC-02-001: Mark Class Attendance
UC-02-002	Teacher	Upload Marks and DMC	UC-02-002: Upload DMC

UC-02-003	Teacher	Manage Meeting Requests	<u>UC-02-003: Manage Meeting Requests</u>
UC-02-004	Teacher	Manage User Registration	<u>UC-02-004: Manage User Registration</u>
UC-03-001	Teacher	Generate TimeTable	<u>UC-03-001: Generate Timetable</u>
UC-03-002	Admin	Approve Leave	<u>UC-03-002: Approve Leave</u>
UC-05-001	Society Head	Generate Emergency Notifications	<u>UC-05-001: Generate Emergency Notification</u>
UC-04-001	Staff	View Survey Results	<u>UC-04-001: View Survey Results</u>

UC-06-001: LOGIN

Element	Description												
ID	UC-06-001												
Name	LOGIN												
Participating Actors	Student, Teacher, Admin, Society Head, Staff												
Goals	A user wants to login into their specific portal to access role-based functionalities.												
Triggers	When a user opens the system URL and clicks the "Login" button or attempts to access a protected module (e.g., Course Registration).												
Pre-conditions	1. The user must have a working device and internet connection. 2. The user must be registered in the system by the Administrator.												
Post-conditions	The user is successfully redirected to their specific dashboard (Student, Teacher, Admin, Society Head, or Staff).												
Basic Flow	<table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>The user selects their role (Student, Teacher, Admin, Society Head, or Staff) from the dropdown menu.</td><td>The system highlights the selected role.</td></tr><tr><td>2</td><td>The user enters their University ID and Password.</td><td>The system validates the input format.</td></tr><tr><td>3</td><td>The user enters the Captcha code shown on the screen.</td><td>The system verifies the Captcha.</td></tr></table>	Step	Actor	System	1	The user selects their role (Student, Teacher, Admin, Society Head, or Staff) from the dropdown menu.	The system highlights the selected role.	2	The user enters their University ID and Password.	The system validates the input format.	3	The user enters the Captcha code shown on the screen.	The system verifies the Captcha.
Step	Actor	System											
1	The user selects their role (Student, Teacher, Admin, Society Head, or Staff) from the dropdown menu.	The system highlights the selected role.											
2	The user enters their University ID and Password.	The system validates the input format.											
3	The user enters the Captcha code shown on the screen.	The system verifies the Captcha.											

	<table><tr><td>4</td><td>The user clicks the "Sign-in" button.</td><td>The system checks the credentials against the selected role and user data in the Database.</td></tr></table>	4	The user clicks the " Sign-in " button.	The system checks the credentials against the selected role and user data in the Database.						
4	The user clicks the " Sign-in " button.	The system checks the credentials against the selected role and user data in the Database.								
Alternate Flows	<table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>The user checks the "Remember Me" checkbox before signing in.</td><td>The system generates a secure token after successful authentication.</td></tr><tr><td>2</td><td>The user closes the browser and re-opens the system URL within 7 days.</td><td>The system detects the valid token and automatically logs the user into their dashboard without requiring credentials.</td></tr></table>	Step	Actor	System	1	The user checks the " Remember Me " checkbox before signing in.	The system generates a secure token after successful authentication.	2	The user closes the browser and re-opens the system URL within 7 days.	The system detects the valid token and automatically logs the user into their dashboard without requiring credentials.
Step	Actor	System								
1	The user checks the " Remember Me " checkbox before signing in.	The system generates a secure token after successful authentication.								
2	The user closes the browser and re-opens the system URL within 7 days.	The system detects the valid token and automatically logs the user into their dashboard without requiring credentials.								
Exceptions	1. Invalid Credentials: System shows "Invalid Username or Password" and clears fields. 2. Role Mismatch: System shows "Role mismatch. Please select the correct role." 3. Captcha Failure: System generates new Captcha and asks user to try again. 4. Account Locked: System locks account for 15 minutes after 5 wrong attempts. 5. Unregistered User: System shows "User not registered. Contact Administrator." 6. Network Failure: System shows "Network error. Please check your connection."									
Qualities	1. Security: Passwords encrypted in database; HTTPS for data									

	transmission. 2. Availability: Login service available 24/7. 3. Performance: Authentication and dashboard load within 2 seconds. 4. Usability: Simple interface with clear error messages. 5. Accessibility: Compatible with screen readers and keyboard navigation.
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UC-06-002: Mark Biometric Attendance

Element	Description						
ID	UC-06-002						
Name	Mark Biometric Attendance						
Participating Actors	Student, Teacher, Admin, Society Head, Staff						
Goals	To record user attendance using fingerprint verification.						
Triggers	User places finger on biometric scanner at campus terminal.						
Pre-conditions	1. User is registered with fingerprint in database. 2. Biometric scanner is working. 3. Attendance time is valid.						
Post-conditions	User attendance is recorded with timestamp.						
Basic Flow	<table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>User tries again after failed attempt.</td><td>System allows up to 3 attempts.</td></tr></table>	Step	Actor	System	1	User tries again after failed attempt.	System allows up to 3 attempts.
Step	Actor	System					
1	User tries again after failed attempt.	System allows up to 3 attempts.					

	<table><tr><td>2</td><td>After 3 failures</td><td>System prompts to use manual login.</td></tr></table>	2	After 3 failures	System prompts to use manual login.
2	After 3 failures	System prompts to use manual login.		
Alternate Flows	Not applicable			
Exceptions	1. Fingerprint Not Recognized: System shows "Fingerprint not recognized. Try again." 2. Duplicate Attendance: System shows "Attendance already recorded today." 3. Scanner Malfunction: System shows "Scanner error. Contact Admin." 4. Network Failure: System shows "Network error. Try again." 5. Timeout: System returns to home screen if no finger placed within 30 seconds.			
Qualities	1. Accuracy: 98% or higher matching rate. 2. Response Time: Verification within 3 seconds. 3. Security: Fingerprint templates stored in encrypted format. 4. Availability: Service available during all class hours. 5. Data Integrity: Attendance cannot be modified without admin approval			

UC-01-001: View Timetable

Element	Description												
ID	UC-01-001												
Name	View Timetable												
Participating Actors	Student, Teacher												
Goals	To view daily or weekly class schedule.												
Triggers	User clicks on "Timetable" option from dashboard.												
Pre-conditions	1. User is logged into the system. 2. Timetable is generated by admin for current semester.												
Post-conditions	User successfully views their class schedule.												
Basic Flow	<table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>User clicks on "View Timetable" from dashboard.</td><td>System displays timetable options (Daily/Weekly).</td></tr><tr><td>2</td><td>User selects Daily or Weekly view.</td><td>System fetches schedule from database.</td></tr><tr><td>3</td><td>User selects the ok option</td><td>System displays timetable with lecture names, timings, and teachers.</td></tr></table>	Step	Actor	System	1	User clicks on "View Timetable" from dashboard.	System displays timetable options (Daily/Weekly).	2	User selects Daily or Weekly view.	System fetches schedule from database.	3	User selects the ok option	System displays timetable with lecture names, timings, and teachers.
Step	Actor	System											
1	User clicks on "View Timetable" from dashboard.	System displays timetable options (Daily/Weekly).											
2	User selects Daily or Weekly view.	System fetches schedule from database.											
3	User selects the ok option	System displays timetable with lecture names, timings, and teachers.											

Alternate Flows	Alternate Flow 1: Teacher View <table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Teacher selects "Teacher" from dropdown.</td><td>System displays timetable showing only lectures assigned to that teacher.</td></tr></table> Alternate Flow 2: No Timetable Found <table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>User tries to view timetable.</td><td>System checks if timetable exists.</td></tr><tr><td>2</td><td>User click the ok button</td><td>If no timetable, system shows "No timetable available for current semester."</td></tr></table>	Step	Actor	System	1	Teacher selects "Teacher" from dropdown.	System displays timetable showing only lectures assigned to that teacher.	Step	Actor	System	1	User tries to view timetable.	System checks if timetable exists.	2	User click the ok button	If no timetable, system shows "No timetable available for current semester."
Step	Actor	System														
1	Teacher selects "Teacher" from dropdown.	System displays timetable showing only lectures assigned to that teacher.														
Step	Actor	System														
1	User tries to view timetable.	System checks if timetable exists.														
2	User click the ok button	If no timetable, system shows "No timetable available for current semester."														
Exceptions	1. No Timetable: System shows "Timetable not generated yet. Contact Admin." 2. Invalid Filter: If user enters wrong section/teacher ID, system shows "No records found." 3. Session Timeout: If user session expires, redirects to login page.															
Qualities	1. Response Time: Timetable loads within 2 seconds. 2. Accuracy: Shows correct schedule based on user role and section. 3. Usability: Clean layout with easy filtering options. 4. Availability: Accessible 24/7 once logged in.															

UC-06-003: View Emergency Notifications

Element	Description												
ID	UC-06-003												
Name	View Emergency Notifications												
Participating Actors	Student, Teacher, Admin, Society Head, Staff												
Goals	To view urgent alerts and announcements.												
Triggers	User clicks on "Notifications" or "Emergency Alerts" from dashboard or homepage.												
Pre-conditions	User is logged into the system (notifications also visible on general system without login).												
Post-conditions	User views list of emergency notifications.												
Basic Flow	<table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>User clicks on "Emergency Notifications" option.</td><td>System fetches all active emergency alerts from database.</td></tr><tr><td>2</td><td>User clicks the ok option</td><td>System displays notifications list with title, date, and time.</td></tr><tr><td>3</td><td>User clicks on specific notification to read details.</td><td>System shows full notification message.</td></tr></table>	Step	Actor	System	1	User clicks on "Emergency Notifications" option.	System fetches all active emergency alerts from database.	2	User clicks the ok option	System displays notifications list with title, date, and time.	3	User clicks on specific notification to read details.	System shows full notification message.
Step	Actor	System											
1	User clicks on "Emergency Notifications" option.	System fetches all active emergency alerts from database.											
2	User clicks the ok option	System displays notifications list with title, date, and time.											
3	User clicks on specific notification to read details.	System shows full notification message.											

Alternate Flows	Alternate Flow 1: General System View (No Login)		
	Step	Actor	System
	1	User opens system homepage without logging in.	System displays scrolling emergency alerts on homepage banner.
	Alternate Flow 2: No Notifications		
	Step	Actor	System
	1	User clicks on notifications.	System checks for active alerts.
	2	User clicks the ok button	If no notifications, system shows "No new notifications."
Exceptions	• Empty List: System shows "No emergency notifications at this time." • Network Error: System shows "Failed to load notifications. Check connection."		
Qualities	• Real-time: Notifications appear immediately after admin sends them. • Priority: Emergency alerts shown at top with red highlight. • Accessibility: Visible on homepage even without login.		

UC-01-002: View Free Classrooms

Element	Description												
ID	UC-01-002												
Name	View Free Classrooms												
Participating Actors	Student, Teacher												
Goals	To find available classrooms for makeup lectures or study.												
Triggers	User clicks on "Free Classrooms" option from dashboard or general system.												
Pre-conditions	1. User is logged into the system. 2. Timetable and room allocations are updated in database.												
Post-conditions	User views list of currently vacant classrooms.												
Basic Flow	<table><tr><th>S t e p</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>User clicks on "View Free Classrooms."</td><td>System displays search options (by time or department).</td></tr><tr><td>2</td><td>User enters preferred time slot (optional).</td><td>System checks database for vacant rooms at that time.</td></tr><tr><td>3</td><td></td><td>System shows list of free classrooms with room</td></tr></table>	S t e p	Actor	System	1	User clicks on "View Free Classrooms."	System displays search options (by time or department).	2	User enters preferred time slot (optional).	System checks database for vacant rooms at that time.	3		System shows list of free classrooms with room
S t e p	Actor	System											
1	User clicks on "View Free Classrooms."	System displays search options (by time or department).											
2	User enters preferred time slot (optional).	System checks database for vacant rooms at that time.											
3		System shows list of free classrooms with room											

	<table><tr><td></td><td></td><td>number, floor, and department.</td></tr><tr><td>4</td><td>User can click "Book" to reserve room (optional).</td><td>System sends booking request to admin for approval.</td></tr></table>			number, floor, and department.	4	User can click "Book" to reserve room (optional).	System sends booking request to admin for approval.									
		number, floor, and department.														
4	User can click "Book" to reserve room (optional).	System sends booking request to admin for approval.														
Alternate Flows	Alternate Flow 1: No Time Specified <table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>User skips time input.</td><td>System displays all rooms currently free at present time.</td></tr></table> Alternate Flow 2: No Rooms Available <table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>User searches for free room.</td><td>System finds no vacant rooms.</td></tr><tr><td>2</td><td>The user clicks the ok option</td><td>System shows "No free classrooms available at this time."</td></tr></table>	Step	Actor	System	1	User skips time input.	System displays all rooms currently free at present time.	Step	Actor	System	1	User searches for free room.	System finds no vacant rooms.	2	The user clicks the ok option	System shows "No free classrooms available at this time."
Step	Actor	System														
1	User skips time input.	System displays all rooms currently free at present time.														
Step	Actor	System														
1	User searches for free room.	System finds no vacant rooms.														
2	The user clicks the ok option	System shows "No free classrooms available at this time."														
Exceptions	1. No Results: System shows "No free classrooms found for selected time." 2. Booking Failed: If room already booked, system shows "Room not															

	available. Please select another." 3. Invalid Time: System shows "Please enter valid time slot."
Qualities	1. Accuracy: Shows real-time room availability based on timetable. 2. Response Time: Search results within 2 seconds. 3. Usefulness: Helps teachers schedule makeup lectures easily.

UC-01-003: View Attendance Analysis

Element	Description												
ID	UC-06												
Name	View Attendance Analysis												
Participating Actors	Student												
Goals	To view personal attendance records and analysis.												
Triggers	Student clicks on "Attendance Analysis" from dashboard.												
Pre-conditions	1. Student is logged into the system. 2. Attendance records exist for the student.												
Post-conditions	Student views attendance graphs and percentages.												
Basic Flow	<table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Student clicks on "View Attendance Analysis."</td><td>System fetches attendance data from database.</td></tr><tr><td>2</td><td>Student clicks on "View Attendance Analysis."</td><td>System displays weekly and monthly attendance graphs.</td></tr><tr><td>3</td><td>Student clicks on "View Attendance Analysis."</td><td>System shows overall attendance percentage.</td></tr></table>	Step	Actor	System	1	Student clicks on "View Attendance Analysis."	System fetches attendance data from database.	2	Student clicks on "View Attendance Analysis."	System displays weekly and monthly attendance graphs.	3	Student clicks on "View Attendance Analysis."	System shows overall attendance percentage.
Step	Actor	System											
1	Student clicks on "View Attendance Analysis."	System fetches attendance data from database.											
2	Student clicks on "View Attendance Analysis."	System displays weekly and monthly attendance graphs.											
3	Student clicks on "View Attendance Analysis."	System shows overall attendance percentage.											

	<table><tr><td>4</td><td>Student clicks on "View Excel Sheets" for detailed record.</td><td>System downloads attendance report in Excel format.</td></tr></table>	4	Student clicks on "View Excel Sheets" for detailed record.	System downloads attendance report in Excel format.						
4	Student clicks on "View Excel Sheets" for detailed record.	System downloads attendance report in Excel format.								
Alternate Flows	Alternate Flow 1: No Data Available <table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Student clicks on attendance analysis.</td><td>System finds no attendance records.</td></tr><tr><td>2</td><td>User clicks on the view button</td><td>System shows "No attendance data found."</td></tr></table>	Step	Actor	System	1	Student clicks on attendance analysis.	System finds no attendance records.	2	User clicks on the view button	System shows "No attendance data found."
Step	Actor	System								
1	Student clicks on attendance analysis.	System finds no attendance records.								
2	User clicks on the view button	System shows "No attendance data found."								
Exceptions	1. No Records: System shows "Attendance not available for this semester." 2. Download Failed: System shows "Error downloading file. Try again." 3. Low Attendance Warning: If attendance below 70%, system highlights in red and shows warning message.									
Qualities	1. Visual Representation: Graphs for easy understanding. 2. Accuracy: Shows correct attendance percentage. 3. Export Option: Allows download of detailed records. 4. Alert System: Automatically warns if attendance is low.									

UC-01-004: Register Courses

Element	Description						
ID	UC-01-004						
Name	Register Courses						
Participating Actors	Student						
Goals	To enroll in semester courses.						
Triggers	Student clicks on "Course Registration" from dashboard during registration period.						
Pre-conditions	1. Student is logged into the system. 2. Registration period is active. 3. Admin has uploaded available courses.						
Post-conditions	Student is enrolled in selected courses.						
Basic Flow	<table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Student clicks on "Register Courses."</td><td>System displays list of available courses with timings and teachers.</td></tr></table>	Step	Actor	System	1	Student clicks on "Register Courses."	System displays list of available courses with timings and teachers.
Step	Actor	System					
1	Student clicks on "Register Courses."	System displays list of available courses with timings and teachers.					

	2	Student enters Course ID or Name for desired course.	System validates course availability.
	3	Student clicks "Register" button.	System checks for schedule conflicts and credit limits.
	4	Student clicks "Register" button.	System confirms enrollment and adds course to student's schedule.
	5	Student clicks "Register" button.	System shows success message: "Course Registered Successfully."
Alternate Flows	Alternate Flow 1: Drop Course		
	Step	Actor	System

	<table><tr><td>1</td><td>Student selects already registered course.</td><td>System shows "Drop Course" option.</td></tr><tr><td>2</td><td>Student clicks "Drop."</td><td>System removes course from student's schedule.</td></tr></table>	1	Student selects already registered course.	System shows "Drop Course" option.	2	Student clicks "Drop."	System removes course from student's schedule.
1	Student selects already registered course.	System shows "Drop Course" option.					
2	Student clicks "Drop."	System removes course from student's schedule.					
Exceptions	1. Schedule Conflict: System shows "Time conflict with already registered course." 2. Registration Closed: System shows "Registration period has ended." 3. Course Full: System shows "Course seats full. Choose another section." 4. Prerequisite Not Met: System shows "You have not completed prerequisite course." 5. Credit Limit Exceeded: System shows "Maximum credit hours reached."						
Qualities	1. Timely Access: Registration available only during specified dates. 2. Conflict Check: Automatically detects time and credit conflicts. 3. Confirmation: Sends email/notification after successful registration. 4. Response Time: Registration completes within 3 seconds.						

UC-01-005: Apply for Leave

Element	Description												
ID	UC-01-005												
Name	Apply for Leave												
Participating Actors	Student, Teacher												
Goals	To submit leave application for approval.												
Triggers	User clicks on "Apply Leave" from dashboard.												
Pre-conditions	1. User is logged into the system. 2. User has valid leave balance (if applicable).												
Post-conditions	Leave request is submitted to concerned authority for approval.												
Basic Flow	<table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>User clicks on "Apply Leave."</td><td>System displays leave application form.</td></tr><tr><td>2</td><td>User selects leave date.</td><td>System validates date (not a holiday/weekend).</td></tr><tr><td>3</td><td>User selects leave type</td><td>System records leave type.</td></tr></table>	Step	Actor	System	1	User clicks on "Apply Leave."	System displays leave application form.	2	User selects leave date.	System validates date (not a holiday/weekend).	3	User selects leave type	System records leave type.
Step	Actor	System											
1	User clicks on "Apply Leave."	System displays leave application form.											
2	User selects leave date.	System validates date (not a holiday/weekend).											
3	User selects leave type	System records leave type.											

	<table><tr><td></td><td>(Half/Full).</td><td></td></tr><tr><td>4</td><td>User enters reason/description.</td><td>System saves the reason.</td></tr><tr><td>5</td><td>User clicks "APPLY" button.</td><td>System submits request and shows "Application Submitted Successfully."</td></tr></table>		(Half/Full).		4	User enters reason/description.	System saves the reason.	5	User clicks "APPLY" button.	System submits request and shows "Application Submitted Successfully."			
	(Half/Full).												
4	User enters reason/description.	System saves the reason.											
5	User clicks "APPLY" button.	System submits request and shows "Application Submitted Successfully."											
Alternate Flows	Alternate Flow 1: Teacher Applying Leave <table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Teacher fills leave form.</td><td>System routes request to HOD/Admin for approval.</td></tr></table> Alternate Flow 2: Student Applying Leave <table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Student fills leave form.</td><td>System routes request to Teacher/Admin for approval.</td></tr></table>	Step	Actor	System	1	Teacher fills leave form.	System routes request to HOD/Admin for approval.	Step	Actor	System	1	Student fills leave form.	System routes request to Teacher/Admin for approval.
Step	Actor	System											
1	Teacher fills leave form.	System routes request to HOD/Admin for approval.											
Step	Actor	System											
1	Student fills leave form.	System routes request to Teacher/Admin for approval.											
Exceptions	1. Invalid Date: System shows "Please select a valid working day."												

	2. Missing Reason: System shows "Please provide reason for leave." 3. Leave Already Applied: System shows "Leave already applied for this date." 4. No Leave Balance: System shows "Insufficient leave balance."
Qualities	1. Tracking: User can check application status (Pending/Approved/Rejected). 2. Notification: Email/alert sent upon approval or rejection. 3. Record Keeping: All leave history maintained for future reference.

UC-01-006: View DMC

Element	Description												
ID	UC-01-006												
Name	View DMC/Results												
Participating Actors	Student/Admin												
Goals	To view Detailed Marks Certificate (DMC) and academic results.												
Triggers	Student clicks on "View DMC" or "Results" from dashboard.												
Pre-conditions	1. Student is logged into the system. 2. Results have been uploaded by teacher/admin.												
Post-conditions	Student views their DMC with semester results.												
Basic Flow	<table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Student clicks on "View DMC."</td><td>System fetches result data from database.</td></tr><tr><td>2</td><td>Student clicks on "View DMC."</td><td>System displays DMC with course names, marks obtained, total marks, and grades.</td></tr><tr><td>3</td><td>Student clicks on "View DMC."</td><td>System shows semester GPA and overall CGPA.</td></tr></table>	Step	Actor	System	1	Student clicks on "View DMC."	System fetches result data from database.	2	Student clicks on "View DMC."	System displays DMC with course names, marks obtained, total marks, and grades.	3	Student clicks on "View DMC."	System shows semester GPA and overall CGPA.
Step	Actor	System											
1	Student clicks on "View DMC."	System fetches result data from database.											
2	Student clicks on "View DMC."	System displays DMC with course names, marks obtained, total marks, and grades.											
3	Student clicks on "View DMC."	System shows semester GPA and overall CGPA.											

	<table><tr><td>4</td><td>Student can download/print DMC (optional).</td><td>System generates PDF for download.</td></tr></table>	4	Student can download/print DMC (optional).	System generates PDF for download.												
4	Student can download/print DMC (optional).	System generates PDF for download.														
Alternate Flows	Alternate Flow 1: Select Semester <table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Student selects specific semester from dropdown.</td><td>System displays DMC for selected semester only.</td></tr></table> Alternate Flow 2: Results Not Announced <table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Student clicks on View DMC.</td><td>System checks if results are published.</td></tr><tr><td>2</td><td>Student clicks on "View DMC."</td><td>If not published, system shows "Results not announced yet."</td></tr></table>	Step	Actor	System	1	Student selects specific semester from dropdown.	System displays DMC for selected semester only.	Step	Actor	System	1	Student clicks on View DMC.	System checks if results are published.	2	Student clicks on "View DMC."	If not published, system shows "Results not announced yet."
Step	Actor	System														
1	Student selects specific semester from dropdown.	System displays DMC for selected semester only.														
Step	Actor	System														
1	Student clicks on View DMC.	System checks if results are published.														
2	Student clicks on "View DMC."	If not published, system shows "Results not announced yet."														
Exceptions	1. No Results Found: System shows "No DMC available for selected semester." 2. Download Error: System shows "Error generating PDF. Try again."															

	3. Server Error: System shows "Failed to load results. Contact Admin."
Qualities	1. Accuracy: Shows correct marks, GPA, and CGPA as per records. 2. Security: Only logged-in student can view their own DMC. 3. Download Option: Allows saving DMC as PDF for future use. 4. Response Time: DMC loads within 3 seconds.

UC-01-007: Manage Fee Challan

Element	Description												
ID	UC-01-007												
Name	Manage Fee Challan												
Participating Actors	Student, Admin												
Goals	Student views and pays fee; Admin generates and manages fee challans.												
Triggers	User clicks on "Fee Challan" or "Fee Management" from dashboard.												
Pre-conditions	1. User is logged into the system. 2. Fee records exist for current semester.												
Post-conditions	Student views fee details; Admin updates fee status.												
Basic Flow	<table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Student clicks on "View Fee Challan."</td><td>System fetches fee record from database.</td></tr><tr><td>2</td><td>Student clicks on "View DMC."</td><td>System displays total fee amount and payment status (Paid/Unpaid).</td></tr><tr><td>3</td><td>Student clicks on "Pay Fee" (if unpaid).</td><td>System shows payment options.</td></tr></table>	Step	Actor	System	1	Student clicks on "View Fee Challan."	System fetches fee record from database.	2	Student clicks on "View DMC."	System displays total fee amount and payment status (Paid/Unpaid).	3	Student clicks on "Pay Fee" (if unpaid).	System shows payment options.
Step	Actor	System											
1	Student clicks on "View Fee Challan."	System fetches fee record from database.											
2	Student clicks on "View DMC."	System displays total fee amount and payment status (Paid/Unpaid).											
3	Student clicks on "Pay Fee" (if unpaid).	System shows payment options.											

	<table><tr><td>4</td><td>Student uploads payment slip after bank transaction.</td><td>System saves slip and updates status to "Under Verification."</td></tr></table>	4	Student uploads payment slip after bank transaction.	System saves slip and updates status to "Under Verification."															
	4	Student uploads payment slip after bank transaction.	System saves slip and updates status to "Under Verification."																
	<table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Admin clicks on "Upload Fee Challan."</td><td>System displays fee management form.</td></tr><tr><td>2</td><td>Admin enters Student Roll No, Total Fee, Scholarship (if any).</td><td>System calculates final amount after discount.</td></tr><tr><td>3</td><td>Admin clicks "UPLOAD."</td><td>System saves challan and makes it visible to student.</td></tr><tr><td>4</td><td>Admin clicks on "View Challan Details."</td><td>System shows all students with fee status.</td></tr><tr><td>5</td><td>Admin verifies payment slip and checks "Paid" box.</td><td>System updates status to "Paid."</td></tr></table>	Step	Actor	System	1	Admin clicks on "Upload Fee Challan."	System displays fee management form.	2	Admin enters Student Roll No, Total Fee, Scholarship (if any).	System calculates final amount after discount.	3	Admin clicks "UPLOAD."	System saves challan and makes it visible to student.	4	Admin clicks on "View Challan Details."	System shows all students with fee status.	5	Admin verifies payment slip and checks "Paid" box.	System updates status to "Paid."
	Step	Actor	System																
	1	Admin clicks on "Upload Fee Challan."	System displays fee management form.																
	2	Admin enters Student Roll No, Total Fee, Scholarship (if any).	System calculates final amount after discount.																
3	Admin clicks "UPLOAD."	System saves challan and makes it visible to student.																	
4	Admin clicks on "View Challan Details."	System shows all students with fee status.																	
5	Admin verifies payment slip and checks "Paid" box.	System updates status to "Paid."																	
Alternate Flows	Alternate Flow 1: Student Views Paid Challan																		

	<table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Student clicks on Fee Challan.</td><td>System shows "Paid" status and payment date.</td></tr></table> Alternate Flow 2: Scholarship Applied <table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Admin enters scholarship amount.</td><td>System automatically calculates discounted fee.</td></tr></table>	Step	Actor	System	1	Student clicks on Fee Challan.	System shows "Paid" status and payment date.	Step	Actor	System	1	Admin enters scholarship amount.	System automatically calculates discounted fee.
Step	Actor	System											
1	Student clicks on Fee Challan.	System shows "Paid" status and payment date.											
Step	Actor	System											
1	Admin enters scholarship amount.	System automatically calculates discounted fee.											
Exceptions	1. No Challan Found: System shows "No fee challan generated for current semester." 2. Invalid Roll No: System shows "Student not found." 3. Payment Failed: System shows "Payment verification failed. Upload clear slip." 4. Duplicate Entry: System shows "Challan already exists for this student."												
Qualities	1. Accuracy: Correct fee calculation with discounts. 2. Verification: Admin verifies payment before marking as paid. 3. Record Keeping: Complete history of fee payments maintained.												

UC-01-008: Register in Society

Element	Description												
ID	UC-01-008												
Name	Register in Society												
Participating Actors	Student, Society Head												
Goals	Student applies for society membership; Society Head approves/rejects request.												
Triggers	Student clicks on "Register in Society" from dashboard.												
Pre-conditions	1. Student is logged into the system. 2. Societies are created by Admin/Society Head.												
Post-conditions	Student's membership request is submitted for approval.												
Basic Flow	<table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Student clicks on "Register in Society."</td><td>System displays list of available societies.</td></tr><tr><td>2</td><td>Student selects a society (e.g., Karekamal, Sports, Drama).</td><td>System shows society details and registration form.</td></tr><tr><td>3</td><td>Student enters Roll No and uploads CV (optional).</td><td>System saves the information.</td></tr></table>	Step	Actor	System	1	Student clicks on "Register in Society."	System displays list of available societies.	2	Student selects a society (e.g., Karekamal, Sports, Drama).	System shows society details and registration form.	3	Student enters Roll No and uploads CV (optional).	System saves the information.
Step	Actor	System											
1	Student clicks on "Register in Society."	System displays list of available societies.											
2	Student selects a society (e.g., Karekamal, Sports, Drama).	System shows society details and registration form.											
3	Student enters Roll No and uploads CV (optional).	System saves the information.											

	4	Student enters description (skills/interests).	System records description.
	5	Student clicks "APPLY."	System submits request and shows "Application Submitted. Awaiting Approval."
(Society Head View)			
	Step	Actor	System
	1	Society Head clicks on "View Membership Requests."	System displays list of pending student applications.
	2	Society Head reviews student details and CV.	System shows complete application.
	3	Society Head clicks "ACCEPT" or "REJECT."	System updates membership status and notifies student.
Alternate Flows	Step	Actor	System

	<table><tr><td>1</td><td>Student tries to apply for same society.</td><td>System checks if already member.</td></tr><tr><td>2</td><td>Student tries to apply for same society.</td><td>If already member, system shows "You are already a member of this society."</td></tr></table>	1	Student tries to apply for same society.	System checks if already member.	2	Student tries to apply for same society.	If already member, system shows "You are already a member of this society."
1	Student tries to apply for same society.	System checks if already member.					
2	Student tries to apply for same society.	If already member, system shows "You are already a member of this society."					
Exceptions	1. No Societies Available: System shows "No societies found for registration." 2. Missing CV: If required, system shows "Please upload CV to apply." 3. Request Already Sent: System shows "You have already applied for this society." 4. Invalid Roll No: System shows "Roll number not found in student records."						
Qualities	1. Tracking: Student can check application status (Pending/Approved/Rejected). 2. Notification: Student receives alert when membership is approved. 3. Document Upload: Supports CV upload in PDF/Image format. 4. Organization: All society members data maintained centrally.						

UC-02-001: Mark Class Attendance

Element	Description												
ID	UC-02-001												
Name	Mark Class Attendance												
Participating Actors	Teacher												
Goals	To record student attendance for a specific class.												
Triggers	Teacher clicks on "Mark Attendance" from dashboard.												
Pre-conditions	1. Teacher is logged into the system. 2. Teacher is assigned to a section/course. 3. Class schedule exists for the day.												
Post-conditions	Student attendance is recorded in the system.												
Basic Flow	<table><tr><th>S t e p</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Teacher clicks on "Mark Attendance."</td><td>System displays list of assigned sections/courses.</td></tr><tr><td>2</td><td>Teacher selects section and date.</td><td>System fetches student list for that section.</td></tr><tr><td>3</td><td>Teacher marks students as Present/Absent.</td><td>System records each student's status.</td></tr></table>	S t e p	Actor	System	1	Teacher clicks on "Mark Attendance."	System displays list of assigned sections/courses.	2	Teacher selects section and date.	System fetches student list for that section.	3	Teacher marks students as Present/Absent.	System records each student's status.
S t e p	Actor	System											
1	Teacher clicks on "Mark Attendance."	System displays list of assigned sections/courses.											
2	Teacher selects section and date.	System fetches student list for that section.											
3	Teacher marks students as Present/Absent.	System records each student's status.											

	<table><tr><td>4</td><td>Teacher clicks "Submit" or "APPLY."</td><td>System saves attendance and shows success message.</td></tr><tr><td>5</td><td></td><td>System updates attendance records for all students.</td></tr></table>	4	Teacher clicks "Submit" or "APPLY."	System saves attendance and shows success message.	5		System updates attendance records for all students.									
4	Teacher clicks "Submit" or "APPLY."	System saves attendance and shows success message.														
5		System updates attendance records for all students.														
Alternate Flows	Alternate Flow 1: Bulk Upload <table><tr><td>S t e p</td><td>Actor</td><td>System</td></tr><tr><td>1</td><td>Teacher clicks "Enter Here to get Excel Sheet."</td><td>System generates Excel template with student list.</td></tr><tr><td>2</td><td>Teacher marks attendance in Excel and uploads.</td><td>System reads file and updates records.</td></tr></table> Alternate Flow 2: Edit Attendance <table><tr><td>S t e p</td><td>Actor</td><td>System</td></tr><tr><td>1</td><td>Teacher selects a date with already marked</td><td>System displays existing</td></tr></table>	S t e p	Actor	System	1	Teacher clicks "Enter Here to get Excel Sheet."	System generates Excel template with student list.	2	Teacher marks attendance in Excel and uploads.	System reads file and updates records.	S t e p	Actor	System	1	Teacher selects a date with already marked	System displays existing
S t e p	Actor	System														
1	Teacher clicks "Enter Here to get Excel Sheet."	System generates Excel template with student list.														
2	Teacher marks attendance in Excel and uploads.	System reads file and updates records.														
S t e p	Actor	System														
1	Teacher selects a date with already marked	System displays existing														

	<table><tr><td></td><td>attendance.</td><td>attendance.</td></tr><tr><td>2</td><td>Teacher makes changes and resubmits.</td><td>System updates records with new data.</td></tr></table>		attendance.	attendance.	2	Teacher makes changes and resubmits.	System updates records with new data.
	attendance.	attendance.					
2	Teacher makes changes and resubmits.	System updates records with new data.					
Exceptions	1. No Student List: System shows "No students enrolled in this section." 2. Invalid Date: System shows "Cannot mark attendance for future dates." 3. Attendance Already Marked: System shows "Attendance already recorded for this date. Edit if needed." 4. Empty Submission: System shows "Please mark attendance before submitting." 5. File Upload Error: System shows "Invalid file format. Upload Excel file only."						
Qualities	1. Accuracy: Correctly marks each student's attendance. 2. Flexibility: Manual marking or Excel bulk upload options. 3. Real-time Update: Attendance visible to students immediately. 4. History: Previous attendance records can be viewed and edited.						

UC-02-002: Upload DMC

Element	Description
ID	UC-02-002
Name	Upload DMC
Participating Actors	Teacher
Goals	To upload student marks and generate Detailed Marks Certificate.
Triggers	Teacher clicks on "Upload DMC" from dashboard.
Pre-conditions	1. Teacher is logged into the system. 2. Teacher is assigned to a course/section. 3. Examinations are completed.
Post-conditions	Student marks are saved and DMC is generated.

Basic Flow

Step	Actor	System
1	Teacher clicks on "Upload DMC."	System displays course/section selection.
2	Teacher selects section.	System shows student list of that section.
3	Teacher selects student Roll No.	System displays marks entry form.
4	Teacher enters marks for Quiz 1, Mids, Quiz 2, Final.	System validates marks (within max limits).
5	Teacher clicks "UPLOAD."	System saves marks and calculates total, percentage, and grade.
6		System generates DMC and makes it visible to student.

**Alternate
Flows**

Alternate Flow 1: Add Optional Subjects

Step	Actor	System
1	Teacher clicks "Add Optional."	System adds extra fields for additional assessments.

Alternate Flow 2: Bulk Upload

Step	Actor	System
1	Teacher downloads Excel template.	System generates template with student list.
2	Teacher enters all marks in Excel and uploads.	System reads file and updates all students at once.

Exceptions	1. Invalid Marks: System shows "Marks exceed maximum limit. Please check." 2. Missing Entry: System shows "Please fill all required fields." 3. Student Not Found: System shows "Roll number does not exist in this section." 4. Duplicate Entry: System shows "Marks already uploaded for this student. Edit if needed." 5. File Format Error: System shows "Invalid file. Use provided template."
Qualities	1. Accuracy: Automatic grade calculation based on marks. 2. Efficiency: Bulk upload option for faster entry. 3. Security: Only assigned teacher can upload marks. 4. Timely Access: Students can view DMC immediately after upload.

UC-02-003: Manage Meeting Requests

Element	Description
ID	UC-02-003
Name	Manage Meeting Requests
Participating Actors	Student, Teacher
Goals	Student requests meeting with teacher; Teacher accepts or rejects request.
Triggers	Student clicks on "Request Meeting" from dashboard.
Pre-conditions	1. Both student and teacher are logged into the system. 2. Teacher has set available slots (optional).
Post-conditions	Meeting is scheduled if teacher accepts request.

Basic Flow

Step	Actor	System
1	Student clicks on "Request Meeting."	System displays teacher list.
2	Student selects teacher from list.	System shows teacher's available slots (if any).
3	Student selects date and time.	System checks for conflicts with student's class schedule.
4	Student enters meeting purpose/desc ription.	System saves description.
5	Student clicks "APPLY."	System submits request and shows "Meeting request sent. Awaiting approval."

Basic Flow (Teacher View)

Step	Actor	System
1	Teacher clicks on "View Meeting Requests."	System displays list of pending requests with student Roll No and purpose.
2	Teacher reviews request details.	System shows complete request.
3	Teacher clicks "Accept" or "Reject."	System updates request status and notifies student.
4	Teacher clicks "Accept" or "Reject."	If accepted, meeting is added to both calendars.

Alternate Flows	Alternate Flow 1: Reschedule by Teacher <table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Teacher cannot accept requested time.</td><td>Teacher selects "Reschedule" and suggests new time.</td></tr><tr><td>2</td><td>Student receives notification.</td><td>System shows new time proposal to student.</td></tr><tr><td>3</td><td>Student accepts or rejects new time.</td><td>System updates status accordingly.</td></tr></table>	Step	Actor	System	1	Teacher cannot accept requested time.	Teacher selects "Reschedule" and suggests new time.	2	Student receives notification.	System shows new time proposal to student.	3	Student accepts or rejects new time.	System updates status accordingly.
Step	Actor	System											
1	Teacher cannot accept requested time.	Teacher selects "Reschedule" and suggests new time.											
2	Student receives notification.	System shows new time proposal to student.											
3	Student accepts or rejects new time.	System updates status accordingly.											
Exceptions	· No Slots Available: System shows "Teacher has no available slots on selected date." · Schedule Conflict: System shows "You have a class at this time. Select different slot." · Request Already Sent: System shows "You already have a pending request for this teacher." · Teacher Not Available: System shows "Teacher is on leave. Select another date." · Missing Description: System shows "Please enter meeting purpose."												

Qualities	· Conflict-Free: Checks student and teacher schedules automatically. · Notification: Both parties receive email/alert for request and response. · Tracking: Student can check request status (Pending/Accepted/Rejected). · Calendar Integration: Accepted meetings appear in both calendars.
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UC-02-004: Manage User Registration

Element	Description
ID	UC-02-004
Name	Manage User Registration
Participating Actors	Admin
Goals	To register new users (students, teachers, staff) in the system.
Triggers	Admin clicks on "Register User" from dashboard.
Pre-conditions	1. Admin is logged into the system. 2. User details are ready for registration.
Post-conditions	New user is created and can login to the system.

Basic Flow

Step	Actor	System
1	Admin clicks on "Register Student" or "Register User."	System displays registration form.
2	Admin selects user role (Student/Teacher/Staff).	System shows role-specific fields.
3	Admin enters Roll No/Employee ID, Name, Department, Section, etc.	System validates input format.
4	Admin clicks "REGISTER."	System creates user account with default password.
5	Admin clicks "REGISTER."	System shows "User Registered Successfully" with login details.

Alternate Flows	Alternate Flow 1: Bulk Registration <table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Admin downloads Excel template.</td><td>System generates template.</td></tr><tr><td>2</td><td>Admin fills multiple users and uploads.</td><td>System registers all users at once.</td></tr></table> Alternate Flow 2: Edit User <table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Admin searches for existing user.</td><td>System displays user details.</td></tr><tr><td>2</td><td>Admin updates information and saves.</td><td>System updates user record.</td></tr></table>	Step	Actor	System	1	Admin downloads Excel template.	System generates template.	2	Admin fills multiple users and uploads.	System registers all users at once.	Step	Actor	System	1	Admin searches for existing user.	System displays user details.	2	Admin updates information and saves.	System updates user record.
	Step	Actor	System																
1	Admin downloads Excel template.	System generates template.																	
2	Admin fills multiple users and uploads.	System registers all users at once.																	
Step	Actor	System																	
1	Admin searches for existing user.	System displays user details.																	
2	Admin updates information and saves.	System updates user record.																	
Exceptions	1. Duplicate Entry: System shows "User with this Roll No/ID already exists." 2. Invalid Format: System shows "Please enter valid data in all fields." 3. Missing Fields: System shows "All fields are required." 4. File Error: System shows "Invalid file format. Use Excel template."																		

Qualities	1. Efficiency: Bulk registration option available. 2. Security: Default password requires change on first login. 3. Organization: All users stored with proper roles and departments.
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UC-03-001: Generate Timetable

Element	Description
ID	UC-03-001
Name	Generate Timetable
Participating Actors	Admin
Goals	To create and manage class schedules for sections and teachers.
Triggers	Admin clicks on "Generate Timetable" from dashboard.
Pre-conditions	1. Admin is logged into the system. 2. Courses, teachers, and sections are already registered.
Post-conditions	Timetable is created and visible to students and teachers.

Basic Flow

Step	Actor	System
1	Admin clicks on "Generate Timetable."	System displays timetable creation form.
2	Admin selects section and day.	System shows empty timetable slots.
3	Admin enters Lecture No, Start Time, End Time.	System saves time slots.
4	Admin assigns Course and Teacher to each slot.	System validates no conflicts.
5	Admin clicks "Generate" or "Save."	System creates timetable and publishes it.
6	Admin clicks "Generate" or "Save."	System sends notification to concerned students and teachers.

Alternate Flows	Alternate Flow 1: Edit Timetable									
	<table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Admin selects existing timetable.</td><td>System displays current schedule.</td></tr><tr><td>2</td><td>Admin makes changes and saves.</td><td>System updates timetable.</td></tr></table>	Step	Actor	System	1	Admin selects existing timetable.	System displays current schedule.	2	Admin makes changes and saves.	System updates timetable.
	Step	Actor	System							
	1	Admin selects existing timetable.	System displays current schedule.							
2	Admin makes changes and saves.	System updates timetable.								
Alternate Flow 2: View as Student/Teacher										
<table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Admin selects "Student" or "Teacher" view from dropdown.</td><td>System displays timetable from selected role perspective.</td></tr></table>	Step	Actor	System	1	Admin selects "Student" or "Teacher" view from dropdown.	System displays timetable from selected role perspective.				
Step	Actor	System								
1	Admin selects "Student" or "Teacher" view from dropdown.	System displays timetable from selected role perspective.								
Exceptions	1. Teacher Conflict: System shows "Teacher already assigned to another class at this time." 2. Room Conflict: System shows "Room already in use for selected time." 3. Missing Data: System shows "Please assign teacher and course to all slots." 4. Overlapping Slots: System shows "Lecture timings cannot overlap."									

Qualities	1. Conflict-Free: Automatically detects teacher and room conflicts. 2. Flexibility: Easy to edit and update when needed. 3. Visibility: Available to students and teachers immediately after generation. 4. Organization: Separate timetables for each section and teacher.
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UC-03-002: Approve Leave

Element	Description
ID	UC-03-002
Name	Approve Leave
Participating Actors	Admin, HOD
Goals	To review and approve/reject leave applications from students and teachers.
Triggers	Admin clicks on "Approve Leave" from dashboard.
Pre-conditions	1. Admin is logged into the system. 2. Leave applications are submitted by users.
Post-conditions	Leave status is updated (Approved/Rejected) and applicant is notified.

Basic Flow			
	Step	Actor	System
	1	Admin clicks on "Approve Leave."	System displays list of pending leave applications.
	2	Admin views applications with ID, Date, Type, and Reason.	System shows complete details of each request.
	3	Admin selects an application to review.	System displays full application.
	4	Admin checks the "Approve" checkbox or clicks "Reject."	System records decision.
	5	Admin checks the "Approve" checkbox or clicks "Reject."	System updates leave status and notifies applicant.

Alternate Flows	Alternate Flow 1: Filter by Type <table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Admin selects filter (Student/Teacher).</td><td>System shows only selected category applications.</td></tr></table> Alternate Flow 2: Reject with Reason <table><tr><th>Step</th><th>Actor</th><th>System</th></tr><tr><td>1</td><td>Admin clicks "Reject."</td><td>System asks for rejection reason.</td></tr><tr><td>2</td><td>Admin enters reason.</td><td>System saves reason and notifies applicant.</td></tr></table>	Step	Actor	System	1	Admin selects filter (Student/Teacher).	System shows only selected category applications.	Step	Actor	System	1	Admin clicks "Reject."	System asks for rejection reason.	2	Admin enters reason.	System saves reason and notifies applicant.
	Step	Actor	System													
	1	Admin selects filter (Student/Teacher).	System shows only selected category applications.													
Step	Actor	System														
1	Admin clicks "Reject."	System asks for rejection reason.														
2	Admin enters reason.	System saves reason and notifies applicant.														
Exception	1. No Pending Applications: System shows "No pending leave requests." 2. Already Processed: System shows "This leave request is already processed." 3. Invalid Selection: System shows "Please select an application to process."															
Qualities	1. Efficiency: All pending requests in one place. 2. Transparency: Applicant can see status and reason if rejected. 3. Record Keeping: All leave history maintained. 4. Notification: Automatic alerts sent on status change.															

UC-05-001: Generate Emergency Notification

Element	Description		
ID	UC-03-003		
Name	Generate Emergency Notification		
Participating Actors	Admin, Teacher, Society Head		
Goals	To broadcast urgent alerts and announcements to users.		
Triggers	User clicks on "Generate Emergency Notification" from dashboard.		
Pre-conditions	1. User is logged into the system with appropriate role. 2. There is an urgent announcement to broadcast.		
Post-conditions	Emergency notification is sent and visible to all relevant users.		
Basic Flow			
	Step	Actor	System
	1	User clicks on "Generate Emergency Notification."	System displays notification creation form.
	2	User selects notification type (Emergency/General).	System prepares appropriate template.

	3	User enters notification message.	System saves message.
	4	User selects target audience (All/Students/Teachers/Specific Society).	System records audience selection.
	5	User clicks "Send."	System broadcasts notification immediately.
	6	User clicks "Send."	System shows "Notification Sent Successfully."
Alternate Flows	Alternate Flow 1: Use Template		
	Step	Actor	System
	1	User selects from pre-defined templates (e.g., "Class Cancelled," "University Closed").	System auto-fills message.
	2	User edits if needed and sends.	System broadcasts notification.
	Alternate Flow 2: Schedule Notification		
	Step	Actor	System

	<table><tr><td>1</td><td>User sets future date and time.</td><td>System saves notification and sends at scheduled time.</td></tr></table>	1	User sets future date and time.	System saves notification and sends at scheduled time.
1	User sets future date and time.	System saves notification and sends at scheduled time.		
Exceptions	1. Empty Message: System shows "Please enter notification message." 2. No Audience Selected: System shows "Please select target audience." 3. Network Error: System shows "Failed to send notification. Try again."			
Qualities	1. Real-time: Notifications appear instantly on user dashboards. 2. Priority: Emergency alerts shown with red highlight at top. 3. Targeted: Can send to specific user groups. 4. Persistence: Important notifications remain visible until dismissed.			

UC-04-001: View Survey Results

Element	Description
ID	UC-04-001
Name	View Survey Results
Participating Actors	Admin
Goals	To analyze feedback and survey responses submitted by students for teachers.
Triggers	Admin clicks on "View Survey" from dashboard.
Pre-conditions	1. Admin is logged into the system. 2. Students have submitted teacher surveys/feedback.
Post-conditions	Admin views survey results and statistics.

Basic Flow

Step	Actor	System
1	Admin clicks on "View Survey" or "View Survey Results."	System displays search options.
2	Admin enters Teacher ID to view specific teacher's feedback.	System fetches all survey responses for that teacher.
3		System displays graphical statistics (charts/graphs) showing ratings.
4	Admin enters Student Roll No (optional).	System shows specific student's feedback.
5		System displays detailed comments and descriptions.

Alternate Flows	Alternate Flow 1: View All Teachers Summary		
	Step	Actor	System
	1	Admin selects "Overall Summary."	System shows comparison charts of all teachers.
	Alternate Flow 2: Export Results		
	Step	Actor	System
	1	Admin clicks "Export."	System generates Excel/PDF report of survey results.
Exceptions	· No Data Found: System shows "No survey results available for selected teacher." · Invalid ID: System shows "Teacher ID not found." · Empty Results: System shows "No surveys submitted yet."		
Qualities	1. Visual Representation: Graphs and charts for easy analysis. 2. Anonymity: Student identities are hidden in results. 3. Comprehensive: Shows both ratings and comments. 4. Export Option: Reports can be downloaded for record-keeping.		